

Protocol

This trial protocol has been provided by the authors to give readers additional information about their work.

Protocol for: Sweeney CJ, Chen Y-H, Carducci M, et al. Chemohormonal therapy in metastatic hormone-sensitive prostate cancer. *N Engl J Med* 2015;373:737-46. DOI: 10.1056/NEJMoa1503747

This supplement contains the following items:

- 1. Original protocol, final protocol, summary of changes.**
- 2. Original statistical design, amendments, final statistical design**



Eastern Cooperative Oncology Group

Coordinating Center

Frontier Science

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Jean MacDonald, Director of Research Operations
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FILE COPY

TO: ECOG Clinical Research Associates and Investigators with Patients for E3805
FROM: ECOG Coordinating Center
DATE: July 28, 2006
SUBJECT: ACTIVATION OF PROTOCOL E3805 - *CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer.*

This study has been reviewed and approved by the Central IRB
This study is now active.

DRUGS:

ECOG has entered into a clinical trial study agreement with Sanofi-Aventis Pharmaceuticals for E3805. Sanofi-Aventis Pharmaceuticals is providing the docetaxel (IND exempt) to participating institutions for use in this protocol.

After patient randomization, a supply of docetaxel may be obtained. Investigators must fax a signed and completed "E3805 Clinical Supplies Shipping Form" (Appendix VII) to the ECOG Coordinating Center, ATTN: Drug Team at 617-632-2063. Do not send drug requests directly to Sanofi-Aventis Pharmaceuticals. Allow 4-5 business days for receipt of the drug.

The following drugs are commercially available: LHRH agonists (such as leuprolide, goserelin, triptorelin) and Antiandrogens (flutamide and bicalutamide).

IND STATUS:

Docetaxel is a commercially marketed product which has been approved by the FDA for: the treatment of locally advanced or metastatic breast cancer that returns after any prior chemotherapy; locally advanced or metastatic non-small cell lung cancer that recurs after prior platinum-based chemotherapy; and in combination with prednisone for the treatment of advanced metastatic prostate cancer. However its use in patients eligible for this protocol with prostate cancer is not currently approved by the FDA. When used as directed by this protocol, docetaxel is classified as an "unapproved use of an approved agent" and, by definition, considered as an investigational agent. However, while it is not an indication currently approved by the FDA, the use of docetaxel in this protocol is exempt from the requirements of an IND as described under Title 21 CFR 312.2(b).

Institutions should use the package insert that accompanies the drug for information about the docetaxel.

SAFETY ALERTS:

Drug sponsors are obliged to forward all Serious Adverse Events (SAE) that occur on this protocol to the FDA. Therefore, the ECOG Coordinating Center will copy the drug sponsors on all Serious Adverse Event (SAE) reports received. Physicians will review the SAE report and if further clarification or questions regarding the patient status, outcome or treatment is required, the company will contact the institution directly. The ECOG Coordinating center should be copied on all correspondence between the company and the institution.

Should any SAE report received qualify as an amendment to the approved labeling relating to the safety information of docetaxel, the drug sponsor will forward a safety alert report to ECOG. ECOG will disseminate these safety alert reports to all ECOG investigators in the bimonthly group mailings. These reports should be forwarded to your Institutional Review Board (IRB)/Ethics Committee (EC) within 90 days for review. Reporting instructions are provided with each safety alert. These safety alerts and any correspondence to your IRB/EC should be maintained in your E3805 study files.

RANDOMIZATION:

ECOG Institutions

The following documents must be on file at the CTSU Regulatory Office/Coalition of National Cancer Cooperative Groups before any institution may enter a patient.

- CTSU Regulatory Transmittal Form
- Copy of the institution's IRB-approved informed consent document and written justification for
 - any changes made to the informed consent
- One of the following
 - CTSU IRB Certification Form OR
 - Signed HHS 310 Form OR
 - IRB Approval Letter

The investigator will register patients via the ECOG webpage, 24 hours a day, 7 days a week, using the Web-based Patient Registration Program (<https://webreg.ecog.org>).

RESEARCH LABORATORY STUDY SUBMISSIONS:

The submission of tumor tissue (section 10) and blood samples (sections 11.1 and 11.2) are requested for the correlative studies outlined in section 11. Samples are to be submitted from patients who have consented to participate in the research laboratory studies or banking for future research [see pgs 13-16 of the Model Consent, Appendix I].

QUALITY OF LIFE COMPONENT:

Quality of life will be evaluated 5 times during this study. The QOL assessment is mandatory. Please see Section 5.1.3 for the assessment schedule.

FORMS:

The forms to be submitted for this protocol are listed in the E3805 Forms Packet. It is the responsibility of the participating institution to copy the enclosed forms and maintain a supply of available forms for data submission.

At the request of the Patient Representative Committee we have added a Patient Thank You Letter (template) as Appendix III. Please take a moment to review the letter and consider using it at your institution.

Records Retention

FDA regulations (21 CFR 312.62) require clinical investigators to retain all trial-related documentation, including source documents, long enough to allow the sponsor to use the data to support marketing applications.

This study will be used in support of a US marketing application (New Drug Application), all records pertaining to the trial (including source documents) must be maintained for:

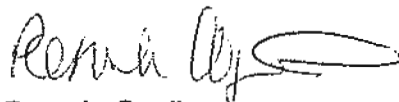
- two years after the FDA approves the marketing application, or
- two years after the FDA disapproves the application for the indication being studied,
or
- two years after the FDA is notified by the sponsor of the discontinuation of trials and that an application will not be submitted.

Please contact the ECOG Coordinating Center prior to destroying any source documents.

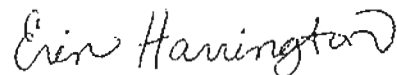
TOXICITY AND ADVERSE EVENT REPORTING:

This study will utilize the CTCAE version 3.0 for toxicity and Adverse Event reporting. A copy of the CTCAE version 3.0 should be available at your institution. It can be downloaded from the CTEP home page (<http://ctep.cancer.gov>). All appropriate treatment areas should have access to a copy of the CTCAE version 3.0.

If you have any questions regarding this protocol, including questions on eligibility, please contact the ECOG Study Chair, Dr. Christopher Sweeney, at (317) 274-3515 or the Study Chair Liaison, Anne Foster, R.N., at (317) 274-0951.



Pamela Coglianò
Manager, Protocol Development



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Eastern Cooperative Oncology Group

CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer

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STUDY STATISTICIAN: Wei Wang, Ph.D.
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PROSTATE SUBCOMMITTEE CHAIR: Michael Carducci, M.D.
OUTCOMES SUBCOMMITTEE CO-CHAIR: David Cella, Ph.D.
QOL CO-CHAIR: Linda Patrick-Miller, Ph.D.

Version Date: July 28, 2006

STUDY PARTICIPANTS

ECOG Entire Group

Patient enrollments from institutions that are not aligned with ECOG will be conducted via the NCI Cancer Trials Support Unit (CTSU) and all data should be sent to CTSU Data Operations unless otherwise specified in the CTSU logistical appendix.

ACTIVATION DATE

July 28, 2006

Addendum # 1 – Incorporated Prior to Activation

Update #1 – Incorporated Prior to Activation

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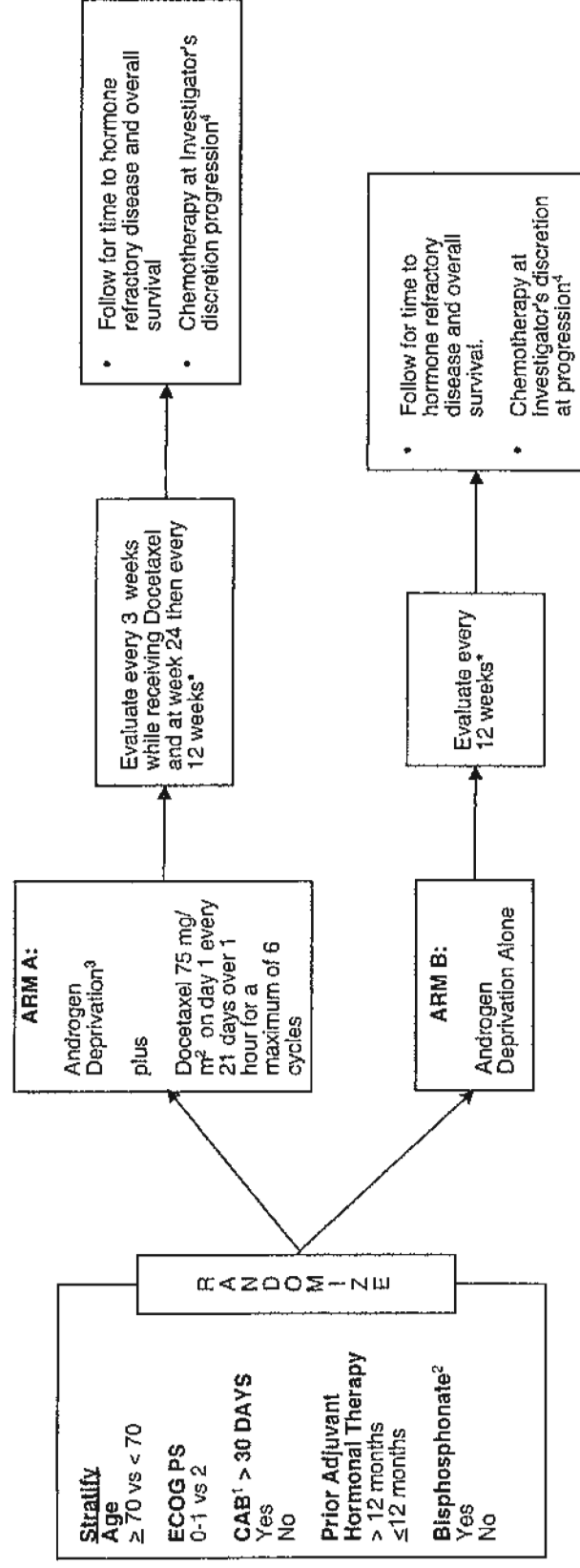
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CANCER TRIALS SUPPORT UNIT (CTSU) ADDRESS AND CONTACT INFORMATION

To submit site registration documents:	For patient enrollments:	To mail study forms or data:
CTSU Regulatory Office 1818 Market Street, Suite 1100 Philadelphia, PA 19103 Phone - 1-888-823-5923 Fax - 215-569-0206	CTSU Data Operations Center Voice Mail - 1-888-462-3009 Fax - 1-888-691-8039 [For CTSU patient enrollments that must be completed within approximately one hour or other extenuating circumstances, call 301-704-2376. Please use the 1-888-462-3009 number for ALL other CTSU patient enrollments.]	Westat CTSU Data Operations Center 1441 W. Montgomery Avenue Rockville, MD 20850-2062
For patient eligibility or treatment-related questions: Contact the Study PI of the Coordinating Group. The option remains to contact CTSU Help Desk for assistance in obtaining a response from the Group.		
All other questions (including forms-specific questions) should be communicated by phone or e-mail to the CTSU Help Desk at: CTSU General Information Line - 1-888-823-5923, or ctsucontact@westat.com . All calls and correspondence will be triaged to the appropriate CTSU representative.		
The CTSU Public Web site is located at: www.ctsu.org		
The CTSU Registered Member Web site is located at http://members.ctsu.org		

NOTE: CTSU logistical and registration information is found in Appendix IV.

Schema



1 cycle = 21 days
Accrual goal: 568

All patients will have a PSA every 12 weeks. If PSA test rises $> 50\%$ of lowest level recorded since starting hormonal therapy, progression will need to be confirmed by a repeat PSA test at least 4 weeks later.

Extensive Disease: Metastases involving any bony structure beyond the vertebral column and pelvis bone and any involvement of viscera. There must be four or more lesions. Patients with disease limited to vertebral column and/or pelvis alone with or without lymph node or lymph node only disease involvement are not eligible for this trial.

1 **CAB:** Combined Androgen Blockade. Investigators are not obliged to use CAB but must declare whether it will be continued beyond testosterone surge with LHRH agonists (i.e. 30 days) when starting LHRH agonist.

2 **Bisphosphonate:** Dose shown to decrease skeletal related events (e.g. zoledronic acid every 3 to 4 weeks) (Please see Section 5.4.2 for Supportive Care Guidelines relating to bisphosphonates).

3 **Hormonal Therapy:** Combined Androgen Blockade or monotherapy as surgical or medical castration at investigator's discretion. Investigators are reminded to use combined androgen blockade for four weeks if starting LHRH agonist therapy in men with conditions that could be made worse by an initial rise in testosterone (e.g.: impending spinal cord compression.) Hormonal therapy can be commenced up to 90 days prior to randomization.

Premedication is required to decrease or prevent acute anaphylactoid reactions to docetaxel, and to decrease the severity or delay the onset of late-occurring fluid retention problems. Dexamethasone 8 mg po 12, 3 and 1 hour prior to docetaxel infusion. Use of diphenhydramine is optional.
Docetaxel: 75 mg/m² of docetaxel will be given on day 1 every 21 days.

4 Type of therapy at time of hormone refractory disease is not being dictated but investigators are encouraged to treat with docetaxel every three weeks on BOTH ARMS so the effect of early versus delayed chemotherapy on overall survival can be assessed. If the investigator deems that the patient is not a suitable candidate for docetaxel chemotherapy, the reason for this decision will be recorded. All therapy for hormone refractory disease will be recorded, including anti-androgen withdrawal, second-line hormonal therapy and chemotherapy. If a patient has more than one therapy, then all therapies will be recorded. Patients will be followed until development of HRPC and for overall survival.

1. Introduction

1.1 Background

Prostate cancer afflicts approximately 230,000 men per year and results in approximately 30,000 deaths (1). Although the PSA (prostate specific antigen) test has been able to diagnose prostate cancer at an earlier stage, it is unclear whether the overall mortality has decreased. The current treatment for hormone naive metastatic prostate cancer is hormone ablation either by LHRH analogue therapy or orchiectomy as monotherapy or in combination with an antiandrogen. Survival varies depending on the extent of disease at commencement of therapy. With the advent of the PSA test many patients are commenced on hormonal therapy at a very early stage and subjected to the long-term effects of androgen ablation including osteoporosis. However, if patients with an asymptomatic rising PSA after definitive local therapy are observed until they develop overt metastatic disease (i.e. evident by imaging techniques), the median time from PSA relapse to clinical progression is approximately 8 years (2). In the pre-PSA era studies relied upon bone scan and CT scans to document the presence of metastatic disease. The median overall survival for men commencing androgen ablation with clinically evident metastatic disease (i.e. not PSA only disease) is about 30 months (3). This information is derived from the meta-analysis of 27 randomized trials with 8,275 men comparing castration (medical or surgery) alone against combined androgen blockade (3). Currently, patients with overt metastases are placed solely on androgen ablation and followed until symptomatic progression. Most men are continued on androgen deprivation indefinitely. Once hormonal therapy is no longer effective, chemotherapy is employed with palliative intent and can prolong overall survival slightly. The purpose of this proposed trial is to determine whether instituting chemotherapy when starting hormonal therapy can delay the time to progression to a clinically meaningful degree without effecting quality of life in men with extensive disease. This is defined as metastases involving appendicular skeleton (including ribs and clavicle) with or without axial involvement and/or visceral metastases. These men have a poor prognosis with a median time to PSA progression of about only 10 months and median time to clinical progression (e.g. worsening bone metastases) of about 14 months. This was seen in the previous ECOG led phase III trial which accrued 1,387 patient in five years of which 75% had extensive disease (4). This was also observed in the SWOG led Intergroup trial in the same population (5). In contrast patients with minimal disease have a 22 month median time to PSA progression with androgen ablation alone and median time to clinical progression of close to four years. The short time to clinical progression also results in men with extensive disease having a median overall survival of only 24 months compared to about five years for patients with minimal disease. Clearly, men with extensive disease are in urgent need of more effective therapies.

Until recently no therapy for hormone refractory prostate cancer (HRPC) had been shown to confer a survival advantage although numerous approaches improved the quality of life of these patients. These include radio-pharmaceuticals (strontium, samarium) (6), suramin (7), and low dose corticosteroids (8). Mitoxantrone combined with a low dose corticosteroid was the standard treatment in the 1990's based on two randomized studies that compared this combination to the corticosteroid alone (9, 10). In both studies, there was a palliative benefit as evidenced by a decrease in pain or opiate requirements but no survival benefit for patients who received mitoxantrone.

Recently, two large phase III trials showed a survival benefit for men with hormone refractory prostate cancer treated with a docetaxel-based regimen (11, 12). One is the SWOG 9916 trial of docetaxel plus estramustine versus mitoxantrone plus prednisone. Docetaxel was dosed at 60 mg/m² every 3 weeks plus 5 days of oral estramustine. The lower docetaxel dose was to minimize toxicities seen at 70 mg/m². Patients were escalated from 60 mg/m² to 70 mg/m² if no grade 3 or 4 toxicities occurred in the first cycle. The other is a multinational trial (TAX327) of single agent docetaxel given either weekly (30 mg/m²) or

every three weeks (75 mg/m²) versus mitoxantrone plus prednisone. In the latter study prednisone 5 mg twice per day was dosed continuously. In both studies every three week docetaxel was more effective in terms of serological response and overall survival. The mitoxantrone plus prednisone arm in the SWOG study had a PSA response of 27% versus 50% with every three week docetaxel plus estramustine ($p < 0.001$) and the corresponding result in the TAX327 trial for the mitoxantrone plus prednisone arm was 32% versus 45% in the every 3 week docetaxel arm ($P < 0.001$). The overall survival also favored docetaxel based therapy with the median survival for mitoxantrone plus prednisone versus every three week docetaxel being 16 versus 18 months in the SWOG study ($p = 0.008$) and 16.5 versus 18.9 months ($p = 0.009$) in TAX327. Docetaxel every three weeks without estramustine, is therefore the optimal regimen. The weekly regimen did not provide any significant benefits.

This trial aims to evaluate whether early use of active chemotherapy (docetaxel) in men with extensive disease in combination with hormonal therapy can delay the time to development of progression/hormone refractory disease and ultimately increase overall survival. Given the fact patients are going to be starting hormonal therapy (androgen ablation) daily prednisone will not be given with the docetaxel in this protocol and will thus minimize the toxicity of chronic low dose steroid exposure. This is expected to make the regimen better tolerated for the 18 weeks it is planned to be administered and will remove this as a confounding variable.

A theory in support of why early chemotherapy may be more beneficial when starting hormonal therapy is that chemotherapy may be more effective on small volume disease that has not been eradicated by androgen deprivation. This has the potential to delay the time to progression and hence prolong disease control. An argument against combining chemotherapy with hormonal therapy is that the androgen deprivation takes the cells out of cycle and may cause the chemotherapy to be less effective. A randomized phase III trial is required to answer this question. A quality of life analysis will be incorporated to determine whether the postulated clinically meaningful increase in disease control is not associated with an overall decrement in quality of life. A clinically significant increase in disease control is considered to be an increase in the median overall survival from 24 to 32 months in men with extensive and hence poor prognosis metastatic prostate cancer. This is considered to be of a magnitude large enough that would alter the current standard of care if quality of life was not impaired.

1.2 Rationale for selected approach and trial design

In summary, docetaxel based chemotherapy has shown significant activity in hormone refractory prostate cancer. The therapy to be taken forward to the phase III trial of patients in patients STARTING hormonal therapy has been chosen based on the results of the phase III trials in the hormone refractory setting. A randomized phase III trial is required to definitively answer the question of whether early chemotherapy plus androgen ablation can significantly increase the duration of cancer control versus androgen ablation alone.

1.3 Supporting Preliminary Data

To date, only one trial has been completed and analyzed addressing the question of chemohormonal therapy as primary therapy for metastatic hormone sensitive prostate cancer. The MDACC group, lead by Dr. Millikan compared androgen deprivation to androgen deprivation with ketoconazole, adriamycin, vinblastine and estramustine (KAVE). It is reported that the median time to PSA progression in patients with high volume metastases was 10 months in the 64 patients with androgen deprivation alone and 19 months in those with chemohormonal therapy. For the 50 patients with low volume metastases, there was no improvement in time to PSA progression as it was 22 months in both arms with about 24 patients per arm. At the time of this analysis there was no improvement in overall survival in either group (Personal Communication: Randall Millikan).

From this study we at least know that two relatively ineffective chemotherapy agents (adriamycin and vinblastine) prolonged time to PSA progression in patients with high volume metastatic burden which equates to the patients with "extensive" disease. Docetaxel holds even more promise based on the data showing improvement over mitoxantrone plus prednisone in HRPC. Another theoretical advantage for using docetaxel is that it can abrogate BCL-2's anti-apoptotic properties (13) which has been shown to be involved in the emergence of androgen insensitive clones (14-18). It has also been reported that retreatment after a "drug holiday" with docetaxel, if no progression on docetaxel, can result in significant responses (19). Thus, retreatment with docetaxel at the time of progression has the possibility of compounding its own clinical benefit by adding to the total time of disease control when re-instituted when a patient is hormone refractory.

1.4 Rationale for Correlative Studies

In addition to asking the question of whether chemotherapy at the time of starting hormonal therapy prolongs survival, this clinical trial provides a unique opportunity to collect specimens (tumor specimens and blood) that will be annotated with clinical information. The intent of the genetic studies will be to identify genes in the patients' germline DNA and/or tumor that predict response and toxicity. The sequential serum analyses will allow for proteins to be studied longitudinally and thus further facilitate potential biomarker discovery.

2. Objectives

2.1 Primary Objective

- 2.1.1 To evaluate the ability of early chemotherapy to improve overall survival in men commencing androgen deprivation for metastatic prostate cancer.

2.2 Secondary Objectives

- 2.2.1 To determine whether early chemotherapy can increase the time to clinical progression (radiographic or symptomatic deterioration due to disease) over hormonal therapy alone.
- 2.2.2 To determine whether early chemotherapy can increase the time to development of hormone refractory disease over hormonal therapy alone.
- 2.2.3 To determine whether early chemotherapy can increase time to serological progression over hormonal therapy alone.
- 2.2.4 To determine rates of biochemical response at 6 months and 12 months in the chemohormonal arm versus the hormonal therapy alone arm.
- 2.2.5 To determine the frequency of adverse events and the tolerability of chemotherapy combined with hormonal therapy versus hormonal therapy alone.
- 2.2.6 To determine whether the postulated clinically meaningful increase in disease control is associated with an alteration in overall quality of life using the Functional Assessment of Cancer Therapy-Prostate questionnaire (FACT-P self-administered).
- 2.2.7 To determine the ability of PSA changes to be a surrogate for clinical benefit from therapy and overall survival.

2.3 Tertiary Objectives

- 2.3.1 To determine if there are proteins differentially translated from the genome in hormone sensitive prostate cancer, prostate cancer that has responded to hormonal therapy and hormone refractory prostate cancer.
- 2.3.2 To determine the frequency of polymorphisms of enzymes involved in steroid metabolism and other carcinogenic processes.
- 2.3.3 To determine if the amount and frequency of certain carcinogenic proteins such as CXCR4 and Manganese Superoxide Dismutase can be correlated with a poor prognosis.

3. Selection of Patients

Each of the criteria in the following section must be met in order for a patient to be considered eligible for this study. Use the spaces provided to confirm a patient's eligibility. For each patient, this section should be photocopied, completed and maintained in the patient's chart.

In calculating days of tests and measurements, the day a test or measurement is done is considered Day 0. Therefore, if a test is done on a Monday, the Monday four weeks later would be considered Day 28.

ECOG Patient No. _____

Patient's Initials (L, F, M) _____

NOTE: All questions regarding eligibility should be directed to the ECOG Coordinating Center at (617) 632-3610.

NOTE: Institutions may use the eligibility checklist as source documentation if it has been reviewed, signed, and dated prior to randomization by the treating physician.

3.1 Patients must have histologically or cytologically confirmed prostate cancer with EXTENSIVE metastatic disease and have been on androgen deprivation therapy for less than 90 days. Hormonal therapy must not have commenced more than 90 days prior to randomization.

Disease stage: _____

Rarely pathology is not available and if clinical situation confirms prostate cancer (such as response to androgen ablation) pathology is not required and patient can be enrolled after discussed with study chair.

3.2 Definition of extensive disease: Metastases involving at least one lesion in any bony structure beyond the vertebral column and pelvic bone or any involvement with viscera. In the absence of visceral lesions, there must be 4 or more bone lesions. Patients with disease limited to vertebral column and/or pelvis alone with or without lymph node or lymph node only disease involvement are not eligible for this trial.

NOTE: Radiological studies identifying measurable or non-measurable disease must be obtained within 6 weeks (42 days) of starting androgen deprivation therapy (note: this is not protocol therapy).

3.3 Patients are not eligible if the PSA has risen greater than 50% from its lowest point since the beginning of androgen deprivation therapy to the registration for protocol therapy.

3.4 Patients must have adequate organ function within 4 weeks prior to randomization and evidenced by:

3.4.1 Absolute Neutrophil Count $\geq 1500/\text{mm}^3$

ANC: _____ Date of test: _____

3.4.2 Platelet count $\geq 100,000/\text{mm}^3$

Platelet count: _____ Date of test: _____

3.4.3 Total bilirubin $\leq \text{ULN}$

Total bilirubin: _____ Date of test: _____

3.4.4 ALT $\leq 2.0 \times$ upper limit of normal

ALT: _____ Date of test: _____

3.4.5 Creatinine clearance of ≥ 30 mL/min. Creatinine clearance (CrCl) should be calculated at screening using the Cockcroft-Gault formula:

Creatinine clearance for males (mL/min) = $(140 - \text{age})(\text{body weight in kg}) / (72 \times (\text{serum creatinine in mg/dl}))$

Creatinine clearance: _____ Date of test: _____

3.4.6 PT, $\text{INR} \leq 1.5$ X upper limit of normal (except if on therapeutic anti-coagulation).

PT: _____ Date of test: _____

INR: _____ Date of test: _____

3.4.7 PTT ≤ 1.5 X upper limit of normal (except if on therapeutic anti-coagulation).

PTT: _____ Date of test: _____

NOTE: All values must be obtained within 4 weeks prior to beginning protocol therapy.

3.5 If a patient has had major surgery, the patient must be longer than 4 weeks post major surgery and recovered from all toxicity prior to beginning protocol therapy.

3.6 Patients must have discontinued hormonal therapy in the adjuvant and/or neoadjuvant setting 12 months prior to beginning protocol therapy, AND must not have exceeded 24 months of therapy AND have shown to have no evidence of disease (PSA < 0.1 ng/dL after prostatectomy plus hormonal therapy and < 0.5 ng/dL and not have doubled above nadir after radiation therapy plus hormonal therapy) at least 12 months after completing adjuvant or neoadjuvant hormonal therapy. Patients with prior chemotherapy in the adjuvant or neoadjuvant setting are ineligible. The last depot injection must have expired by the time the 24 month mark was reached.

(e.g., patient completed 24 months of adjuvant therapy with 8 three month depot LHRH agonists injections – last dose given 8/03 – 24 months completed 11/03. Patient is eligible if no evidence of disease at this time and commenced hormonal therapy for metastatic disease on or after 11/05).

3.7 Patients must have ECOG performance status of 0- 2. (NB: Patients with PS 2 are only eligible if the decline in PS is due to metastatic prostate cancer).

3.8 Patients must be at least 18 years of age.

3.9 Patients must not have participated in another clinical trial within 30 days prior to randomization. Patients may participate in non-therapeutic trials.

3.10 Patients must not have prior history of malignancy in the past 5 years with the exception of basal cell and squamous cell carcinoma of the skin. Other malignancies that are considered to have a low potential to progress (eg: grade 2, T1A TCC) may be enrolled if approved by principal investigator.

3.11 Peripheral neuropathy must be $\leq$ grade 1.

3.12 Patients with a history of severe hypersensitivity reaction to Docetaxel® or other drugs formulated with polysorbate 80 must be excluded.

3.13 No active cardiac disease defined as active angina, symptomatic congestive heart failure, or myocardial infarction within previous six months.

3.14 Patients may be enrolled if they have had prior palliative radiation therapy. However, this has to have been commenced within 30 days of starting androgen deprivation.

4. Randomization Procedures

Submitting Regulatory Documents

Before an ECOG Institution may enter patients, protocol specific regulatory documents must be submitted to the CTSU Regulatory Office at the following address:

CTSU Regulatory Office
Coalition of National Cancer Cooperative Groups
1818 Market Street, Suite 1100
Philadelphia, PA 19103
FAX: (215) 569-0206

Required Protocol Specific Regulatory Documents

1. CTSU Regulatory Transmittal Form.
2. Copy of IRB Informed Consent Document.

NOTE: Any deletion or substantive modification of information concerning risks or alternative procedures contained in the sample informed consent document must be justified in writing by the investigator and approved by the IRB.

3. A. CTSU IRB Certification Form.
Or
B. HHS 310 Form.
Or
C. IRB Approval Letter

NOTE: The above submissions must include the following details:

- Indicate all sites approved for the protocol under an assurance number.
- OHRP assurance number of reviewing IRB
- Full protocol title and number
- Version Date
- Type of review (full board vs. expedited)
- Date of review.
- Signature of IRB official

The CTSU encourages you to link to the following RSS2.0 webpage so that more information on RSS2.0 as well as the submission forms can be accessed http://www.ctsu.org/rss2_page.asp. If you have questions regarding regulatory document submission, please telephone the CTSU Help Desk at 1-888-823-5923 or E-mail CTSUContact@westat.com. Monday through Friday, 9:00am - 6:00pm.

Patients must not start protocol treatment prior to randomization.

Treatment should start within five working days after randomization.

NOTE: Please refer to Appendix IV for CTSU registration guidelines.

Institutions may register eligible patients to this study via the ECOG webpage 24 hours a day, 7 days a week, using the Web-based Patient Registration Program (<https://webreg.ecog.org>). If you need assistance or have questions, please telephone the Central Randomization Desk at the ECOG Coordinating Center at (617) 632-2022, Monday through Friday 9:00am – 5:00pm Eastern Time. Please note that a password is required to use this program. The following information will be requested:

4.1 Protocol Number

4.2 Investigator Identification

4.2.1 Institution and affiliate name

4.2.2 Investigator's name

4.3 Patient Identification

4.3.1 Patient's initials and chart number

4.3.2 Patient's Social Security number

4.3.3 Patient demographics

4.3.3.1 Sex

4.3.3.2 Birth date (mm/yyyy)

4.3.3.3 Race

4.3.3.4 Ethnicity

4.3.3.5 Nine-digit ZIP code

4.3.3.6 Method of payment

4.4 Eligibility Verification

Patients must meet all of the eligibility requirements listed in Section 3. An eligibility checklist has been appended to the protocol. A confirmation of registration will be forwarded by the ECOG Coordinating Center.

4.5 Stratification Factors

4.5.1 Age

4.5.1.1 ≥ 70

4.5.1.2 < 70

4.5.2 ECOG Performance Status

4.5.2.1 PS 0-1

4.5.2.2 PS 2

4.5.3 CAB > 30 Days

4.5.3.1 Yes

4.5.3.2 No

4.5.4 Prior Adjuvant Hormonal Therapy

4.5.4.1 > 12 months

4.5.4.2 ≤ 12 months

4.5.5 Bisphosphonate

4.5.5.1 Yes

4.5.5.2 No

4.6 Additional Requirements

4.6.1 Patients must provide a signed and dated, written informed consent form.

4.6.2 Correlative samples should be submitted for analysis as outlined in Sections 10 and 11.

4.7 Instructions for Patients who Do Not Start Assigned Protocol Treatment

If a patient does not receive any assigned protocol treatment, baseline and follow-up data will still be collected and must be submitted according to the instructions in the E3805 Forms Packet. Document the reason for not starting protocol treatment on the Off Treatment Form. Also report the date and type of the first non-protocol treatment that the patient receives.

5. Treatment Plan

5.1 Administration Schedule

5.1.1 Arm A: Androgen Deprivation plus Docetaxel

5.1.1.1 Androgen Deprivation Therapy

Options for androgen-deprivation (i.e., hormone ablation) therapy include the following:

- LHRH agonist or antagonist therapy
- Surgical castration

NOTE: Patients can be randomized to the study within 5 working days before beginning docetaxel and may begin androgen-deprivation therapy up to 90 days before beginning protocol therapy.

Use of combined androgen blockade (medical or surgical castration combined with antiandrogen treatment) is to be instituted at the investigator's discretion.

Antiandrogens (e.g., bicalutamide or flutamide) may be used in addition to androgen-deprivation therapy, but may not be used as the sole hormonal therapy. Anti-androgens combined with 5-alpha reductase inhibitors are also not allowed. An investigator may administer this class of drug as a single agent ≤ 7 days before medical castration to cover the testosterone surge associated with LHRH agonists. *The 90-day window of androgen-deprivation or hormonal therapy starts with commencement of either the antiandrogen or LHRH therapy.*

5.1.1.2 Docetaxel

75 mg/m² of docetaxel will be given on day 1 every 21 days over one hour for up to 6 cycles (1 cycle = 21 days).

5.1.1.3 Premedication for Docetaxel

Premedication is required to decrease or prevent acute anaphylactoid reactions to docetaxel, and to decrease the severity or delay the onset of late-occurring fluid retention problems. Dexamethasone 8 mg po approximately 12, 3 and 1 hour prior to docetaxel infusion. Use of diphenhydramine is optional.

5.1.1.4 Calcium Carbonate

Patients will receive concomitant treatment with calcium carbonate at a dose of at least 500 mg orally per day every evening (any product with amount stated on the bottle and verified by investigator, e.g., TumsTM). Calcium is best absorbed when taken with meals.

5.1.1.5 Vitamin D

Patients will receive concomitant treatment with vitamin D by administration of any multivitamin containing at least 400 IU of vitamin D (any product with amount stated on the bottle and verified by investigator).

For guidance on use of bisphosphonates, please see Section 5.4.2 (Supportive Care).

At the time of hormone refractory disease, investigators are encouraged to treat with docetaxel therapy every three weeks ON BOTH ARMS so the effect of early versus delayed chemotherapy on overall survival can be performed. If the investigator deems that the patient is not a suitable candidate for docetaxel chemotherapy, the reason for this decision will be recorded. All therapy for hormone refractory disease will be recorded including anti-androgen withdrawal, second-line hormonal therapy and chemotherapy. If a patient has more than one therapy, then all therapies will be recorded. Patients will be followed until development of HRPc and for overall survival.

5.1.2 Arm B: Androgen Deprivation Alone

5.1.2.1 Androgen Deprivation Therapy

Options for androgen-deprivation (i.e., hormone ablation) therapy include the following:

- LHRH agonist or antagonist therapy
- Surgical castration

NOTE: Patients may begin androgen-deprivation therapy up to 90 days before beginning protocol therapy.

Use of combined androgen blockade (medical or surgical castration combined with antiandrogen treatment) is to be instituted at the investigator's discretion.

Antiandrogens (e.g., bicalutamide or flutamide) may be used in addition to androgen-deprivation therapy, but may not be used as the sole hormonal therapy. Anti-androgens combined with 5-alpha reductase inhibitors are also not allowed. An investigator may administer this class of drug as a single agent ≤ 7 days before medical castration to cover the testosterone surge associated with LHRH agonists. *The 90-day window of androgen-deprivation or hormonal therapy starts with commencement of either the antiandrogen or LHRH therapy.*

5.1.2.2 Calcium Carbonate

Patients will receive concomitant treatment with calcium carbonate at a dose of at least 500 mg orally per day every evening (any product with amount stated on the bottle and verified by investigator, e.g., TumsTM). Calcium is best absorbed when taken with meals.

5.1.2.3 Vitamin D

Patients will receive concomitant treatment with vitamin D by administration of any multivitamin containing at least 400 IU of vitamin D (any product with amount stated on the bottle and verified by investigator).

For guidance on use of bisphosphonates, please see Section 5.4.2 (Supportive Care).

At the time of hormone refractory disease, investigators are encouraged to treat with docetaxel therapy every three weeks ON BOTH ARMS so the effect of early versus delayed chemotherapy on overall survival can be performed. If the investigator deems that the patient is not a suitable candidate for docetaxel chemotherapy, the reason for this decision will be recorded. All therapy for hormone refractory disease will be recorded including anti-androgen withdrawal, second-line hormonal therapy and chemotherapy. If a patient has more than one therapy, then all therapies will be

recorded. Patients will be followed until development of HRPC and for overall survival.

5.1.3 For Arms A and B: Quality of Life (QOL)

Quality of Life will be measured at the following timepoints:

- Pretreatment (within 1 week prior to therapy)
- Week 12
- Week 24
- Week 36
- Week 48

Administration of the FACT-P, FACIT-F Fatigue Subscale, FACT-Taxane Subscale and the Brief Pain inventory will allow us to examine overall QOL, as well as specific treatment-associated and disease-associated changes in QOL. The questionnaire should be administered to patients by staff during visits. Differences between pre-treatment and subsequent scores will be analyzed using appropriate statistical tests.

5.1.3.1 The questionnaire must be administered at the timepoints listed above unless the patient refuses. The patient should be instructed to respond to the questionnaire in terms of his experience during the time frame specified on each questionnaire.

5.1.3.2 The patient should be asked to read the instructions at the beginning of each questionnaire and complete all of the items. It is permissible to assist the patient with the completion of the questionnaire as long as the staff person does not influence the patient's responses.

5.1.3.3 The questionnaire must be reviewed by the protocol nurse or research coordinator as soon as the patient completes them to ensure all items were marked appropriately. If more than one answer was marked, the patient should be asked to choose the answer which best reflects how he is feeling. If a question was not answered, the patient should be asked if he would like to answer it. The patient should always have the option to refuse. If the patient refuses, it should be indicated on the questionnaire that he declined to answer the item.

5.1.3.4 Completed questionnaires must be returned to the ECOG Coordinating Center. If a questionnaire cannot be completed by the patient, the reason should be noted on the Assessment Compliance Form and the form should be returned to the ECOG Coordinating Center.

5.1.3.5 If a patient misses an appointment on the scheduled date, the questionnaire may be completed by telephone on the appointed date or at the time the appointment is rescheduled. If the missed date is on a treatments date, the QOL assessment will be done when the patient comes for the rescheduled treatment.

5.1.3.6 If a patient cannot complete the questionnaire because he is too sick, this should be documented on the Assessment Compliance Form.

5.2 Dose Modifications

There will be no dose modifications for androgen deprivation. Intermittent hormonal therapy is not allowed.

For docetaxel, no more than two dose modifications should be allowed for any patient. If a patient requires a third reduction of docetaxel, they will be followed every 12 weeks and treated with androgen deprivation alone. Dose adjustments are to be made according to the system showing the greatest degree of toxicity. Toxicities will be graded using the NCI Common Terminology Criteria for Adverse Events (CTCAE), version 3.0. If toxicity occurs, the appropriate treatment will be used to ameliorate signs and symptoms including antiemetics for nausea and vomiting, anti-diarrheals for diarrhea and anti-pyretics and anti-histamines for drug fever before toxicity grade is determined.

Dose adjustments for toxicity should be made according to the guidelines that follow. If a dose is reduced due to toxicity the dose will not be re-escalated back to starting level. Treatment may be delayed no more than three weeks to allow recovery from toxicity. If treatment must be delayed longer than three weeks from scheduled day of dosing, patient will be treated with androgen deprivation alone.

Dose Level	Docetaxel (mg/m ²)
Level 0	75 mg/m ²
Level - 1	65 mg/m ²
Level - 2	55 mg/m ²

5.2.1 Dose Modifications for Myelosuppression

Dose modifications to be made based on granulocyte and/or platelet count drawn prior to planned treatment (can be done day prior to planned dose):

Docetaxel	Granulocytes/mm ³ Day 1 of treatment	Platelet/mm ³ Day 1 of treatment
No change	> 1,500 ^ψ	or > 100,000
Reduce one dose level #	1,000 to 1,499	or 75,000 to 99,999
Delay dosing one week ⁺	<1000	or < 75,000

If a dose reduction is made, maintain the lower dose for all subsequent cycles

Three dose reductions will require that the patient discontinue docetaxel and continue on androgen ablation alone.

* If planned day 1 dose must be delayed for three consecutive weeks, discontinue docetaxel and continue on androgen ablation alone.

+ If a dose is held, the patient will be retreated with a one level dose reduction once granulocyte count has recovered to at least 1,500/mm³ and platelet count has recovered to at least 100,000/mm³.

ψ Management of prolonged grade 4 neutropenia and neutropenic fever- In order to maximize dose intensity, patients with afebrile Grade 4 neutropenia ≥ 7 days or Grade ≥ 3 neutropenia associated with fever (one reading of oral temperature > 38.5°C, or three readings of oral temperature >38.0°C in a 24-hour period) should be retreated after recovery to an absolute neutrophil count to 1,500 granulocytes/mm³ but with a one level dose reduction. The fever must have resolved and if an infection is identified, it must be adequately treated and have clinically resolved before restarting therapy. If prior bacteremia, blood cultures must be negative on recheck. Patient can continue with chemotherapy dosing while on antibiotics.

5.2.2 Dose Modification for Hepatic Dysfunction

ALT and Bilirubin will be evaluated pre-study and prior to Day 1 of cycles 1-6:

Patients who develop abnormal liver function tests for any reason while on the study will have the following dose reductions:

Dose Modifications for Abnormal Liver Function

Bilirubin		ALT/ SGPT	Action
> ULN	or	> 5 x ULN	Wait ≤ 3 weeks. If recovered*, reduce docetaxel dose by one dose level. If not, discontinue docetaxel.
≤ ULN	and	> 3 x ULN	Reduce docetaxel dose by one dose level

* Recovery is < 3X ULN for ALT/SGPT and WNL for bilirubin. Dose modifications are based on SGPT alone due to confounding effect of bone disease on SGOT levels.

5.2.3 Dose Modification for Stomatitis

If stomatitis ≥ grade 2 is present on day 1 of any cycle, docetaxel should be held until stomatitis has resolved. If Grade 3/4 stomatitis occurs at any time, the dose of docetaxel will be reduced one dose level for all subsequent doses. If a second event is incurred, docetaxel will be reduced one more dose level.

5.2.4 Dose Modification for Peripheral Neuropathy

If ≥ Grade 3, the patient will discontinue docetaxel.

If Grade 2, the Docetaxel will be held and the patient should be retreated upon recovery to a ≤ Grade 1 toxicity with a dose reduction of Docetaxel by one level.

If Grade 2 or greater neurotoxicity persists for more than 3 weeks, the patient will discontinue docetaxel.

5.2.5 Hypersensitivity Reactions for Docetaxel

See Appendix V for treatment of hypersensitivity reactions. Treatment should be discontinued for Grade 4 hypersensitivity reactions. There are no dose reductions for hypersensitivity reactions.

Grade 4 Hypersensitivity is defined as a reaction that is life threatening and requires pressor and/or ventilator support or shock associated with acidemia and impairing vital organ function due to tissue hypoperfusion.

Patients with two episodes of Grade 3 Hypersensitivity reactions or one Grade 4 Hypersensitivity reaction are to discontinue docetaxel.

5.2.6 Diarrhea

If patients experience >grade 2 diarrhea and concurrent grade 3 or 4 neutropenia, hold Docetaxel until ANC>1000/mm³ and diarrhea ≤ grade 2.

If patients experience significant diarrhea (>3 loose stools/24hrs over baseline), they should be treated prophylactically in subsequent cycles with loperamide or diphenoxylate. If patient experiences >grade 2 diarrhea despite prophylaxis, Docetaxel should be reduced one dose level. If patients experience > grade 2 diarrhea despite prophylaxis AND dose reduction, they should discontinue docetaxel.

5.2.7 Delay of Therapy:

If docetaxel has to be delayed for more than 3 weeks from planned day of dosing because of any toxicity, then the patient is to be treated with androgen deprivation alone.

5.2.8 Other Toxic Effects Thought to be related to docetaxel:

If toxicities $\leq$ Grade 2, manage the subject symptomatically if possible, and retreat without dose reduction.

If toxicities $\geq$ Grade 3 and clinically significant (not mentioned above), docetaxel should be withheld (except for anemia as patients can be transfused) until resolution to $\leq$ Grade 1 or baseline if baseline was greater than Grade 1, then reinstituted, if medically appropriate, at a dose reduction of one level.

5.3 Adverse Event Reporting Requirements

5.3.1 Purpose

Adverse event data collection and reporting, which are required as part of every clinical trial, are done to ensure the safety of patients enrolled in the studies as well as those who will enroll in future studies using similar agents. Adverse events are reported in a routine manner at scheduled times during a trial (please refer to the E3805 Forms Packet for the list of forms with directions for routine adverse event reporting). Additionally, certain adverse events must be reported in an expedited manner for more timely monitoring of patient safety and care. The following sections provide information about expedited reporting.

5.3.2 Determination of reporting requirements

Reporting requirements may include the following considerations: 1) whether the patient has received an investigational or commercial agent; 2) the characteristics of the adverse event including the grade (severity), the relationship to the study therapy (attribution), and the prior experience (expectedness) of the adverse event; 3) the phase (1, 2, or 3) of the trial; and 4) whether or not hospitalization or prolongation of hospitalization was associated with the event.

Commercial agents are those agents not provided under an IND but obtained instead from a commercial source. The NCI, rather than a commercial distributor, may on some occasions distribute commercial agents for a trial.

Steps to determine if an adverse event is to be reported in an expedited manner:

Step 1: *Identify the type of event using the NCI Common Terminology Criteria for Adverse Events (CTCAE) Version 3.0.* The CTCAE provides descriptive terminology and a grading scale for each adverse event listed. A copy of the CTCAE can be downloaded from the CTEP home page (<http://ctep.cancer.gov>). Additionally, if assistance is needed, the NCI has an Index to the CTCAE that provides help for classifying and locating terms. All appropriate treatment locations should have access to a copy of the CTCAE.

Step 2: *Grade the event using the NCI CTCAE.*

Step 3: *Determine whether the adverse event is related to the protocol therapy (investigational or commercial).* Attribution categories are as follows: Unrelated, Unlikely, Possible, Probable, and Definite.

Step 4: Determine the prior experience of the adverse event. Expected events are those that have been previously identified as resulting from administration of the agent. An adverse event is considered *unexpected*, for expedited reporting purposes only, when either the type of event or the severity of the event is **NOT** listed in:

- **Arm A and B** – the drug package insert or protocol

Step 5: Review Section 5.3.6 for E3805 and/or ECOG specific requirements for expedited reporting of specific adverse events that require special monitoring.

NOTE: For general questions regarding expedited reporting requirements, please contact the NCI Medical Help Desk: 301-897-7497.

5.3.3 Reporting methods

- **Arm A and B** – This study requires that expedited adverse event reporting use the NCI's Adverse Expedited Reporting System (AdEERS). The NCI's guidelines for AdEERS can be found at <http://ctep.cancer.gov>. For questions regarding the use of the AdEERS application, please contact the NCI Technical Help Desk: 301-840-8202.

An AdEERS report must be submitted to ECOG and the appropriate regulatory agencies by one of the following methods:

- Electronically submit the report via the AdEERS Web-based application located at <http://ctep.cancer.gov>

or
- Fax the completed NCI Adverse Event Expedited Report – Single Agent or Multiple Agents paper template located at <http://ctep.cancer.gov> to ECOG (617-632-2990), Attention: AE and the FDA (800-332-0178).

NOTE: Paper copies of AdEERS reports will only be accepted if the AdEERS system is down. Once the system is restored, a report submitted on a paper template must be entered into the AdEERS system by the original submitter of the report at the site.

Any supporting or follow up documentation must be faxed to ECOG (617-632-2990), Attention: AE. In addition, supporting or follow up documentation must be faxed to the FDA (800)-332-0178).

5.3.4 When to report an event in an expedited manner

When an adverse event requires expedited reporting, submit a full AdEERS report within the timeframes outlined in Section 5.3.6.

NOTE: Adverse events that meet the reporting requirements in Section 5.3.6 and occur within 30 days of the last dose of protocol treatment must be reported on an expedited adverse event report form (using AdEERS). For any adverse events that occur more than 30 days after the last dose of treatment, only those that have an attribution of possibly, probably, or definitely AND meet the reporting requirements in Section 5.3.6 must be reported on an expedited adverse event report form (using AdEERS).

5.3.5 Other recipients of adverse event reports

ECOG will forward AdEERS reports to the appropriate regulatory agencies and pharmaceutical company, if applicable.

Adverse events determined to be reportable must also be reported by the institution, according to the local policy and procedures, to the Institutional Review Board responsible for oversight of the patient.

5.3.6 Expedited reporting for commercial agents

Commercial reporting requirements are provided below. The commercial agent used in arms A and B of this study is Docetaxel.

Expedited reporting requirements for adverse events experienced by patients on arm(s) with commercial agents only – Arms A and B					
Attribution	Grade 4		Grade 5 ^a		ECOG and Protocol-Specific Requirements
	Unexpected	Expected	Unexpected	Expected	See footnote (b) for special requirements.
Unrelated or Unlikely			7 calendar days	7 calendar days	
Possible, Probable, Definite	7 calendar days		7 calendar days	7 calendar days	
7 Calendar Days: Indicates a full AdEERS report is to be submitted within 7 calendar days of learning of the event.					
a This includes all deaths within 30 days of the last dose of treatment regardless of attribution. NOTE: Any death that occurs > 30 days after the last dose of treatment and is attributed possibly, probably, or definitely to the treatment must be reported within 7 calendar days of learning of the event.					
b Protocol-specific expedited reporting requirements: The adverse events listed below also require expedited reporting for this trial:					
Serious Events: Any event following treatment that results in <u>persistent or significant disabilities/incapacities, congenital anomalies, or birth defects</u> must be reported via AdEERS within 7 calendar days of learning of the event. For instructions on how to specifically report these events via AdEERS, please contact the NCI Medical Help Desk at 301-897-7497.					

5.3.7 Reporting secondary AML/MDS/ALL

All cases of acute myeloid leukemia (AML), myelodysplastic syndrome (MDS), and acute lymphocytic leukemia (ALL) that occur in patients on NCI-sponsored trials following their chemotherapy for cancer must be reported to ECOG. Submit the following information within 30 days of an AML/MDS/ALL diagnosis occurring after treatment for cancer on NCI-sponsored trials:

- a completed NCI/CTEP Secondary AML/MDS/ALL Report Form (*do not use AdEERS*);
- a copy of the pathology report confirming the AML/MDS/ALL; and
- a copy of the cytogenetics report (if available).

ECOG will forward copies to the Investigational Drug Branch (IDB) of the NCI Cancer Therapy Evaluation Program (CTEP).

NOTE: If a patient has been enrolled in more than one NCI-sponsored study, the NCI/CTEP Secondary AML/MDS/ALL Report Form must be submitted for the most recent trial. ECOG must be provided with a copy of the report even if ECOG was not the patient's most recent trial.

5.3.8 Reporting of other second primary cancers

All cases of new primary cancers that occur on ECOG protocols during or after protocol must be reported to ECOG, according to the follow up schedule outlined in the E3805 Forms Packet, on the ECOG Second Primary Form within 30 days of diagnosis, regardless of relationship to protocol treatment. This form is not for use for reporting recurrence or development of metastatic disease. A copy of the pathology report should be sent, if available.

NOTE: Once data regarding survival and remission status are no longer required by the protocol, no follow-up data should be submitted, including the NCI AML/MDS/ALL form and ECOG Second Primary Form.

Submit AML/MDS/ALL and Second Primary information to:

ECOG Coordinating Center
FSTRF
900 Commonwealth Avenue
Boston, MA 02215

5.4 Supportive Care

- 5.4.1 All supportive measures consistent with optimal patient care will be given throughout the study.
- 5.4.2 It is unknown whether bisphosphonates at doses shown to prevent skeletal complication in hormone refractory disease is beneficial when a patient is just starting hormonal therapy. If a patient is to be treated with high dose bisphosphonates (not osteoporosis prevention dosing) then the investigator must declare this at the beginning to ensure patients treated in this manner are equally stratified to both arms. Once a patient is stratified, bisphosphonates at doses to prevent skeletal related events will be prohibited for patients who were not stratified by use of these agents. These patients can be treated with this class of drug at these doses once a skeletal related event occurs or once a patient develops hormone refractory disease. Osteoporosis prevention doses can be used at the investigator's discretion.
- 5.4.3 Medications to treat treatment-related side effects, such as nausea, hot flashes, diarrhea, are to be given at the investigator's discretion.

5.5 Duration of Therapy

Chemotherapy will be given for a maximum of six (6) cycles to patients on Arm A.

Patients will receive protocol therapy unless:

- 5.5.1 Extraordinary Medical Circumstances: If at any time the constraints of this protocol are detrimental to the patient's health, protocol treatment should be discontinued. In this event submit forms according to the instructions in the E3805 Forms Packet.
- 5.5.2 Patient withdraws consent.

5.5.3 Unacceptable toxicity

At the time of hormone refractory disease investigators are encouraged to treat with docetaxel every three weeks ON BOTH ARMS so the effect of early versus delayed chemotherapy on overall survival can be performed. If the investigator deems that the patient is not a suitable candidate for docetaxel chemotherapy, the reason for this decision will be recorded. All therapy for hormone refractory disease will be recorded including anti-androgen withdrawal, second-line hormonal therapy and chemotherapy. If a patient has more than one therapy then all therapies will be recorded. Patients will be followed until development of HRPC and for overall survival.

5.6 Duration of Follow-up

For this protocol, all patients, including those who discontinue chemotherapy early, will be followed until development of hormone refractory disease and for survival for 10 years from the date of registration. All patients must also be followed through completion of all protocol therapy.

6. Measurement of Effect

6.1 Solid Tumor Response Criteria (RECIST)

6.1.1 Malignant Disease Evaluation

To assess objective response, it is necessary to estimate the overall tumor burden at baseline to which subsequent measurements will be compared. Measurable disease is defined by the presence of at least one measurable lesion.

All measurements should be recorded in metric notation by use of a ruler or calipers. The same method of assessment and the same technique should be used to characterize each identified lesion at baseline and during follow-up. All baseline evaluations should be performed as closely as possible to the beginning of treatment and **never more than four weeks** before registration.

The term evaluable in reference to measurability will not be used because it does not provide additional meaning or accuracy.

At baseline, tumor lesions will be characterized as either measurable or non-measurable.

6.1.1.1 Measurable

Lesions that can be accurately measured in at least one dimension (longest diameter to be recorded) as > 20 mm (2.0 cm) with conventional techniques or as > 10 mm (1.0 cm) with **spiral** CT scan.

If the measurable disease is restricted to a solitary lesion, its neoplastic nature should be confirmed by cytology/histology.

6.1.1.2 Non-Measurable

All other lesions, including small lesions [longest diameter < 20 mm (2.0 cm) with conventional techniques or < 10 mm (1.0 cm) with **spiral** CT scan] and truly non-measurable lesions.

Lesions considered to be truly non-measurable include the following: bone lesions, leptomeningeal disease, ascites, pleural/pericardial effusion, inflammatory breast disease, lymphangitis cutis/pulmonis, abdominal masses that are not confirmed and followed by imaging techniques, and cystic lesions.

Masses growing in a previously radiated field (e.g. prostatic fossa mass) are measurable if documented to have increased in size since XRT.

6.1.2 Definitions of Response

6.1.2.1 Target Lesions

All measurable lesions up to a maximum of five lesions per organ and 10 lesions in total, representative of all involved organs. Target lesions should be selected on the basis of their size (those with the longest diameters) and their suitability for accurate repeated measurements.

The sum of the longest diameters of all target lesions will be calculated at baseline and reported as the *baseline sum longest diameter*. The *sum longest diameter* will be used to characterize the objective tumor response. For lesions measurable in 2 or 3 dimensions, always report the longest diameter at the time of each assessment.

6.1.2.2 Complete Response (CR)

The disappearance of all target lesions. To be assigned a status of complete response, changes in tumor measurements must be confirmed by repeat assessments performed **no less than four weeks** after the criteria for response are first met.

6.1.2.3 Partial Response (PR)

At least a 30% decrease in the sum of the longest diameters of target lesions, taking as reference the *baseline sum longest diameter*. To be assigned a status of partial response, changes in tumor measurements must be confirmed by repeat assessments performed **no less than four weeks** after the criteria for response are first met.

6.1.2.4 Progressive Disease (PD)

At least a 20% increase in the sum of the longest diameters of target lesions, taking as reference the *smallest sum longest diameter* recorded since the baseline measurements, or the appearance of one or more new lesion(s).

6.1.2.5 Stable Disease (SD)

Neither sufficient shrinkage to qualify for partial response nor sufficient increase to qualify for progressive disease. To be assigned a status of stable disease, measurements must have met the stable disease criteria at least once after study entry at a minimum interval of at least 8 weeks.

6.1.3 Nontarget Lesions

All other lesions or sites of disease. Measurements of these lesions are not required, but the presence or absence of each should be noted throughout follow-up.

6.1.3.1 Complete Response (CR)

The disappearance of all nontarget lesions and normalization of tumor marker levels, if applicable. To be assigned a status of complete response, changes in tumor measurements must be confirmed by repeat assessments performed **no less than four weeks** after the criteria for response are first met.

6.1.3.2 Incomplete Response/Stable Disease (SD)

The persistence of one or more nontarget lesion(s) and/or the maintenance of tumor marker levels above the normal limits. To be assigned a status of stable disease, measurements must have met the stable disease criteria at least once after study entry at a minimum interval of at least 8 weeks.

6.1.3.3 Progressive Disease (PD)

The appearance of one or more new lesion(s) and/or unequivocal progression of existing nontarget lesions.

6.1.4 Symptomatic Deterioration

Patients with a global deterioration of health status requiring discontinuation of treatment without objective evidence of disease progression at that time should be classified as having symptomatic deterioration

6.2 Evaluation of Patient's Best Overall Response

The best overall response is the best response recorded from registration until disease progression/recurrence, taking as reference for progressive disease the smallest measurements recorded since registration. The table below provides overall responses for all possible combinations of tumor responses in target and nontarget lesions, with or without new lesions.

To be assigned a status of complete or partial response, changes in tumor measurements must be confirmed by repeat assessments performed **no less than four weeks** after the criteria for response are first met.

To be assigned a status of stable disease, measurements must have met the stable disease criteria at least once after study entry at a minimum interval of at least 8 weeks.

Overall Response for all Possible Combinations of Tumor Response

NOTE: For this protocol, measurable disease response is not being evaluated. However, time to radiographic progression is being evaluated and the following table will apply.

Target Lesions	Nontarget Lesions	New Lesions	Overall Response
CR	CR	No	CR
CR	Incomplete response/SD	No	PR
PR	Non-PD	No	PR
SD	Non-PD	No	SD
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

CR = complete response; PR = partial response; SD = stable disease; PD = progressive disease

6.2.1 First Documentation of Response

The time between initiation of therapy and first documentation of PR or CR.

6.2.2 Confirmation of Response

To be assigned a status of complete or partial response, changes in tumor measurements must be confirmed by repeat assessments performed **no less than four weeks** after the criteria for response are first met.

6.2.3 Duration of Response

Duration of overall response – the period measured from the time that measurement criteria are met for complete or partial response (whichever status is recorded first) until the first date that recurrent or progressive disease is objectively documented; taking as reference the smallest measurements recorded since treatment started.

6.2.3.1 Duration of Overall Complete Response

The period measured from the time measurement criteria are met for complete response until the first date that recurrent disease is objectively documented.

6.2.3.2 Duration of Stable Disease

A measurement from registration until the criteria for disease progression is met, taking as reference the smallest measurements recorded since registration. To be assigned a status of stable disease, measurements must have met the stable disease criteria at least once after study entry at a minimum interval or at least 8 weeks.

6.2.4 Methods of Measurement

Imaging based evaluation is preferred to evaluation by clinical examination. The same imaging modality must be used throughout the study to measure disease.

6.2.4.1 CT and MRI

CT and magnetic resonance imaging (MRI) are the best currently available and most reproducible methods for measuring target lesions. Conventional CT and MRI should be performed with contiguous cuts of 10 mm or less in slice thickness. Spiral CT should be performed by use of a 5 mm contiguous reconstruction algorithm. This specification applies to tumors of the chest, abdomen, and pelvis, while head and neck tumors, and those of the extremities require specific procedures.

6.2.4.2 Chest X-Ray

Lesions on chest x-ray are acceptable as measurable lesions when they are clearly defined and surrounded by an aerated lung. However, CT is preferable.

6.2.4.3 Clinical Examination

Clinically detected lesions will only be considered measurable when they are superficial (e.g., skin nodules and palpable lymph nodes). For skin lesions, documentation by color photography, including a ruler to estimate size of the lesion, is recommended. Photographs should be retained at the institution.

6.2.4.4 Cytology and Histology

Cytologic and histologic techniques can be used to differentiate between complete and partial response in rare cases (e.g., after treatment to differentiate residual benign lesions and residual malignant lesions in germ cell tumors). Cytologic confirmation of the neoplastic nature of any effusion that appears or worsens during treatment is required when the measurable tumor has met response or stable disease criteria.

6.2.4.5 Endoscopy and Laparoscopy

Endoscopy and laparoscopy have not been fully or widely validated, so their use should be limited to validation studies in specialized institutions, and to confirming complete histopathologic response when biopsy specimens have been obtained.

6.2.4.6 Ultrasound

Ultrasound may be used only as an alternative to clinical measurements for superficial palpable lymph nodes, subcutaneous lesions, and thyroid nodules, and for confirming complete disappearance of superficial lesions usually assessed by clinical examination.

6.3 Serological Response

6.3.1 Response According to PSA Levels

PSA levels will be assessed before hormone therapy, at time of randomization, then every 12 weeks thereafter. Patients PSA will be assigned a response according to the following criteria:

Complete Serological Response – PSA level less than 0.2 ng/mL measured for 2 consecutive measurements at least 4 weeks apart.

Serological Partial Response (PR) – decline of PSA value, referenced to the pre-androgen therapy level, by greater than or equal to 50% for 2 consecutive measurements at least 4 weeks apart.

Serological Progression (PD) – increase in PSA to more than 50% of nadir taking as reference the lowest recorded PSA level since starting androgen deprivation therapy. Two consecutive increases must be documented with each measurement obtained at least 2 weeks apart. The date of the first recorded increase will be deemed the date of progression. If a patient achieves a PSA less than 4 ng/mL, progression will only be deemed to occur once the level rises above 4 ng/mL and has risen 50% above the level before commencing androgen deprivation therapy.

6.3.2 Definitions of Serological Response

The following definitions of serological responses apply in this study:

First Documentation of Response – The period measured from the initiation of therapy until first documentation of partial or complete response.

Confirmation of Response – To be assigned a status of complete or partial response, changes in PSA level must be confirmed by repeat assessments performed *no less than 4 weeks* after the criteria for response are first met.

Duration of Response – The period measured from the time that measurement criteria are met for complete or partial response (whichever status is recorded first) until the first date that progressive disease is objectively documented, taking as reference the lowest PSA level recorded since androgen deprivation therapy started.

Duration of Complete Response – The period measured from the time that measurement criteria are met for complete response until the first date that progressive disease is objectively documented.

Time to Serological Progression – The period measured from the initiation of therapy until at least a 50% increase in PSA level, taking as reference the lowest PSA level recorded since the treatment started. The date of the first recorded increase will be deemed the date of progression. If a patient achieves a PSA less than 4 ng/mL, progression will only be deemed to occur once the level rises above 4 ng/mL and has risen 50% above baseline.

6.4 Endpoint Definitions

6.4.1 Overall Survival

Survival will be measured as the period from randomization until death due to any cause.

6.4.2 Time to Hormone Refractory Disease:

The interval between randomization and the date of documented clinical or serological progression with testosterone less than 50 ng/dL or death from any cause.

6.4.3 Time to Clinical Progression:

The interval between randomization and the date of documented clinical progression or death from any cause. Clinical progression is defined as increasing symptomatic bone metastases, progression per RECIST criteria or clinical deterioration due to cancer per investigator's opinion.

6.4.4 Time to Serological Progression:

The interval between randomization and the date of documented PSA progression or death from any cause.

6.5 Quality of Life (FACT-P)

As an additional secondary endpoint, Quality of Life, including benefit and adverse effects of the treatment and the disease, will be measured in all patients.

6.5.1 Overall Quality of Life

The Functional Assessment of Cancer Therapy-Prostate (FACT- P) is a self-report measure of both general and disease-specific quality of life (40). The FACT-P (version 4) contains 39 likert items distributed over 5 subscales: Physical (7 items), Social (7 items), Emotional (6 items), and Functional (7 items) well-being, and the additional concerns related to Prostate Cancer Scale (12 items). The FACT-P is scored by summing responses to the items in each subscale. The subscale scores are then combined to create a total QOL score. A trial outcome index (TOI) can also be calculated by summing the Physical and Functional Well-being Scales and the Prostate Cancer Scale. The FACT-P requires only a sixth-grade reading level and can be completed in 8 – 10 minutes (41).

The FACT-P scales have been widely used to evaluate quality of life in clinical trials for prostate cancer treatments. Internal consistency of the FACT-P (version 4) subscales ranges from .85 to .89. Concurrent, construct and discriminant validity has been documented (41). Sensitivity to change in performance status and PSA score over two months has been demonstrated for the Physical Well-being, Functional Well-being, Prostate Cancer Scales, the Trial Outcome Index, and the total FACT-P (version 4) scale (41).

The FACT-P instrument has been validated in samples of patients with extensive disease prostate cancer (41), and used to assess quality of life in studies of patients with extensive disease prostate cancer (42). However, treatment trials in this patient population have not always demonstrated a significant change in global measures of QOL, such as the FACT-P and the PROSQOLI (43,44). It has been hypothesized that for patients with extensive disease prostate cancer who are undergoing treatment, QOL is the product of both the alleviation of disease-related symptoms and the introduction of treatment-related symptoms (44). Thus, a thoughtful evaluation of the impact of the study treatment on QOL in this sample will require that in addition to overall QOL, disease-related symptomatology and treatment-associated symptomology symptoms must be assessed, as well.

6.5.2 Treatment-associated and Disease-associated Symptoms

The FACT-Taxane instrument is a sub-scale developed to supplement QOL assessment by the FACT scales. It is a 16-item scale that assesses patients' experience of the symptoms most commonly associated with taxane treatment, and has been validated for this purpose (40). The FACT-Taxane subscale has been demonstrated to have excellent internal consistency reliability, validity and responsiveness to change over time (40). Significant decreases in the FACT-Taxane scale have been demonstrated to coincide with the emergence of neurotoxicity in patients undergoing taxane treatment and the relationship of the FACT-taxane score to duration of taxane treatment is linear (40).

Approximately 90% of patients undergoing taxane treatment experience anemia (45). A clinical symptom associated with anemia, fatigue is also a prevalent correlate of extensive disease prostate cancer (44,46). Thus, fatigue, a negative correlate of QOL in cancer patients in general, as a correlate of both extensive disease prostate cancer and the treatment in Arm A, can be expected to play a particularly important role in the evaluation of QOL in this study.

Fatigue will be evaluated using the 13-item Fatigue Subscale of the FACIT-F, which was developed to assess the physical experience of fatigue in cancer patients (47,48). Summed with the four core subscales of the FACT-P (Physical, Social, Emotional and Functional) it creates the Functional Assessment of Cancer Therapy-Fatigue (FACIT-F) score, a measure of both the physical experience and the functional consequences of fatigue (47,48). The Fatigue Subscale and the FACIT-F exhibit strong convergent and discriminant validity, internal consistency, test-retest reliability, and sensitivity to group differences in performance status (47,48).

Pain is the dominant disease-associated symptom experienced by men with metastatic prostate cancer. Therefore, reduction in pain has been the most significant palliative endpoint in treatment trials in this population. Pain will be assessed using the Brief Pain Inventory (BPI) Short Form (49). The BPI is a rapidly and easily completed self-report instrument, designed to measure pain intensity, interference with daily functioning, and relief. Reliability and validity of the BPI are well-established (49, 50).

7. Study Parameters

7.1 Therapeutic Parameters

1. Prestudy scans and x-rays used to assess all measurable or non-measurable sites of disease must be done within 6 weeks of starting hormonal therapy.
2. Prestudy CBC (with differential and platelet count) should be done ≤ 4 weeks before randomization.
3. All required prestudy chemistries, as outlined in Section 3.3, should be done ≤ 4 weeks before randomization – unless specifically required on Day 1 as per protocol.

Visit/ Assessment	Baseline	Day 1, Cycle 1	Day 1, Cycles 2-6	Follow-up; Every 12 weeks ⁶	Hormone Refractory
Arm A					
Physical Examination, Vital Signs	X	X ²	X ⁵	X	
CBC with diff, Platelets ¹	X	X ²	X ¹		
Creatinine clearance, PTT, PT, INR	X	X ²			
Total Bilirubin, ALT	X	X ²	X ¹		
Arm B					
Physical Examination, Vital Signs	X	X ²		X	
CBC with diff, platelets ¹	X				
Creatinine clearance, PTT, PT, INR	X				
Both Arms A and B					
Medical History, ECOG PS	X				
CT Abdomen/Pelvis	X				X
Chest X-ray/CT chest ³	X				X
Bone Scan	X				X
Creatinine Clearance	X				
FACT-P, FACT-Taxane, FACIT-F, and the BPI	X		X ⁷	X ⁸	X
PSA	X		X	X	X
Testosterone					X ⁴
Biological Sample Submissions	Refer to Section 7.2				

- 1 CBCs (with differential and platelet count which includes WBC, ANC, Platelets, Hgb, and Hct) and ALT and total bilirubin which are required for protocol therapy on days 1 of cycles 2-6, must be done day prior to or day of treatment with docetaxel. If holidays or weather prevent treatment on day 1 as planned, day 1 will be the next feasible day of treatment and should be no more than a 7 day delay.
- 2 Does not need to be repeated if within 7 days of screening PE.
- 3 All patients must at least have a CXR prior to randomization to assess status of disease. CT chest at investigator's discretion. CXR within 6 weeks of starting hormonal therapy is acceptable.
- 4 To document castrate level at progression if patient has not had a bilateral orchiectomy.
- 5 Physical examinations can be done up to 72 hours prior to dosing of docetaxel.
- 6 Follow up every 12 weeks will continue beyond the time of HRPc for overall survival and will require physical examination and PSA measurements and recording of all subsequent therapy.
- 7 For both Arms A and B, submit at week 12.
- 8 For both Arms A and B, submit at weeks 24, 36 and 48.

7.2 Biological Sample Submissions

Samples for the correlative studies should be submitted as outlined in Sections 10 and 11. Collection and submission of samples is to be limited to those patients who have given written consent to participate in the correlative studies or banking or both.

NOTE: Institutions outside of the United States and Canada are excluded from submitting fresh samples because of the costs and problems associated with international shipping.

Biological material		NUMBER OF TUBES OR SAMPLES		
		Baseline	Week 24	Progression
Original Diagnostic paraffin embedded tumor block ²		X		
Peripheral Blood ^{1,3,4}	5 mL EDTA lavender top	2	1	1
Serum ^{1,4}	10 mL red top	1	1	1

1. Samples are to be collected after registration, prior to treatment.
2. Materials are to be submitted as described in Section 10.
3. If baseline lavender top tube samples are not taken, please submit at any alternative time point. Please note on Material Submission Form week number and day samples were taken.
4. Kits are available to order and will include the EDTA Lavender top tubes and Red top tube. To order kits, fax Kit Order Form (Appendix VI) to Kristina Kirkpatrick at (317) 921-2053.

8. Drug Formulation and Procurement

8.1 Docetaxel

Please refer to the FDA approved package insert for additional information.

8.1.1 Other Names

Taxotere, RP 56976, NSC #628503, IND # 40466.

8.1.2 Classification

Antimicrotubule agent.

8.1.3 Mode of Action

Docetaxel, a semisynthetic analog of taxol, promotes the assembly of tubulin and inhibits microtubule depolymerization. Bundles of microtubules accumulate and interfere with cell division.

8.1.4 Storage and Stability

Store intact vials between 2° and 25°C (36° and 77°F). Retain in the original package to protect from bright light. The final dilution (in either 0.9% sodium chloride or 5% Dextrose solution) is stable for 4 hours if stored between 2° and 25°C (36° and 77°F).

8.1.5 Dose Specifics

75 mg/m² of docetaxel will be given on day 1 every 21 days for a maximum of six (6) cycles.

8.1.6 Preparation

Just prior to use, allow the docetaxel vial to reach room temperature for 5 minutes. Add the entire contents of the ethanol diluent vial and mix by gently rotating the vial for 45 seconds. Allow to stand for 5 minutes at room temperature, and check that the solution is homogeneous and clear (persistent foam is normal). The resulting solution contains 10 mg/mL of docetaxel. Please note that the solution contains 15% overfill. Dosing amounts should be based in the concentration per extractable volume, not the total volume of the vial. The desired dose is diluted in D5W or NS. The volume of the infusion should be adjusted in order to have a final docetaxel concentration of between 0.3 mg/mL and 0.74 mg/mL. Non-PVC-containing intravenous infusion bags and administration sets should be used to avoid patient exposure to the plasticizer DEHP.

8.1.7 Administration

Docetaxel will be administered to patients as an IV infusion over approximately one hour.

8.1.8 Incompatibilities

Intravenous bags and administration sets containing DEHP (di-[2-ethylexyl] phthalate). No further information available.

8.1.9 Availability

Docetaxel is a commercially marketed product which has been approved by the FDA for: the treatment of locally advanced or metastatic breast cancer that returns after any prior chemotherapy; locally advanced or metastatic non-small cell lung cancer that recurs after prior platinum-based chemotherapy; and in combination with prednisone for the treatment of advanced metastatic prostate cancer. However its use in patients eligible for this protocol with prostate cancer is not currently

approved by the FDA. When used as directed by this protocol, docetaxel is classified as an "unapproved use of an approved agent" and, by definition, considered as an investigational agent. However, while it is not an indication currently approved by the FDA, the use of docetaxel in this protocol is exempt from the requirements of an IND as described under Title 21 CFR 312.2(b).

NOTE: During the course of this study, if the FDA approves docetaxel for use in patients with prostate cancer, then the patient may have to pay for the amount of drug needed to complete the study.

Docetaxel (Taxotere[®]) is available and distributed free of charge from Sanofi-Aventis Pharmaceuticals. The vials of docetaxel will be packaged as 80 mg/2 mL vials. The initial shipment will be 6 vials, enough for 3 cycles. Vials of diluent will also be included.

After patient randomization, a supply of docetaxel may be obtained. Investigators must fax a signed and completed "E3805 Clinical Supplies Shipping Form" (Appendix VII) to the ECOG Coordinating Center, ATTN: Drug Team at 617-632-2063. Do not send drug requests directly to Sanofi-Aventis Pharmaceuticals. Allow 4-5 business days for receipt of the drug. See below for re-ordering instructions.

During the Study

There will be no weekend or holiday delivery of drugs. All deliveries will be made by express carrier.

Receipt of drug shipment: Complete the "Receipt From" section of the shipping form and fax to Debra Dawson at 908-243-7745.

Reorders: Each institution is responsible for reordering study drug. Complete and fax the E3805 Clinical Supplies Shipping Form (Appendix VII) to the ECOG Coordinating Center at 617-632-2063. Please allow 4-5 business days for processing reorders.

Drug Inventory Records: The Investigator or a responsible party, designated by the investigator, must maintain careful record of the inventory and dispensation of all drugs received. Please use the NCI Drug Accountability Record Form or equivalent. (See the NCI Investigators Handbook for Procedures for Drug Accountability and Storage.) The unused docetaxel must be accounted for.

At The Completion of The Study:

At the completion of the study, all used and unused drugs may be destroyed at the site, in accordance with the applicable regulations, guidelines and institution's procedures. Please maintain appropriate records of the disposal, including dates, quantities and method of destruction.

8.1.10 Side Effects

1. Cardiac: arrhythmias, pericardial effusions.
2. Hematologic: dose-related neutropenia, leukopenia, thrombocytopenia, anemia, hypoglycemia, hypernatremia.
3. Gastrointestinal: nausea and vomiting, diarrhea, oral mucositis, pancreatitis, esophagitis.
4. Neurologic: reversible dyesthesias or paresthesias, peripheral neuropathy, mild or moderate lethargy or somnolence, headache, seizures.
5. Hypersensitivity: hypersensitivity (local or general skin rash, flushing, pruritus, drug-fever, chills and rigors, low back pain), severe anaphylactoid reactions (flushing with hypo- or hypertension, with or without dyspnea).

6. Dermatologic: alopecia, desquamation following localized pruriginous maculopapular eruption, skin erythema with edema, extravasation reaction (erythema, swelling, tenderness, pustules), reversible peripheral phlebitis, nail changes.
7. Hepatic: increased transaminase, alkaline phosphatase, bilirubin; hepatic failure; hepatic drug reaction.
8. Pulmonary: dyspnea with restrictive pulmonary syndrome, pleural effusions.
9. Other: asthenia, dysgeusia, anorexia, conjunctivitis, arthralgia, muscle aches, myopathy, peripheral edema, fluid retention syndrome, ascites.

8.1.11 Nursing Implications

1. Monitor CBC and platelet count prior to drug administration.
2. Symptom management of expected nausea, vomiting, and mucositis.
3. Advise patients of possible hair loss.
4. Monitor for signs and symptoms of hypersensitivity reactions. Insure that recommended premedications are given.
5. Monitor liver function tests.
6. Evaluate site regularly for signs of infiltration.
7. Monitor for symptoms of peripheral neuropathy.
8. Monitor for signs of fluid retention and cutaneous reactions.

8.1.12 References

Investigator's Brochure: Docetaxel. Rhone-Poulenc Rorer, June 14, 1995.

Pazdur R, *et al.* Phase I trial of Taxotere; Five-day schedule. *J Natl Cancer Inst* 84:1781-8, 1992.

Pazdur R, *et al.* Phase I trial of Taxotere (RP56976). *Proc Am Soc Clin Oncol* 11:111, 1992.

Burris H, *et al.* A phase I clinical trial of Taxotere as a 6-hour infusion repeated every 21 days in patients with refractory solid tumors. *Proc Am Soc Clin Oncol* 11:137, 1992.

Bissett D, *et al.* Phase I study of Taxotere (RP56976) given as a 24-hour infusion. *Proc Am Assoc Cancer Res* 33:526, 1992.

Bruno R, *et al.* Clinical pharmacology of Taxotere (RP56976) given as a 1-2 hour infusion every 2-3 weeks. *Proc Am Assoc Cancer Res* 33:261, 1992.

DeValeriola D, *et al.* Phase I pharmacokinetic study of Taxotere (RP56976) administered as a weekly infusion. *Proc Am Assoc Cancer Res* 33:1563, 1992.

8.2 LHRH agonists (such as leuprolide, goserelin, triptorelin)

Please refer to the FDA approved package insert for additional information.

8.2.1 Description:

LHRH agonists are long-acting analogs of the native LHRH peptide and are effective at reducing serum testosterone. Analog approved by the FDA can be used in this study.

8.2.2 Supply:

Commercial available.

8.2.3 Administration:

LHRH analogs are administered with a variety of techniques such as subcutaneously, intramuscularly, or insertion. The manufacturer's instructions should be followed.

8.2.4 Toxicity:

The most common side effect is vasomotor hot flashes; edema, gynecomastia, bone pain, thrombosis, and gastrointestinal disturbances. Other side effects include impotence and loss of libido, weight gain, depression, dizziness, anemia, skin redness, pain at the injection site, unusual taste in the mouth, increased thirst and urination, and rarely allergic generalized rash and difficulty breathing.

8.3 Antiandrogens (flutamide and bicalutamide)

Please refer to the FDA approved package insert for additional information.

8.3.1 Availability:

Commercial available.

8.3.2 Administration:

The drugs are administered orally as prescribed.

8.3.3 Toxicity (flutamide):

Diarrhea and anemia. A high percentage of patients treated with flutamide alone developed gynecomastia within 2 to 8 months. There have been post marketing reports of hospitalization, and rarely, death due to liver failure in patients taking flutamide. Other side effects include elevated transaminase levels, jaundice, hepatic encephalopathy, and death related to acute hepatic failure, which is reversible after prompt discontinuation in some patients; impotence and loss of libido, fatigue, and rarely photosensitivity.

8.3.4 Toxicities (bicalutamide):

Breast tenderness, breast swelling, and hot flashes. When bicalutamide 50 mg was given in combination with an LHRH analog, the LHRH analog adverse event profile predominated with a high incidence of hot flashes (53%), and relatively low incidences of gynecomastia (4.7%) and breast pain (3.2%). Other side effects include impotence and loss of libido, fatigue, and rarely photosensitivity.

9. Statistical Considerations

9.1 Primary Endpoint

The primary clinical endpoint is the difference across treatments in overall survival. Patients will be stratified based on age, ECOG PS, use of combined androgen blockade and bisphosphonates, no subgroup analyses are planned. Primary endpoint will be studied for all intent-to-treat patients.

Secondary endpoints include PSA response, change of PSA over time, time to hormone refractory disease, time to clinical progression and time to PSA progression. All patients who receive treatment, regardless of eligibility, will be evaluated for toxicity.

9.2 Sample Size and Accrual

Once the study is fully launched, we plan to accrue 12 patients per month for 45 months, and to follow all patients for 36 months. We anticipate it will take 12 months to fully activate enough sites to reach 12 patients per month. Under these assumptions, the study will require 540 eligible patients (270 eligible patients per arm). Assuming 5% of ineligibility rate, 28 additional patients are planned for accrual, for a total of 568 patients. The study will have about 80% power to detect a 33.3% improvement in median survival time from 24 months in arm B to 32 months in arm A. A log-rank test will be used, with a one-sided 0.025 significance level.

Interim analyses will be performed for all semi-annual ECOG Data Monitoring Committee (DMC) meetings beginning when approximately 25% of the planned full information has occurred, about 29 months after the study opens, continuing every 6 months until either criterion for early stopping are met or full information is reached. Twenty-five percent of the full information under the alternative hypothesis is equal to 101 failures. Full information will be achieved when 408 eligible patients have died. The final analysis is scheduled for this time point. This study will be monitored for early stopping in favor of the null hypothesis using repeated confidence interval methodology similar to that described by Jennison and Turnbull, 1990 (Statistical Science 5:299-317). At each interim analysis the nominal $(1-2 \times \alpha)$ confidence interval on the overall survival hazard ratio comparing placebo arm to the vaccine arm will be computed, where α is the nominal one-sided significance level of the use function boundary at the information fraction for the particular analysis time. If the confidence interval does not contain the target alternative of 1.33, then the data monitoring committee may consider closing an arm of the trial or stopping the study early for overall lack of treatment differences. Under the accrual and failure rate assumptions above, the following table gives the interim analysis times, information times, and the corresponding futility boundaries and truncated O'Brien Fleming boundaries (P. O'Brien and T. Fleming, 1979, Biometrics 35:549-556). Information from interim analyses will be reviewed by the DMC, who may decide to stop the study early if efficacy or futility has been demonstrated. Because of delays in initiation of accrual and delays in data submission and processing, it is likely that the actual analysis times will be 6-12 months later.

Repeated Analyses	Real Time	Information Time	Number of Failures		P-value	Lower Boundary	Upper Boundary	Number of Patients
			under H0	under H1				
1	29	0.25	112.37	101.10	0.0005	-1.85	3.29	348
2	35	0.35	155.78	140.83	0.0003	-1.60	3.29	420
3	41	0.45	203.74	185.04	0.0003	-1.35	3.29	492
4	47	0.57	254.85	232.48	0.0018	-0.66	2.83	540
5	53	0.68	300.24	275.34	0.0034	-0.20	2.56	540
6	59	0.76	338.41	312.13	0.0040	0.11	2.40	540
7	65	0.84	370.51	343.70	0.0042	0.35	2.29	540
8	71	0.91	397.49	370.83	0.0041	0.53	2.21	540
9	77	0.97	420.18	394.14	0.0039	0.67	2.15	540
10	81	1.00	433.26	407.84	0.0024	0.74	2.13	540

9.3 Randomization and Stratification

Randomization will be performed with the following stratifications: 1) age: ≥ 70 vs. < 70 ; 2) ECOG performance score: 0-1 vs. 2; and 3) intended use of combined androgen blockade greater than 30 days from start of therapy; and 4) prior adjuvant hormonal therapy > 12 months vs. ≤ 12 months and 5) bisphosphonate use.

The method of permuted blocks will be used for subject randomization. No per-site treatment-allocation balance will be implemented.

9.4 Quality of Life Analysis

The primary quality of life objective is to describe quality of life over time within each arm. Quality of life will be evaluated at baseline (within 1 week prior to therapy), week 12, week 24, week 36 and week 48. The primary comparison of interest will be the comparison between baseline (pre-treatment) and week 12 for scores on the FACT-P instrument. Treatment-associated and disease-associated changes in QOL will also be compared between baseline and week 12 for scores on the FACIT-F Fatigue Subscale, FACT-Taxane Subscale and the Brief Pain inventory. A comparative analysis will also be performed between arms A and B at each time point.

There are two sources of missing assessments in this study. The first source of missing assessments is among patients who are alive but do not complete the assessment, for reasons such as missed appointments, staff forgetting to administer the instrument, or patient refusal. We assume that this missingness is random, and will analyze the remaining questionnaires under this assumption. We expect only about 2% of the assessments to be missing at baseline, 10% by week 12, and about 30% to be missing by 24 weeks. The other type of missingness is due to patients who will die during the study. If the median survival is 2-3 years and we assume that deaths follow an exponential distribution, we would expect about 5%-8% of the patients to have died by week 12. Because quality of life is likely to be related to survival, this censoring may be non-ignorable. For this reason, the comparison of Week 12 to baseline will be considered the primary analysis.

We expect 85% to be alive and have completed baseline and week 12 assessments. A paired t-test will be used to test the null hypothesis that the change in quality of life score is

not equal to 0 for each arm. Assuming 230 patients on each arm with measures on both days and a two-sided test with $\alpha=0.05$, we have 90% power to detect a difference of 0.22 standard deviation. This translates into a change of about 3 points assuming the standard deviation of the difference is 14.5 points, which was the standard deviation of the difference in QOL scores between baseline and week 8 from patients in E1898. Differences in quality of life between treatment arms will also be of interest. A t-test will be used to test for pairwise differences between treatment arms at week 12. Assuming a two-sided test with $\alpha=0.05$, the study will have 90% power to detect a difference of 0.3 standard deviation between two treatment arms. This translates into a change of about 4.4 points assuming the common standard deviation is 14.5 points. Methods developed by Schluchter (1992) (20) will be used to jointly model quality of life and survival data adjusting for non-ignorable missing data due to death or patient withdrawal. Similar analyses will be performed on the FACIT-F Fatigue Subscale scores, FACT-Taxane Subscale score and the Brief Pain inventory scores.

9.5 Gender and Ethnicity

Based on previous data from *E1898* the anticipated accrual in subgroups defined by gender and race is:

Ethnic Category	Gender		
	Females	Males	Total
Hispanic or Latino	0	0	0
Not Hispanic or Latino	0	568	568
Ethnic Category: Total of all subjects	0	568	568
Racial Category			
American Indian or Alaskan Native	0	0	0
Asian	0	7	7
Black or African American	0	52	52
Native Hawaiian or other Pacific Islander	0	0	0
White	0	509	509
Racial Category: Total of all subjects	0	568	568

10. Pathology Review

Pathology should be submitted for use in the correlative study described in Section 11. Samples for the correlative studies should be submitted only from patients who have given written consent for the use of their samples for these purposes.

When a patient is registered to receive protocol therapy, the submitting pathologist and clinical research associate should refer to Appendix II (Pathology Submission Guidelines), for submission requirements.

10.1 The materials required for this protocol are:

10.1.1 Forms:

- ECOG Pathology Material Submission Form (#638), Parts A & B completed. Please identify the clinical status of the submitted material (i.e., pretreatment as opposed to remission and relapse).
- A copy of the surgical pathology report, if available.

10.1.2 Biological Material Submission

10.1.2.1 Correlative Studies

- One tumor block from the original diagnostic biopsy
- If a tumor block is unavailable, a minimum of five (5), preferably 10, unstained positively charged slides (3-4 microns) and one (1) H & E slide may be substituted.

NOTE: A copy of the completed submission form will be sent to the Coordinating Center by the Pathology Coordinating Office.

10.2 Shipping Procedures

10.2.1 Submission Schedule

The original diagnostic block or slides should be submitted within one month of registration.

Submit materials to:

ECOG Pathology Coordinating Office
Robert H. Lurie Comprehensive Cancer Center
of Northwestern University Medical School
Olson Pavilion – Room 8501
710 North Fairbanks Court
Chicago, IL 60611
Tel: (312) 503-3384
FAX: (312) 503-3385

10.3 Pathology Coordinating Office: Sample processing and Routing

The appropriate materials for the correlative studies will be distributed to Dr. Christopher Sweeney for analysis.

10.4 Banking

Residual material from the blocks/slides submitted will be retained at the ECOG Central Repository for possible use in future ECOG approved studies. Any residual blocks will be available for purposes of individual patient management on specific written request. If future use is denied or withdrawn by the patient, the samples will be removed from consideration for use in any future study.

11. Correlative Studies

NOTE: An informed consent MUST be signed prior to the submission of any material for any correlative study, including mandatory diagnostic reviews. Samples for optional correlative studies should be submitted only from patients who have given written consent for the use of their samples for these purposes.

NOTE: Kits are available. They will include the EDTA Lavender top tube, Red top tube, and shipping materials. Upon institutional IRB approval, complete Kit Order Form (Appendix VI) and fax to Kristina Kirkpatrick at (317) 921-2053. Kits will NOT be shipped overnight courier unless expressly stated on the form that it is needed within 24 hours de to patient enrollment.

NOTE: An E3805 Material Submission Form #2377 must be submitted with each submission. All samples, sample treatments, and collection time points are to be indicated on the form(s). The original E3805 Material Submission Form #2377 should accompany the samples to the central laboratory(ies). Please forward a copy of the original form to the ECOG Coordinating Center (Attn: Correlative Sciences) at the same time it is sent to the central laboratory.

NOTE: Institutions outside of the United States and Canada are excluded from submitting fresh samples because of the costs and problems associated with international shipping.

Submission Summary:

Biological material		NUMBER OF TUBES OR SAMPLES		
		Baseline	Week 24	Progression
Tumor Block Original Diagnostic paraffin embedded ²		X		
Peripheral Blood ^{1,3}	5 mL EDTA lavender top tube	2	1	1
Serum ¹	10 mL red top tube	1	1	1

1. Samples are to be collected after registration, prior to treatment.
2. Materials are to be submitted as described in Section 10 and Appendix II.
3. If baseline lavender top tube samples are not taken, please submit at any alternative time point. Please note on Material Submission Form week number and day samples were taken.

11.1 Sample Preparation

Sample preparation questions may be directed to Kristina Kirkpatrick at (317) 921-2050 or by email at kriskirk@iupui.edu

11.1.1 Tumor Tissue

If available, original diagnostic paraffin embedded tumor block is requested. If tumor block is not available a minimum of five (5) unstained, positively charged slides and one H & E slide may be substituted. Materials are to be submitted as outlined in Section 10, within 1 month of registration.

11.1.2 Blood Draws

Submission schedule is outlined in the table above. Kits are available. They will include the EDTA Lavender top tubes, red top tube, shipping containers and pre-printed shipping labels. Upon institutional IRB approval, complete Kit Order Form (Appendix VI) and fax to Kristina Kirkpatrick at (317) 921-2053. Kits will NOT be shipped by overnight courier unless expressly stated on the form that it is needed within 24 hours due to patient enrollment.

11.1.2.1 Red Top Tube

SERUM SAMPLES MUST BE FROZEN WITHIN ONE HOUR OF BLOOD DRAW.

Draw approximately 10 ml of blood into the red top tube, allow to coagulate for 30 minutes then centrifuge it at 3500 rpm for 30 minutes. Without delay, divide the serum into three (3) cryovials in about 1.5 ml aliquots then freeze immediately at -70°C until time of shipment. Samples MUST be shipped by overnight courier as soon as possible, DO NOT ALLOW SAMPLES TO THAW. If a -70°C freezer is not available serum samples may be stored at -20°C.

11.1.2.2 EDTA Lavender Top Tubes

11.1.2.2.1 Whole Blood Processing

Draw 5 mL of whole blood into one of the lavender top tubes, shake gently, and then separate sample equally into two (2) cryovials. Sample may be stored at -20° to -70°C and shipped with aliquots from serum sample.

11.1.2.2.2 Plasma Processing

SAMPLE PROCESSING MUST BE STARTED WITHIN 30 MINUTES OF BLOOD DRAW.

Draw approximately 5 mL into the second lavender top tube, invert gently then centrifuge it at 3500 rpm for 30 minutes. Without delay, divide plasma into three cryovials in ~ 1.0 ml aliquots, and then freeze the samples immediately at -70°C. Samples MUST be shipped as soon as possible and sent by overnight courier, DO NOT ALLOW SAMPLES TO THAW. If a -70°C freezer is not available plasma samples may be stored at -20°C until shipped.

11.2 Shipping Guidelines

Complete the appropriate time points pre-printed cryovial label provided in the kit (baseline, week 24 or hormone refractory) using the Sharpie pen provided. Fill in the patient initials, date, and ECOG patient ID number then placed on relevant cryovial. If multiple patients' samples are shipped together, bag each patients' samples separately using provided baggies. The appropriate Material Submission form must accompany the samples.

Samples are to be shipped on at least 5 pounds dry ice, in the shipping containers provided. Ship Monday through Thursday only (to arrive Tuesday through Friday). Shipments that arrive on Saturdays and holidays will NOT be accepted. Dr. Sweeney's laboratory must be contacted PRIOR to shipping, email notification to: Raja Shanmugam at rshanmug@iupui.edu and Kristina Kirkpatrick at kriskirk@iupui.edu or call her at (317) 921-2050.

Ship samples to:

Sweeney Lab
Attn: Raja Shanmugam
1044 West Walnut Street R4-169
Indianapolis, IN 46202

NOTE: If your facility does not have dry-ice, contact Kristina Kirkpatrick to review potential sources.

11.3 Proteomic Study

Two-dimensional polyacrylamide gel electrophoresis (2D PAGE) has been the principal tool for the separation and simultaneous analysis of multiple proteins (20). This methodology, which is able to resolve numerous proteins in one experiment, provides the highest resolution in protein separation. Although the resolving power of 2D PAGE remains unchallenged, the high sensitivity, speed and reproducibility of mass spectrometry-based approaches have boosted their application in all aspects of protein analysis, including profiling, discovery and identification (21, 22), creating what is now called "proteomics." Another technology, Surface Enhanced Laser Desorption /Ionization Protein Chip Mass Spectrometry (SELDI) (23) has brought mass protein screening to the field of proteomics. We will use the combined technologies of 2D-PAGE, and SELDI mass spectrometry (MS) to identify protein biomarkers in the serum that distinguish hormone sensitive and hormone resistant prostate cancer. Once discovered, markers will be identified using tandem MS. Drs Hickey and Malkas at Indiana University have had extensive experience with this technology and will perform this analysis at Indiana University (24).

SELDI technology uses patented ProteinChip arrays coated with a chemical to affinity capture polypeptide molecules from complex mixtures. Various types of protein chips with different capturing matrices such as hydrophobic, ionic, and metal binding are available to optimize protein-binding affinities. Also, ProteinChips with preactivated surfaces allow covalent immobilization of antibodies, receptors, DNA, glycoproteins, etc. for specific affinity capture. Retained proteins are subsequently analyzed by time of flight (TOF) mass spectrometry. With the aid of SELDI software, a retentate map is generated depicting the mass/charge spectrum of proteins captured on the chip's surface. The spectra can then be combined or compared to elucidate changes of the protein profiles between samples. The serum samples resulting from the clinical trial will be processed on each of three different types of chip surfaces with affinity for hydrophobic, ionic or metal binding. Samples will be analyzed in duplicate to confirm reproducibility of the protein profiles. Protein peaks will be labeled and normalized for total ion current to account for variation in ionization efficiencies. Using SELDI software, peak mass and intensity level, information will be exported and analyzed by the cumulative probability and decision tree learning algorithms. This experiment will not only determine if we have discovered novel potential biomarkers, but it will provide us with the relative expression patterns of biomarkers at the different time points being evaluated. Novel therapies that modulate the expression of these proteins can then be developed and (for example) possibly prevent development of hormone refractory prostate cancer.

A high throughput proteomic analysis system for prostate cancer using patient sera will be developed for this study. The methodology to be used is based on that shown to be able to discriminate sera from patients with early or advanced ovarian cancer from normal sera (25). The analysis will be based on a system for the separation and profiling of serum proteins by SELDI-TOF. The position of an individual protein or peptide within a readout spectrum corresponds to the mass dependent time-of-flight to the detector. Resulting protein patterns will then be analyzed as clusters or groupings that are similar or not similar. A set of known "training" samples will be used to segregate the data into two types of clusters: those containing samples from patients known to have hormone refractory prostate cancer (n=100) and clusters containing samples from patients with hormone sensitive prostate cancer (n=100). After "training" a computer algorithm, the pattern of an unknown sample will be diagnostically classified by its similarity to the diseased or unaffected clusters found in the training set. Training the computer will be done by means of computer-simulated Darwinian evolution in which the outcome is "survival of the fittest" subset of data in a sample. The algorithm starts by randomly selecting many proteomic patterns within the training data for analysis. Each chosen pattern is tested to see how well it can discriminate affected from unaffected individuals in the training set, by cluster analysis. Successful proteomic patterns are kept and

recombined, whereas unsuccessful patterns are discarded. Ultimately, a best pattern emerges after many iterations by the algorithm. This pattern, which best segregates the training sets, is used to diagnostically classify unknown samples. The discriminatory power of this system for the detection of early ovarian cancer is highly encouraging, revealing 100% sensitivity and 98% specificity for a small study population of 67 control women with non-malignant disease, 18 patients with stage I ovarian cancer and 32 patients with Stage II, III or IV ovarian cancer. Based on this and unpublished data showing specificity of the serum proteomic pattern analysis for ovarian compared to prostate cancer, we anticipate these studies may reveal a similar power for detection of specific protein expressions that will differentiate hormone sensitive and hormone refractory prostate cancer.

Potential pitfalls to the SELDI analyses are that the resolution and mass accuracy of the TOF mass analyzer does not allow reliable protein identification, and large molecular weight proteins are hard to identify. To solve these problems we will digest the proteins retained on the surface of the chip and analyze the resulting peptides using an LCQ Advantage ion trap MS. LCQ Advantage identifies proteins retained on the chip's surface by generating internal sequencing data. Identification of high molecular weight markers will also be accomplished using 2D-PAGE.

2D PAGE/Mass Spectrometry (MS) analysis will be performed to enhance the detection of potential markers in the samples produced in the SELDI experiments. Specifically, we will identify a specific protein profile of interest and use this technology to identify the proteins of interest. This is done to offset some of the weaknesses of the current SELDI technology; namely the inability to both resolve polypeptides greater than 70kDa and to identify the resulting proteins/peptides. 2D-PAGE analyses will be performed on the samples and the resolved polypeptides will be visualized using colloidal Coomassie blue dye or silver stain. Polypeptide migration patterns will be analyzed by Phoretix 2D Advanced Software, identifying differences in protein migration and expression. Polypeptide "spots" will be excised from the gel, digested with trypsin, and analyzed using the LCQ Advantage ion trap MS. Analysis of the peptides will identify the most abundant proteins present in the spots by generating internal sequence information on the peptides. This type of MS can also identify post-translational modifications, and identify the exact amino acid residues in which they reside. Low expression level of some proteins as compared to other polypeptides limit the usefulness of this technique. Differences in protein migration and expression in serum proteins from patients with hormone sensitive and hormone refractory disease will be analyzed using a CCD camera (3000 x more sensitive than the human eye, (NIH proteomics workshop, 2001)) and Phoretix 2D Advanced Software. Markers identified by differences in the migration and expression profile by 2D-PAGE will be identified by analysis using an LCQ ion trap MS. If needed approaches that increase the likelihood of identifying markers present in low abundance. First, cellular fractionation and chromatographic approaches will reduce the numbers of proteins resolved by 2D-PAGE greatly increasing the probability of visualizing a low abundance biomarker. Also, the resolving power of 2D-PAGE can be greatly increased by using smaller pH gradients (i.e., pH range 7-10 as compared to 3-10), and finally, the use of "ultrasensitive" protein stains like Sypro ruby red increases the detection limits of 2D-PAGE and therefore increases the probability of identifying a low abundance biomarker.

In summary, the serum samples that result from the clinical trial will be subjected to SELDI ProteinChip MS and 2D PAGE analyses, and potential biomarkers will be identified by MS/MS. This approach will permit a comprehensive search for changes in polypeptide profiles that are present at the various stages of prostate cancer. The hypothesis to be tested is that there are proteins differentially translated from the genome in hormone sensitive prostate cancer, prostate cancer that has responded to hormonal therapy and hormone refractory prostate cancer. This analysis will be performed at the Indiana Cancer Pavilion under the direction of Dr. Christopher Sweeney.

The following definitions will be applied:

The "hormone sensitive" patients will be defined as patients who have met eligibility criteria. It is known that 90% of these patients will have a PSA response. Patients who do not respond (>50% decline in PSA per Section 6.3) to androgen ablation will be removed from the analysis.

"Prostate cancer that has responded to hormonal therapy" will be defined in patients with a PSA that has met PSA criteria for PSA response (per Section 6.3) and is maintained at the 6 month mark from start of protocol therapy. In some cases, patients will have been on therapy for up to 9 months (90 day window for androgen ablation prior to enrollment). Patients treated with androgen ablation alone will be analyzed separately and in combination with the patients who received chemotherapy.

Serum from patients with "hormone refractory prostate cancer" will be obtained from patients who have provided serum at time of hormone refractory disease (see Section 6.4.2 for definition).

To ensure reproducibility the following analyses will be performed:

All samples will be run in duplicate. A run-in pilot study on samples from 70 patients with HRPC has been conducted at Indiana University and the optimized conditions from these experiments will be applied to the samples from this protocol. A normal serum control sample will be added to one spot on each chip to serve as quality control for chip integrity, process integrity, and mass spectrometer function. All samples and controls will be added randomly to the chips. Samples of replicates will be run in separate batches to assess inter-assay variability. Samples from different "disease states" (e.g. hormone sensitive and hormone refractory) will be intermingled to avoid assay batch effects.

Planned Analysis - Training Set/Test-set Approach

Samples from the Hoosier Oncology Group trial "A phase III, randomized, double-blind, placebo-controlled trial evaluating the ability of risedronate to prevent skeletal related events in patients with metastatic prostate cancer commencing hormonal therapy" will serve as the training set. Samples are being collected at the same timepoints as E3805. Classifiers will be completely specified based on data from this study (As of May 2005, 50 samples have been collected). It is anticipated there will be approximately 100 patients from the HOG study to develop the "training set" and 100 patients from E3805 to form the "test set". Using the test set of size 100, overall accuracy of the classifier can be estimated to within approximately +/- 10%.

11.3.1 Sample Submission

Serum aliquots submitted as outlined in Section 11.1.2 will be used for this study. No additional samples are requested.

11.4 Genomic Analysis

In addition to analyzing the proteome, an analysis of the genome may also provide insight into the causes of prostate cancer and outcomes from treatment. We acknowledge that analyses of genetic polymorphisms will initially be exploratory. However, the potential value of obtaining the DNA and performing the analyses will provide the basis for further directed research. We will make use of Dr David Flockhart's (Indiana University) extensive experience expertise in this endeavor (34-38). The same precautions for protecting the anonymity of the patients and obtaining consent to perform the studies will be adhered to as those described for the proteomic analysis. We will obtain ethnicity information on all patients as polymorphisms often differ between ethnic groups. Two examples of the analyses that can be performed are the following:

Pharmacogenetic analysis of drug metabolism and transport genes

Cytochrome p450 3A4 and *mdr* genes metabolize and transport chemotherapeutics such as docetaxel respectively. As such variants of these genes may predict for altered efficacy and toxicity. Preliminary data from a Hoosier Oncology Group trial assessing two docetaxel based regimens for hormone refractory prostate cancer found that patients with the more functional ABCG2 (breast cancer resistance protein – BCRP) gene with C/C at position 421 have a lower chance of being alive beyond 15 months (27% of 44 patients) compared with those with the less functional transporter (C/A). In the later group, 4 of 6 patients were alive beyond 15 months ($p=0.05$). It is therefore hypothesized that, the presence of a less functional germline transporter predicts for less efflux of cytotoxic drugs from cancer cells and better survival. It should be recognized this was from an analysis of only 52 patients. Therefore a study such as E3805 with a larger number of patients is needed to confirm or refute the significance of this finding. The C/A variant is found in about 25% of Caucasians. Therefore, approximately 70 patients (25% of 270) treated with docetaxel when commencing hormonal therapy will have this variant. It is hypothesized that this group will have a longer time to hormone refractory disease than those with the C/C genotype (approximately 200 patients) as the latter group will have a more effective drug transporter to remove docetaxel from the tumor. The time to progression of patients with C/A and C/C genotypes treated with androgen ablation alone will be analyzed and compared with their counterparts on the chemohormonal therapy arm. An example of the statistical power that will be available to address pharmacogenetic questions such as the one detailed with the ABCG2 gene is the following. Assume that the C/A genotype occurs in approximately 25% of patients and it confers a better overall survival for patients receiving docetaxel. With genotype results from 270 patients receiving docetaxel, there will be approximately 90% power to detect a hazard ratio of 1.8 when comparing overall survival between C/C and C/A groups using a two-sided 0.05 level logrank test and assuming a median overall survival of 32 months in the docetaxel arm and accrual over 45 months with 36 months additional follow-up.

Analysis of genetic polymorphisms that may alter the activity of enzymes involved in activation or deactivation of steroid hormones

The rationale for this is the following: The development of prostate cancer is possibly modulated by steroid hormones and many steroids and environmental carcinogens are subject to cytochrome P450 mediated metabolism which can generate reactive metabolites and may promote tumor initiation. Polymorphisms in the enzymes that are specifically involved in the metabolism of steroids may therefore be a specific risk factor for this disease. Some studies have suggested that this may in fact be the case (26-33). We will access the DNA of patients enrolled onto this trial and determine the frequency of polymorphisms of enzymes that are possibly involved in the genesis of prostate cancer. This will then be compared with frequency of these polymorphisms that will be identified from a screened population (samples and analyses will be obtained from a screening study that is being conducted Indiana University). The analysis in the screening study will employ a case-control design. The frequency of polymorphisms in E3805 will then be compared informally with the results from the screening study. The demographics and ethnicity of patients from both E3805 and the screening study will be known and allow investigators to investigate the generalizability of the data from the two studies. One published example of a polymorphism of interest is an analysis in a Japanese population of the cytochrome p450 1B1 gene. This gene encodes for an enzyme involved in the hydroxylation of steroid hormones. The odds ratio for developing prostate cancer if the genotype T/T at codon 119 was 4.02 compared to the reference genotype of G/G (95% confidence interval: 1.73-9.38 - $P<0.001$). Validating the frequency and significance of this polymorphism in a study of patients with metastatic disease and in what we expect to be predominately Caucasian population is required.

This analysis will be performed at the Indiana Cancer Pavilion under the direction of Dr. Christopher Sweeney.

11.4.1 Sample Submission

DNA will be extracted and analyzed from the lavender top tube blood samples submitted as outlined in Section 11.1.2. No additional samples are requested.

11.5 Nuclear factor kappa B and Anti-apoptotic proteins.

There has been very little tissue obtained in a standardized fashion with clinical data from patients with metastatic disease. It has been noted that the expression of nuclear factor kappa B has been reported to be over-expressed in organ confined prostate cancer (39). However, its effect on prognosis is unknown at this time. CXCR4 and Manganese superoxide dismutase are proteins under the control of nuclear factor kappa B and are associated with metastasis and avoidance of apoptosis respectively. Other proteins of interest under control of NFkB that will be analyzed include TRAF-1, Bcl-XL, and cIAP2. The amount and frequency of expression of these proteins as well as BCL-2 will be analyzed in the samples from patients on this trial (with metastatic disease) and compared with samples from patients who never developed metastatic disease and will also be correlated with nuclear factor kappa B expression. This analysis will be performed under the direction of Dr. Christopher Sweeney.

11.5.1 Sample Submission

The original diagnostic tumor blocks that were submitted as outlined in Section 10 will be used for this study. No additional samples are requested.

11.6 Banking

The residuals and/or derivatives of the samples submitted for the correlative studies described in Section 11 will be retained at the ECOG Central Repository for possible use in future ECOG approved studies. Any residual blocks will be available for purposes of individual patient management on request. If future use is denied or withdrawn by the patient, the samples will be removed from consideration for use in any future study.

11.7 Sample Inventory Submission Guidelines

Inventories of all samples collected, aliquoted, and used on the above mentioned laboratory correlative study(ies) will be submitted to the ECOG Coordinating Center on a quarterly basis. The quarterly cut-off dates are March 31, June 30, September 30 and December 31. Data is due at the ECOG Coordinating Center 1 week after these cut-off dates. Inventories will be submitted electronically or by diskette by any laboratory holding and/or using any specimens associated with this study. Electronic submissions should be submitted to ecog.labdata@jimmy.harvard.edu. All other correspondence should be addressed to the attention of the Translational Science Team.

11.8 Lab Data Transfer Guidelines

The data and results collected on the above mentioned correlative study(ies) will be submitted to the ECOG Coordinating Center by the central laboratory(ies) within three months of completion of the laboratory studies and after the results have been collated. Electronic submissions should be submitted to ecog.labdata@jimmy.harvard.edu. All other correspondence should be addressed to the attention of the Translational Science Team.

Please refer to the E3805 Forms Packet for copies of all forms. The E3805 Forms Packet may be downloaded by accessing the ECOG World Wide Web Home Page (<http://www.ecog.org>). Forms must be submitted to the ECOG Coordinating Center, FSTRF, 900 Commonwealth Avenue, Boston, MA 02215 (ATTN: DATA).

This study will be monitored by the CTEP Data Update System (CDUS) version 3.0. Cumulative CDUS data will be submitted quarterly from the ECOG Coordinating Center to CTEP by electronic means.

12. Records to Be Kept

Please refer to the E3805 Forms Packet for the forms submission schedule and copies of all forms. The E3805 Forms Packet may be downloaded by accessing the ECOG World Wide Web Home Page (<http://www.ecog.org>). Forms must be submitted to the ECOG Coordinating Center, FSTRF, 900 Commonwealth Avenue, Boston, MA 02215 (ATTN: DATA).

This study will be monitored by the CTEP Data Update System (CDUS) version 3.0. Cumulative CDUS data will be submitted quarterly from the ECOG Coordinating Center to CTEP by electronic means.

13. Patient Consent and Peer Judgment

Current FDA, NCI, state, federal and institutional regulations concerning informed consent will be followed.

14. References

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**CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for
Extensive Disease Prostate Cancer**

Version Date: 7/28/06

Appendix I

**Informed Consent Template for Cancer Treatment Trials
(English Language)**

This is a clinical trial, a type of research study. Your doctor will explain the clinical trial to you. Clinical trials include only people who choose to take part. Please take your time to make your decision about taking part. You may discuss your decision with your friends and family. You can also discuss it with your health care team. If you have any questions, you can ask your doctor for more explanation.

You are being asked to take part in this study because you have extensive metastatic prostate cancer. Extensive is defined as disease beyond the pelvic and spinal bones with a total of four or more lesions or any involvement of the liver or lung.

WHY IS THIS STUDY BEING DONE?

The purpose of this study is to determine whether giving docetaxel chemotherapy, at the time you are starting hormonal therapy and when the hormonal therapy is no longer working, is more effective than giving it only at the time the hormonal therapy is no longer working. Hormonal therapy refers to drugs or surgical procedures such as an orchiectomy used to lower your testosterone. This puts prostate cancer into remission in most patients as the testosterone is like a fuel for the cancer. It is the standard to give this as the only treatment for your cancer. Normally chemotherapy is reserved for when a patient's cancer starts to grow again despite having a low testosterone level. In this study you will either get docetaxel when you start your hormonal therapy or when/if your cancer grows with a low testosterone level. If you have chemotherapy when you start hormones for the first time, you may get chemotherapy when/if your cancer grows back with a low testosterone level.

If your disease worsens while you are on hormone treatments, your doctor will consider your next treatment very carefully. This protocol has asked but not required that your doctor use docetaxel (a type of chemotherapy) once your disease worsens, even if you had received docetaxel when you started hormone therapy. Your doctor may also try another hormone treatment before starting you on docetaxel chemotherapy. The purpose of this study is to determine whether receiving docetaxel both when you start hormone therapy (or within 90 days of beginning hormonal therapy) as well as when your disease worsens, is better than only receiving docetaxel when your disease gets worse. This trial will let us know which approach is more effective in treating your disease.

HOW MANY PEOPLE WILL TAKE PART IN THE STUDY?

About 568 people will take part in this study.

WHAT WILL HAPPEN IF I TAKE PART IN THIS RESEARCH STUDY?

BEFORE YOU BEGIN THE STUDY

You will need to have the following exams, tests or procedures to find out if you can be in the study. These exams, tests or procedures are part of regular cancer care and may be done even if you do not join the study. If you have had some of them recently, they may not need to be repeated. This will be up to your doctor.

- Physical Examination with medical history and vital signs
- Blood tests
- Scans of the body (bone scan, chest scan, abdomen/pelvis scan)

You will be "randomized" into one of the study groups described below. Randomization means that you are put into a group by chance. A computer program will place you in one of the study groups. Neither you nor your doctor can choose the group you will be in. You will have a *one in two* chance of being placed in the group that receives chemotherapy when starting hormonal therapy (Group A). Group B patients will be treated with hormonal therapy alone. Hormonal therapy refers to lowering of your testosterone and can be done by removal of testicles or injections ("shots"). Chemotherapy is normally given at the time the cancer learns to grow without testosterone. This study is trying to see if people's cancer can be controlled for longer if chemotherapy is given when starting hormonal therapy and what the effect the chemotherapy has on a patient's quality of life.

DURING THE STUDY

If the exams, tests and procedures show that you can be in the study, and you choose to take part, then you will need the following tests and procedures. They are part of regular cancer care.

If you are to receive chemotherapy immediately, you will have the following done every 3 weeks, prior to your infusion:

- Physical Examination, vital signs
- Blood tests

If you are to receive hormonal therapy alone, you will have the following tests every 12 weeks:

- Physical Examination, vital signs
- Blood test

WHEN YOU ARE FINISHED TAKING CHEMOTHERAPY

You will be followed every 12 weeks for a doctor visit to assess the status of your health and your cancer.

STUDY CHART

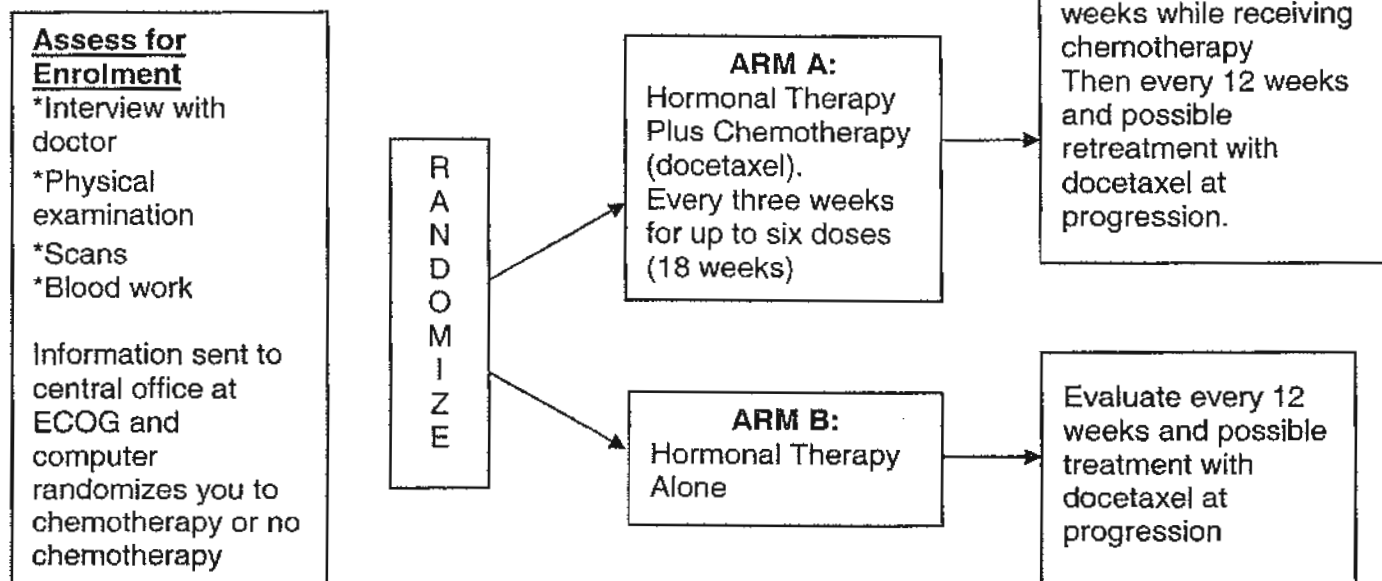
Study Chart

You will receive hormonal therapy (treatments to lower testosterone) continuously and if you are randomized to receive chemotherapy when starting hormonal therapy, you will get docetaxel every three weeks for up to 6 doses in this study. This three week period of time is called a cycle. The cycle will be repeated six times. Each cycle is numbered in order. The chart below shows what will happen to you during Cycle 1 and future treatment cycles as explained previously if you are to receive chemotherapy when you start hormonal therapy. The left-hand column shows the day in the cycle and the right-hand column tells you what to do on that day.

Cycle 1-6	
Day	What you do
Just before chemotherapy	Take dexamethasone 12 hours, 3 hours and 1 hour prior to infusion
Day 1	Check into doctor's office, get blood tests and if suitable get chemotherapy

If you are randomized to hormonal therapy alone, you will come to the doctor's office and be followed and receive "hormone" shots if your testicles were not removed. Chemotherapy is to be reserved for when your cancer progresses with a low testosterone.

Study Plan



HOW LONG WILL I BE IN THE STUDY?

We think you will be in the study for several months.

We would like to keep track of your medical condition for 10 years to look at the long-term effects of the study.

CAN I STOP BEING IN THE STUDY?

Yes. You can decide to stop at any time. Tell the doctor if you are thinking about stopping or decide to stop. He or she will tell you how to stop safely.

It is important to tell the doctor if you are thinking about stopping so any risks from the chemotherapy and/or hormonal therapy can be evaluated by your doctor. Another reason to tell your doctor that you are thinking about stopping is to discuss what follow-up care and testing could be most helpful for you.

The doctor may stop you from taking part in this study at any time if he/she believes it is in your best interest; if you do not follow the study rules; or if the study is stopped.

WHAT SIDE EFFECTS OR RISKS CAN I EXPECT FROM BEING IN THE STUDY?

You may have side effects while on the study. Everyone taking part in the study will be watched carefully for any side effects. However, doctors don't know all the side effects that may happen. Side effects may be mild or very serious. Your health care team may give you medicines to help lessen side effects. Many side effects go away soon after you stop taking the chemotherapy and/or hormonal therapy. In some cases, side effects can be serious, long lasting, or may never go away. Death is rare, but possible.

You should talk to your doctor about any side effects that you have while taking part in the study.

Risks and side effects related to the study include those which are:

Docetaxel (Taxotere)

Likely

- Nausea, vomiting or diarrhea
- Lowered white blood cell count (may make you more likely to get an infection)
- Lowered platelets (may make you more likely to bruise or bleed)
- Lowered red blood cell count (may make you feel tired or weak)
- Numbness and pain of the hands and feet
- Hair loss
- Muscle weakness/muscle and joint aches
- Mild to severe allergic reaction at the time the infusion is given
- Nail changes - drying and lines
- Fluid in arms and/or legs
- Changes in liver enzymes, and, rarely, liver damage or failure
- Fever
- Skin rash or dry skin
- Loss of appetite
- Taste changes
- Fatigue (feeling tired)

Less Likely

- Shortness of breath
- Sores in the mouth or throat
- Itching
- Headaches
- Fluid around the heart or the lungs
- Changes in kidney function tests which may lead to stopping docetaxel
- Low blood pressure
- Irregular heart beat (arrhythmias) or heart failure
- Skin irritation, redness, heat, swelling and pain at the site of injection of the medication
- Redness or irritation of the skin at a prior site of radiation therapy
- Irritation of the eye
- Low blood sodium level
- An abnormally high level of the simple sugar glucose in the blood
- Inflammation of the esophagus (the passageway between the mouth and the stomach)

- Inflammation of the pancreas (a small gland behind the stomach that secretes substances such as insulin)

Rare, but serious

- Seizures
- Blood clots
- Damage to the intestines which could be life-threatening

LHRH agonists (such as leuprolide, goserelin, triptorelin)

Likely

- Hot flashes
- Swelling caused by fluid held in the tissues
- Abnormal enlargement of male breasts
- Bone pain, loss of bone density
- Blood clotting within blood vessels.
- Gastrointestinal (stomach) disturbances – diarrhea, nausea, vomiting

Less Likely

- The inability to achieve an erection or have sexual intercourse.
- Loss of sexual desire
- Weight loss
- Weight gain
- Depression
- Increase in liver enzyme
- Dizziness
- Headache
- Decreased number of red blood cells
- Increased thirst and urination
- Unusual taste in mouth

Rare, but Serious

- Rash due to allergic reaction
- Difficulty breathing

Antiandrogens (flutamide and bicalutamide)

Likely

- Abnormal enlargement of male breasts
- Diarrhea
- Nausea, vomiting
- Decreased number of red blood cells
- Breast tenderness and swelling
- Hot flashes
- High blood pressure

Less Likely

- Elevated level of liver enzymes indicating liver injury
- Too much bile in the blood causing a yellow color to the skin, gums, eye, and other tissues
- The inability to achieve an erection or have sexual intercourse.
- Loss of sexual desire
- Fatigue

Rare, but Serious

- Death due to liver failure
- A condition in which the skin is overly sensitive to light

Reproductive risks: You should not become pregnant or father a baby while on this study because the drugs in this study can affect a fetus. Women should not breastfeed a baby while on this study. It is important you understand that you need to use birth control while on this study. Check with your doctor about what kind of birth control methods to use and how long to use them. Some methods might not be approved for use in this study.

For more information about risks and side effects, ask your doctor.

ARE THERE BENEFITS TO TAKING PART IN THE STUDY?

Taking part in this study may or may not make your health better. While doctors hope chemotherapy and/or hormonal therapy will be more useful against cancer compared to the usual treatment, there is no proof of this yet. We do know that the information from this study will help doctors learn more about chemotherapy and/or hormonal therapy as a treatment for cancer. This information could help future cancer patients.

WHAT OTHER CHOICES DO I HAVE IF I DO NOT TAKE PART IN THIS STUDY?

Your other choices may include:

- Getting treatment or care for your cancer without being in a study
- Taking part in another study
- Getting no treatment

Talk to your doctor about your choices before you decide if you will take part in this study.

WILL MY MEDICAL INFORMATION BE KEPT PRIVATE?

The Eastern Cooperative Oncology Group (ECOG) is conducting this study. ECOG is a cancer group that conducts studies for the National Cancer Institute. Your doctor is a member of ECOG or another group that is participating in this study. To help protect your privacy, ECOG has obtained a Confidentiality Certificate from the Department of Health and Human Services (DHHS).

With this Certificate, ECOG cannot be forced (for example, by court subpoena) to disclose information that may identify you in any federal, state or local civil, criminal, administrative, legislative or other proceedings. Disclosure will be necessary, however, upon request of DHHS for audit or program evaluation purposes.

You should understand that a Confidentiality Certificate does not prevent you or a member of your family from voluntarily releasing information about you or your involvement in this research. Note, however, that if an insurer or employer learns about your participation and obtains your consent to receive research information, then ECOG may not use the Certificate of Confidentiality to withhold this information. This means that you and your family must also actively protect your privacy.

Finally, you should understand that your doctor and ECOG are not prevented from taking steps, including reporting to authorities, to prevent serious harm to yourself or others and the Certificate does not prevent the review of your research records under some circumstances by certain organizations for an internal program audit or evaluation. Organizations that may inspect and/or copy your research records for quality assurance and data analysis include groups such as:

- Eastern Cooperative Oncology Group
- National Cancer Institute
- Food and Drug Administration
- Other regulatory agencies and/or their designated representatives
- Drug manufacturers, and/or their representatives
- Central Laboratories
- Cancer Trials Support Unit (CTSU), a research group sponsored by the National Cancer Institute (NCI) to provide greater access to cancer trials

[Note to Local Investigators: The NCI has recommended that the local institution address HIPAA regulations. The regulations may or may not be included in the informed consent form depending on local institutional policy.]

WHAT ARE THE COSTS OF TAKING PART IN THIS STUDY?

You and/or your health plan/ insurance company will need to pay for some or all of the costs of treating your cancer in this study. Some health plans will not pay these costs for people taking part in studies. The docetaxel will be provided free of charge, when starting hormone therapy but will not be provided free of charge if your disease worsens, as this would be standard of care. Check with your health plan or insurance company to find out what they will pay for. Taking part in this study may or may not cost your insurance company more than the cost of getting regular cancer treatment.

You will not be paid for taking part in this study.

For more information on clinical trials and insurance coverage, you can visit the National Cancer Institute's Web site at:

<http://cancer.gov/clinicaltrials/understanding/insurance-coverage>

You can print a copy of the "Clinical Trials and Insurance Coverage" information from this Web site.

Another way to get the information is to call 1-800-4-CANCER (1-800-422-6237) and ask them to send you a free copy.

WHAT HAPPENS IF I AM INJURED BECAUSE I TOOK PART IN THIS STUDY?

It is important that you tell your doctor, _____ *[investigator's name(s)]*, if you feel that you have been injured because of taking part in this study. You can tell the doctor in person or call him/her at _____ *[telephone number]*.

You will get medical treatment if you are injured as a result of taking part in this study. You and/or your health plan will be charged for this treatment. The study will not pay for medical treatment.

WHAT ARE MY RIGHTS IF I TAKE PART IN THIS STUDY?

Taking part in this study is your choice. You may choose either to take part or not to take part in the study. If you decide to take part in this study, you may leave the study at any time. No matter what decision you make, there will be no penalty to you and you will not lose any of your regular benefits. Leaving the study will not affect your medical care. You can still get your medical care from our institution.

We will tell you about new information or changes in the study that may affect your health or your willingness to continue in the study.

In the case of injury resulting from this study, you do not lose any of your legal rights to seek payment by signing this form.

WHO CAN ANSWER MY QUESTIONS ABOUT THE STUDY?

You can talk to your doctor about any questions or concerns you have about this study. Contact your doctor _____ [name(s)] at _____ [telephone number].

For questions about your rights while taking part in this study, call the _____ [name of center] Institutional Review Board (a group of people who review the research to protect your rights) at _____ (telephone number). *[Note to Local Investigator: Contact information for patient representatives or other individuals in a local institution who are not on the IRB or research team but take calls regarding clinical trial questions can be listed here.]*

***You may also call the Operations Office of the NCI Central Institutional Review Board (CIRB) at 888-657-3711 (from the continental US only). [*Only applies to sites using the CIRB.]**

Please note: This section of the informed consent form is about additional research studies that are being done with people who are taking part in the main study. You may take part in these additional studies if you want to. You can still be a part of the main study even if you say 'no' to taking part in any of these additional studies.

You can say "yes" or "no" to each of the following studies. Please mark your choice for each study.

Quality of Life Study

We want to know your view of how your life has been affected by cancer and its treatment. This "Quality of life" study looks at how you are feeling physically and emotionally during your cancer treatment. It also looks at how you are able to carry out your day-to-day activities.

This information will help doctors better understand how patients feel during treatments and what effects the medicines are having. In the future, this information may help patients and doctors as they decide which medicines to use to treat cancer.

You will be asked to complete 5 questionnaires: one on your first visit, one at week 12, one at week 24, one at week 36, and one at week 48. It takes about 15 minutes to fill out each questionnaire.

If any questions make you feel uncomfortable, you may skip those questions and not give an answer.

If you decide to take part in this study, the only thing you will be asked to do is fill out the three questionnaires. You may change your mind about completing the questionnaires at any time.

Just like in the main study, we will do our best to make sure that your personal information will be kept private.

Please circle your answer.

I choose to take part in the Quality of Life Study. I agree to fill out the three Quality of Life Questionnaires.

YES

NO

SCIENTIFIC STUDIES

This study includes one or more laboratory tests that will analyze a small sample of blood and tissue. The tissue sample will be from your diagnostic biopsy, an additional biopsy will not be needed. Blood will be collected by inserting a needle into a vein and drawing 2 to 4 tablespoons of blood. This will be done before you begin treatment, 6 months into your treatment, days 1 and if your disease progresses. The tissue and blood will be sent to central laboratories, where researchers will perform tests in order to better understand prostate cancer. The results from these tests will not be sent to you or your doctor, and they will not be used in planning your care. These tests are only for research purposes.

Making Your Choice

Please read the sentence below and think about your choice. After reading the sentence, circle "Yes" or "No." *No matter what you decide to do, it will not affect your care.* You can participate in the treatment part of the study without participating in the research studies. If you have any questions, please talk to your doctor or nurse, or call our research review board at **ADD TELEPHONE NUMBER**

I agree to participate in the scientific laboratory tests that are being done as part of this study.

Yes No

Please print and sign your name here after you circle your answer.

Your Name:

Your Signature:

Date:

WILL ANY OF THE SAMPLES (E.G., TISSUE) TAKEN FROM ME BE USED FOR OTHER RESEARCH STUDIES?

About Using Tissue for Research

If you participate in the laboratory study associates with this protocol, you will have some blood collected, and the biopsy material from your original diagnostic biopsy will be sent to a central laboratory for analysis.

We would like to keep some of the blood and tissue that is left over for future research. If you agree, this blood and tissue will be kept and may be used in research to learn more about cancer and other diseases. This *[blood and tissue]* will be given only to researchers approved by the Eastern Cooperative Oncology Group (ECOG). Any research done on the tissue also must be approved by the researcher's Institutional Review Board.

Your blood and tissue may be helpful for research, whether you do or do not have cancer. The research that may be done with your blood and tissue will probably not help you. It might help people who have cancer and other diseases in the future.

Reports about the research done with your blood and tissue will not be given to you or to your doctor. These reports will not be put into your health record. The research will not have an effect on your care.

Things to Think About

The choice to let us keep the leftover blood and tissue for future research is up to you. No matter what you decide to do, it will not affect your care, and you may still take part in the Eastern Cooperative Oncology Group study.

If you decide now that your blood and tissue can be kept for research, you can change your mind at any time. Just contact your study doctor and let him or her know that you do not want us to use your blood and tissue]. Then the blood and tissue will no longer be used for research.

In the future, people who do research may need to know more about your health. When the Eastern Cooperative Oncology Group gives them reports about your health, it will not give them your name.

Sometimes blood and tissue is used for genetic research (about disease that are passed on in families). Even if your blood and tissue is used for this kind of research, the results will not be put into your health records.

Your blood and tissue will be used only for research, and it will not be sold. You will not be paid for allowing your leftover blood and tissue to be used in research, even though the research done with your blood and tissue may help to develop new products in the future. Similarly, there will be no cost to you for any blood and tissue collected and stored by the Eastern Cooperative Oncology Group.

It is possible that, at some time in the future, as part of deciding on which therapy to give you, a new test might become available that could be done on some of the blood and tissue] that is now thought of as "leftover." This situation is unusual, but it could happen. In order to see that not all of this leftover blood and tissue is used up, the Eastern Cooperative Oncology Group will take care to see that some of your cancer blood and tissue is stored for 10 years so it is available if you or your doctors should need it.

This will depend upon the amount of leftover blood and tissue that is submitted for this study. However, there may not be any leftover blood and tissue to store.

Benefits

The benefits of research using blood and tissue include; learning more about what causes cancer and other diseases, how to prevent them, how to treat them, and how to cure them.

Risks

There are very few risks to you. The greatest risk is the release of information from your health records. The Eastern Cooperative Oncology Group will protect your records so that your name will be kept private. The chance that this information will be given to someone else is very small.

Making Your Choice

Please read each sentence below and think about your choice. After reading each sentence, circle "Yes" or "No." No matter what you decide to do, it will not affect your care. You can participate in the treatment part of the study without participating in all or part of the blood and tissue research studies. If you have any questions, please talk to your doctor or nurse or call our Institutional Review Board at [ADD TELEPHONE NUMBER].

My blood and tissue may be kept for use in research to learn about, prevent, treat, or cure cancer.	
Yes	No
My blood and tissue may be kept for use in research about other health problems (for example, causes of diabetes, Alzheimer's disease, and heart disease).	
Yes	No
My doctor (or someone from the Eastern Cooperative Oncology Group) may contact me in the future to ask me to take part in more research.	
Yes	No
Please print and sign your name here after you circle your answers.	
Your Name:	
Your Signature:	Date:

WHERE CAN I GET MORE INFORMATION?

You may call the National Cancer Institute's Cancer Information Service at:
1-800-4-CANCER (1-800-422-6237) or TTY: 1-800-332-8615

You may also visit the NCI Web site at <http://cancer.gov/>

For NCI's clinical trials information, go to: <http://cancer.gov/clinicaltrials/>

For NCI's general information about cancer, go to <http://cancer.gov/cancerinfo/>

You will get a copy of this form. If you want more information about this study, ask your doctor.

SIGNATURE

I have been given a copy of all _____ *[insert total of number of pages]* pages of this form. I have read it or it has been read to me. I understand the information and have had my questions answered. I agree to take part in this study.

Participant _____

Date _____

**CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for
Extensive Disease Prostate Cancer**

Appendix II

Pathology Submission Guidelines

The following items are included in Appendix II:

1. Guidelines for Submission of Pathology Materials
(instructional sheet for Clinical Research Associates [CRAs])
2. Instructional memo to submitting pathologists
3. List of Required Materials for E3805
4. ECOG Pathology Submission Form (#638)

Guidelines for Submission of Pathology Materials

The following items should always be included when submitting pathology materials to the ECOG Pathology Coordinating Office:

Institutional Surgical Pathology Report
Pathology materials (see attached List of Required Material)
ECOG Pathology Material Submission Form (# 638)

Instructions:

1. Place the Patient ID label provided by the ECOG Coordinating Center in Part A of the ECOG Pathology Material Submission Form.

If a label is not available, **TYPE or PRINT** the following information in **Part A** of the form:

- Patient's name (last, first)
 - Protocol number
 - Protocol case number (the patient's ECOG sequence number; for intergroup studies, include both the ECOG and other group's sequence numbers)
 - Patient's hospital number
 - Institution
 - Affiliate (if appropriate)
2. Complete blank areas of the pathologist's instructional memo, and forward it, along with the List of Required Material and the ECOG Pathology Material Submission Form, to the appropriate pathologist.
 3. The pathologist should return to you the required pathologic samples and surgical pathology reports, along with the completed ECOG Pathology Material Submission Form (# 638) (Part B completed). If any other reports are required, they should be obtained from the appropriate department at this time.
 4. Keep a copy of the ECOG Pathology Material Submission Form (# 638) for your records (the original should be sent to the PCO).
 5. Double check that ALL required forms, reports, and pathology samples are included in the package to send to the Pathology Coordinating Office (see appropriate List of Required Material).

Pathology specimens submitted for a patient WILL NOT be processed by the Pathology Coordinating Office until all necessary items are received.

6. Mail pathology materials to:

ECOG Pathology Coordinating Office
Robert H. Lurie Comprehensive Cancer Center
of Northwestern University Medical School
Olson Pavilion - Room 8501
710 North Fairbanks Court
Chicago, IL 60611

If you have any questions concerning the above instructions, or if you anticipate any problems in meeting the pathology material submission deadline of one month, contact the Pathology Coordinator at the ECOG Pathology Coordinating Office at Tel: (312) 503-3384 or Fax: (312) 503-3385.

LIST OF REQUIRED MATERIAL

E3805	<i>CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease Prostate Cancer</i>
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Pre-Treatment

1. ECOG Pathology Material Submission Form (# 638) – Parts A & B completed.
2. Institutional pathology report (***must be included with EVERY pathology submission***).
3. Study-specific On-Study Form in addition to the form sent to the ECOG Coordinating Center.
4. Requested pathology materials:
 - One tumor block from the original diagnostic biopsy. If a tumor block is not available, five (5) unstained positively charged slides and one H&E slide may be substituted.

NOTE: Blocks are available for return upon written request for purposes of patient management. However, since blocks are being used for laboratory studies, in some cases the material may be depleted and therefore, the block may not be available for return.



Eastern Cooperative Oncology Group

Coordinating Center

Frontier Science

900 Commonwealth Avenue • Boston, MA 02215

(617) 632-3610 • Fax: (617) 632-2990

Randomization: (617) 632-2022

Jean MacDonald, Director of Research Operations

Mary Steele, Director of Group Administration

Group Chair: Robert L. Comis, M.D.

Group Statistician: Robert Gray, Ph.D.

MEMORANDUM

TO: _____

(Submitting Pathologist)

FROM: Stanley Hamilton, M.D.

Chair

ECOG Laboratory Science and Pathology Committee

DATE: _____

SUBJECT: *Submission of Pathology Materials for E3805*

The patient named on the attached ECOG Pathology Material Submission Form (# 638) has been entered onto an ECOG protocol by _____ (ECOG Investigator). This protocol requires the submission of pathology materials for **laboratory studies**.

Please complete PART B of the Submission Form. Keep a copy for your own records, and return the completed Submission Form, the surgical pathology report(s), the slides and/or blocks, and any other required material (see attached List of Required Material) to the Clinical Research Associate (CRA). The CRA will forward all required pathology material to the ECOG Pathology Coordinating Office.

Blocks and slides submitted for this study will be retained at the ECOG Central Repository for future studies. Paraffin blocks will be returned for purposes of patient management upon written request.

Please note: Since blocks are being *used for laboratory studies*, in some cases the material may be depleted, and, therefore, the block may not be returned.

If you have any questions regarding this request, please feel free to contact the Pathology Coordinating Office at Tel: (312) 503-3384 OR FAX: (312) 503-3385.

The ECOG CRA at your institution is:

Name: _____

Address: _____

Phone: _____

Thank you.

Instructions: This form is a required part of pathology submission. Please complete and submit along with all pathology material and corresponding pathology reports requested by the protocol. See list of required materials as specified in EACH protocol.

ECOG PCO-RL IS FULLY- COMPLIANT WITH DHHS, HIPAA, AND OHRP REGULATIONSTel. 312-503-3384Fax 312-503-3384

PART A: To Be Completed By Data Manager/CRA

DO NOT USE INITIALS – Submit Patient's FULL Name
(The Patient has authorized the use of PHI.)

Date sample sent to ECOG
____/____/____ (M,D,Y)

Data Manager _____

Address _____

Telephone No. () _____

Fax No. () _____

Email address _____

Patient's Name:

Last _____ First _____

ECOG Prot. No. _____ ECOG Patient Seq. No. _____

Participating Group _____ Participating Group Patient ID No. _____

Group _____ Institution _____ PI _____

Step No. _____ Affiliate _____

ECOG Parent Prot. No. _____ Seq. No. _____

PART B: TO BE COMPLETED BY DATA MANAGER/CRA AND SUBMITTING PATHOLOGIST

Complete for Slides/Vials	Status* (See Below)	Date Specimen Collected (M/D/Y)	Disease Site	Number of Slides/Vials	Specimen ID Numbers	Type of Stain	PCO-RL Use Only
							PCO ID Numbers
		/ /					
		/ /					

Complete for Blocks/Punch	Status* (See Below)	Date Specimen Collected (M/D/Y)	Disease Site	Number of Blocks/Punch	Specimen ID Numbers	Fixative	PCO-RL Use Only
							PCO ID Numbers
		/ /					
		/ /					

*Status: Please identify the clinical status of the sample. List **all** that apply:

- Original diagnostic material
- AML/MDS diagnosis
- Pre-protocol treatment biopsy/tissue
- Post-protocol treatment biopsy/tissue
- Post-surgery biopsy/tissue
- Relapse/recurrence
- Remission/response
- Other, specify: _____

Submitting Pathologist _____

Telephone No. () _____

Address _____

Did the patient consent to participate in the storage of samples for future research? Yes No

MATERIAL RETURN (All materials will be retained by the ECOG PCO unless return is requested here.)

Does the submitting institution's policy require the return of any submitted material (blocks, H&E slides, etc.)? Yes No

If so, please indicate which materials must be returned _____

All materials will be returned to the submitting pathologist unless an alternate address is indicated here _____

If materials were not able to be submitted for this protocol and its correlative studies, please circle the reason for non-submission. Attach a formal letter referencing regulations, policy, and/or other explanation. If possible, include a copy of the policy.

Federal/State Regulations _____ Hospital/Institutional Policy _____ Insufficient Tissue _____ Other _____ (Specify) _____

Pathologist of Investigator's Signature _____ (PCO-RL Use Only)PV _____

PART C: ECOG PATHOLOGY COORDINATING OFFICE USE ONLY

Date Sample Received at PCO _____ Date Sent to Reviewer _____ Date Sent to PI/Central Lab _____

Site Compliance % _____ Name of Reviewer _____ PI/Central Lab _____

PCO Comments: _____ Staff Init. _____

Investigator: Keep a copy for your files and submit original form to the destination specified in protocol.

2/05

**CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial
for Extensive Disease Prostate Cancer**

Appendix III

Patient Thank You Letter

We ask that the physician use the template contained in this appendix to prepare a letter thanking the patient for enrolling in this trial. The template is intended as a guide and can be downloaded from the ECOG web site at <http://www.ecog.org>. As this is a personal letter, physicians may elect to further tailor the text to their situation.

This small gesture is a part of a broader program being undertaken by ECOG and the NCI to increase awareness of the importance of clinical trials and improve accrual and follow-through. We appreciate your help in this effort.

[PATIENT NAME]

[DATE]

[PATIENT ADDRESS]

Dear [PATIENT SALUTATION],

Thank you for agreeing to take part in this important clinical program. Programs like this offer a chance to get the best care while helping us make better care available for all patients. Many questions remain unanswered in cancer. With the help of people like you who participate in these programs, we will achieve our goal of effectively treating and ultimately curing cancer.

We believe this program will provide you with high quality, thorough care. Your physician and research staff will maintain very close contact with you. This is important to allow your physician to provide you with the best care while learning as much as possible to help you and other patients.

On behalf of **[INSTITUTION]** and the Eastern Cooperative Oncology Group, we thank you again and look forward to helping you.

Sincerely,

[PHYSICIAN NAME]

CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease Prostate Cancer

Appendix IV

Cancer Trials Support Unit (CTSU) Participation Procedures

Registration/Randomization

Prior to the recruitment of a patient for this study, investigators must be registered members of the CTSU. Each investigator must have an NCI investigator number and must maintain an "active" investigator registration status through the annual submission of a complete investigator registration packet (FDA Form 1572 with original signature, current CV, Supplemental Investigator Data Form with signature, and Financial Disclosure Form with original signature) to the Pharmaceutical Management Branch, CTEP, DCTD, NCI. These forms are available on the CTSU registered member Web site or by calling the PMB at 301-496-5725 Monday through Friday between 8:30 a.m. and 4:30 p.m. Eastern time.

Each CTSU investigator or group of investigators at a clinical site must obtain IRB approval for this protocol and submit IRB approval and supporting documentation to the CTSU Regulatory Office before they can enroll patients. All forms and documents associated with this study can be downloaded from the E3805 Web page on the CTSU registered member Web site (<http://members.ctsuo.org>). Patients can be registered only after pre-treatment evaluation is complete, all eligibility criteria have been met, and all pertinent forms and documents are approved and on file with the CTSU.

Requirements for E3805 site registration:

- CTSU IRB Certification
- IRB/Regulatory Approval Transmittal Sheet

Prestudy requirements for patient enrollment on E3805:

- Patient must meet all inclusion criteria, and no exclusion criteria should apply.
- Patient has signed and dated all applicable consents and authorization forms.
- All baseline laboratory tests and prestudy evaluations performed.
- Patient completes baseline QOL forms prior to treatment start.

CTSU Procedures for Patient Enrollment

Contact the CTSU Patient Registration Office by calling 1-888-462-3009 and leave a voicemail to alert the CTSU Patient Registrar that an enrollment is forthcoming. For immediate registration needs, i.e. within one hour, call the registrar cell phone at 1-301-704-2376. Complete the following forms:

- CTSU Patient Enrollment Transmittal Form
- E3805 Eligibility Checklist

Fax these forms to the CTSU Patient Registrar at 1-888-691-8039 between the hours of 9:00 a.m. and 7:00 p.m., Mon-Fri, Eastern Time (excluding holidays). The CTSU registrar will check the investigator and site information provided to ensure that all regulatory requirements have been met. The registrar will also check the forms for completeness and followup with the site to resolve any discrepancies. Once investigator eligibility is confirmed and enrollment documents deemed complete, the CTSU registrar will contact the ECOG, to obtain assignment of a treatment arm and assignment of a unique patient ID (to be used on all

future forms and correspondence). The CTSU registrar will relay this information to the enrolling site and follow-up with a confirmation via e-mail or fax.

Protocol treatment should start within five working days after randomization.

Data Submission

All case report forms (CRFs) and other documents associated with this study must be downloaded from the E3805 Web page located on the CTSU registered member Web site (<http://members.ctsu.org>). CTSU investigators must use the current version of the protocol-specific E3805 forms and adhere to the E3805 schedule for data submission. CRFs and associated reports must be submitted in the following manner:

- Patient enrollment CRFs should be faxed to the CTSU according to the instructions in the CTSU patient enrollment procedures section of this Appendix.
- Refer to the Special Materials or Substudies section below for instructions on submission of the ECOG Pathology Material Submission Form (#638), E3805 Material Submission Form #2377 and pathology report. Please include the Patient ID, patient initials, and protocol number on all pages of the pathology report and redact the patient's name.
- Original and amended CRFs, reports, and responses to query and delinquency letters must be mailed directly to the CTSU Data Operations Center accompanied by a properly completed CTSU Data Transmittal Form; the CTSU will forward all data submissions to the ECOG.
- Copies of reports that are submitted to the CTSU must include the Patient ID and protocol number on all pages of the report. The patient's name must be redacted.

A CTSU Data Transmittal Form must accompany all post-enrollment data submissions. Data submitted with an improperly completed CTSU Data Transmittal Form or without a CTSU Data Transmittal Form will be returned to the site for corrective action without being processed. The CTSU Data Transmittal Form may only be used when mailing post-enrollment case report forms and reports. Do not use this form when faxing site registration or patient enrollment documentation.

Mail original and amended post-enrollment CRFs, reports, and responses to query and delinquency letters to:

Westat
CTSU Data Operations Center
1441 W. Montgomery Avenue
Rockville, MD 20850-2062

Special Materials or Substudies

Pathology Review:

Submission of pathologic material for correlative studies is optional and will consist of one tumor block from the original diagnostic biopsy or if tumor block is unavailable, a minimum of 5 unstained positively charged slides (3-4 microns) and one H&E slide may be substituted. CTSU investigators should follow the procedures and timelines outlined in protocol Section 10 and Appendix II for submission of these specimens. All specimens, accompanying surgical pathology reports and ECOG Pathology Material Submission Form (#638) transmittals should be shipped to the address identified in Section 10.2 of the protocol. Do *not* send specimens or original forms to the CTSU. Please forward a copy of all transmittals and pathology reports to the CTSU.

Specimen Collection for Correlatives:

Tumor blocks and tumor tissue collected (see 'Pathology Review' section above) and blood samples will be used for correlative studies described in Section 11 of the protocol. Participation in these correlative studies is optional and *requires* patient consent. Kits are available for the collection of samples. Complete the Kit Order Form (Appendix VI) and fax to Kristina Kirkpatrick at (317) 921-2053. All samples (and all associated forms and reports) are to be submitted to the address provided in Section 11.2 of the protocol. All materials and reports must be labeled with the patient initials, protocol number, patient sequence number, and sample type (peripheral blood, plasma, or serum); patient names should be redacted from the pathology report. Specimens should not be submitted to the CTSU, although CTSU should be copied on all transmittal forms (E3805 Material Submission Form #2377 and pathology reports. See protocol Section 11 for submission schedule and further details on sample collection and preparation.

Specimen Banking:

Patients will have the option of consenting to have residuals and/or derivatives of the samples submitted for the above studies for possible future use (see protocol Section 10.4 and 11.6).

Quality of Life:

Administration of the FACT-P, FACIT-F Fatigue Subscale, FACT-Taxane Subscale and the Brief Pain inventory will be used to examine overall QOL. Questionnaires will be administered according to the timepoints outlined in Section 5.1.3 (unless the patient refuses). Please forward all QOL forms to the CTSU. See protocol Section 5.1 for further details.

Adverse Event (AE) Reporting

Assessing and submitting expedited reports

This study will utilize the CTCAE version 3.0 for toxicity and Adverse Event (AE) reporting. A link to the CTCAE guidelines is available on the CTSU registered member Web site. CTSU investigators should assess adverse events according to the instructions and tables in Section 5.3 of the protocol. All reporting should be conducted within the time frames specified in Section 5.3 of the protocol.

CTSU sites must comply with the expectations of their local Institutional Review Board (IRB) regarding submission of documentation of adverse events. Local IRBs must be informed of all reportable serious adverse events.

Expedited reports should be transmitted electronically using the CTEP AdEERS application. A link to the AdEERS application can be found on both the CTSU member homepage and ECOG Web page.

Secondary AML/MDS/ALL reporting:

Per protocol Section 5.2.7, report occurrence of secondary AML, MDS or ALL via the NCI/CTEP AML-MDS Report Form (do not use AdEERS). Submit supporting documentation as outlined in the protocol and forward to ECOG.

Reporting other secondary malignancies:

Upon diagnosis of a secondary cancer, please submit copies of the following to the CTSU within 30 days of diagnosis:

- * ECOG Second Primary Form (#1677)
- * A copy of the pathology report (if available)
- * Any additional supporting documentation

Once received, the CTSU will forward this information to ECOG. Do not send original pathology reports to the CTSU.

Drug Procurement:

CTSU investigators should refer to Section 8.0 for detailed instructions on drug procurement, formulation, storage, administration, and potential toxicities.

Investigational agent: Docetaxel (This drug is a commercially marketed drug, however, when used as directed by this protocol, docetaxel is classified as an "unapproved use of an approved agent" and, by definition, considered as an investigational agent. However, while it is not an indication currently approved by the FDA, the use of docetaxel in this protocol is exempt from the requirements of an IND as described under Title 21 CFR 312.2(b).)

Docetaxel (Taxotere®) is available and distributed free of charge from Sanofi-Aventis Pharmaceuticals.

After patient randomization, a supply of docetaxel may be obtained. Investigators must fax a signed and completed "E3805 Clinical Supplies Shipping Form" (Appendix VI) to the ECOG Coordinating Center, ATTN: Drug Team at 617-632-2063. Do not send drug requests directly to Sanofi-Aventis Pharmaceuticals. Allow 4-5 business days for receipt of the drug. See section 8.0 for further drug information and re-orders.

Commercial agents: LHRH agonists (such as leuprolide, goserelin, triptorelin); Antiandrogens (flutamide and bicalutamide)

Regulatory and Monitoring

Study Audit

To assure compliance with Federal regulatory requirements [CFR 21 parts 50, 54, 56, 312, 314 and HHS 45 CFR 46] and National Cancer Institute (NCI)/ Cancer Therapy Evaluation Program (CTEP) Clinical Trials Monitoring Branch (CTMB) guidelines for the conduct of clinical trials and study data validity, all protocols approved by NCI/CTEP that have patient enrollment through the CTSU are subject to audit.

Responsibility for assignment of the audit will be determined by the site's primary affiliation with a Cooperative Group or CTSU. For Group-aligned sites, the audit of a patient registered through CTSU will become the responsibility of the Group receiving credit for the enrollment. For CTSU Independent Clinical Research Sites (ICRS), the CTSU will coordinate the entire audit process.

Details on audit evaluation components, site selection, patient case selection, materials to be reviewed, site preparation, on-site procedures for review and assessment, and results reporting and follow-up are available for download from the CTSU Operations Manual located on the CTSU Member Web site.

Health Insurance Portability and Accountability Act of 1996 (HIPAA)

The HIPAA Privacy Rule establishes the conditions under which protected health information may be used or disclosed by covered entities for research purposes. Research is defined in the Privacy Rule referenced in HHS 45 CFR 164.501. Templated language addressing NCI-U.S. HIPAA guidelines are provided in the informed consent section of this protocol document; however, authorization for the release of Protected Health Information is considered separate and distinct from the Informed Consent process for participation in this clinical trial.

The HIPAA Privacy Rule does not affect participants from outside the United States. Authorization to release Protected Health Information is NOT required from patients enrolled in clinical trials at non-US sites.

Clinical Data Update System (CDUS) Monitoring

This study will be monitored by the Clinical Data Update System (CDUS) Version 3.0. Cumulative CDUS data will be submitted quarterly to CTEP by electronic means. The sponsoring Group fulfills this reporting obligation by electronically transmitting to CTEP the CDUS data collected from the study-specific case report forms.

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Appendix V

Treatment of Hypersensitivity Reactions

MANAGEMENT OF ACUTE HYPERSENSITIVITY

Severity of Symptoms	Treatment Guidelines
Mild symptoms: localized cutaneous reactions such as mild pruritus, flushing, rash	- consider decreasing the rate of infusion until recovery from symptoms, stay at bedside and monitor patient - then, complete docetaxel infusion at the initial planned rate
Moderate symptoms: any symptom that is not listed above (mild symptoms) or below (severe symptoms) such as generalized pruritus, flushing, rash, dyspnea, hypotension with systolic BP > 80 mm Hg	- interrupt docetaxel infusion - give diphenhydramine 50 mg IV with or without dexamethasone 10 mg IV; monitor patient until resolution of symptoms - resume docetaxel infusion after recovery of symptoms; depending on the physician's assessment of the patient, docetaxel infusion should be resumed at a slower rate, then increased incrementally to the initial planned rate, (<i>e.g. infuse at an 8 hour rate for 5 minutes, then at a 4-h rate for 5 minutes, then at a 2-h rate for 5 minutes, then finally, resume at the 1-h infusion rate</i>) - depending on the intensity of the reaction observed, additional oral or IV premedication with an antihistamine should also be given for the next cycle of treatment, and the rate of infusion should be decreased initially and then increased back to the recommended 1-hour infusion, (<i>e.g. infuse at an 8 hour rate for 5 minutes, then at a 4-h rate for 5 minutes, then at a 2-h rate for 5 minutes, and finally, administer at the 1-h infusion rate</i>)
Severe symptoms: any reaction such as bronchospasm, generalized urticaria, systolic BP ≤ 80mm Hg, angioedema	- immediately discontinue docetaxel infusion - give diphenhydramine 50 mg IV with or without dexamethasone 10 mg IV and/or epinephrine as needed; monitor patient until resolution of symptoms - the same treatment guidelines outlined under moderate symptoms (i.e. the third and fourth bullets) should be followed.
Anaphylaxis (NCI grade 4 reaction)	NO FURTHER STUDY DRUG THERAPY

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Appendix VI

E3805 Collection and Shipping Kit Order Form

Upon institutional IRB approval complete and fax this form as indicated below. Kits will NOT be shipped overnight unless clearly stated on this form that it is needed within 24 hours due to patient enrollment.

Multiple kits will be sent at one time, and are to be stored for use for patients who enroll on this protocol. Any unused kits are to be returned to Dr. Sweeney's lab at the address stated in Section 11.2 of this protocol. Kits will include the EDTA (lavender top) and red top blood collection tubes, shipping containers and pre-printed labels for all three time points. If additional kits are needed, please refax this Kit Order Form and clearly state that you are re-ordering.

TODAYS DATE: _____

Date of Institutional IRB approval: _____

Kits are to be shipped to:

Institution Name: _____

Institution Contact: _____

Phone number for contact: _____

Institution Address: _____

FAX Completed form to Kristina Kirkpatrick at (317) 921-2053.

Questions are to be directed to the Kristina Kirkpatrick by email at kriskirk@iupui.edu or by calling (317) 921-2050.

Comments: _____

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Appendix VII



**E3805 Clinical Supplies Shipping Form
To Investigator Only**

SECTION 1 – TO BE COMPLETED BY SITE

Requested By: _____ Mail stop : _____
Telephone: _____ Fax : _____
Protocol No. : E3805 IND/CTX No. : N/A
Ship to the attention of : _____ Investigator No.: _____
Address : _____ Investigator Name: _____

Telephone: _____ Fax : _____

SECTION 2 All line items MUST be completed by the ECOG COORDINATING CENTER DRUG TEAM

- a) ☐ Investigator shipments: The regulatory requirements are fulfilled at the clinical study center according to valid SOPs
b) ☐ IND studies: Required documents (e.g. protocols, 1572, CV) have been provided to the Regulatory Representative for IND submission/maintenance ☒ Not applicable
c) ☐ Initial shipment ☐ Reshipment
Requestor Name and Title : _____ Date : _____ Signature : _____

REQUESTED SHIP DATE : TO BE COMPLETED BY SITE

Financial Account No. N/A

ECOG Patient Sequence Numbers:

We are pleased to send you the following Investigational Products: COMPLETED BY SA

Number of Patient kits or quantity of bulk drug product	Treatment Period (if applicable)	Product Name/ Strength/ Form	PR or Lot Number	Use by Date

STORAGE CONDITIONS :

PR Number:

Sanofi-aventis Tracking N° :

Number of containers used for shipment:

Carrier/AWB Number :

- ☐ Invoice
☐ Release Certificate and Certificate of Conformance :
☐ Certificate(s) of Analysis

- ☐ Code Breaks Number of set(s) :
☐ Additional document:
☐ Randomization list

Comments :

Packed by :

Checked by :

Actual Shipment Date :

ACKNOWLEDGEMENT OF RECEIPT – TO BE COMPLETED BY SITE

Please acknowledge receipt of the shipment by checking the contents and signing approval below:

Number of containers received:

Contents received intact: Yes ☐ No ☐; If No, please explain: _____

Date received: _____ Signature : _____ Print Name: _____

Please fax acknowledgement of receipt to Aventis Attention: Debra Dawson at fax No. 908-243-7745
(local contact)

** PLEASE READ IMPORTANT INSTRUCTIONS **
This registration worksheet must be completed and all questions must be answered by the institution prior to patient registration. The eligibility checklist found in the protocol must be completed prior to patient registration and kept in the patient's chart. ECOG institutions are responsible for monitoring patient eligibility at study entry. The registration worksheet and eligibility checklist must be signed by treating M.D. and kept in the patient files at the institution if the checklist is to be used as source documentation. Do not submit these forms to the ECOG Coordinating Center.

5. Enter the patient's initials.

Enter first initial: _____
Enter last initial: _____

10. Has the eligibility checklist been completed?

- 1. No [INELIGIBLE]
- 2. Yes [next question]
- 3. Unknown [INELIGIBLE]

20. Is the patient eligible?

- 1. No [INELIGIBLE]
- 2. Yes [next question]
- 3. Unknown [INELIGIBLE]

30. Did patient give signed and dated written informed consent?

- 1. No [INELIGIBLE]
- 2. Yes [next question]
- 3. Unknown [INELIGIBLE]

40. On what date was consent signed?

enter date (mm/dd/yyyy): _____

50. Calculate: the number of days from the value of question 40 to TODAY

enter result: _____

- less than 0, INELIGIBLE
- greater than or equal to 0, next question.
- unknown , INELIGIBLE

60. Has HIPAA permission (authorization) for release of personal health information to ECOG been signed by patient?

- 1. No [INELIGIBLE]
- 2. Yes [next question]
- 3. Exempt (non-USA participant) [question 90]
- 4. Unknown [INELIGIBLE]

70. What is date authorization was signed?

NOTE: This date must be day of or before registration.

enter date (mm/dd/yyyy): _____

80. Calculate: the number of days from the value of question 70 to TODAY

enter result: _____

- less than 0, INELIGIBLE
- greater than or equal to 0, next question.
- unknown , INELIGIBLE

90. Enter the patient's age.

enter value: _____

- less than 18, INELIGIBLE
- greater than or equal to 18, next question.
- unknown , INELIGIBLE

100. What is the patient's current ECOG performance status value?

- 0: Patient fully active, able to carry on all pre-disease performance without restriction.
 1: Patient restricted in physically strenuous activity, but ambulatory and able to carry out work of a light or sedentary nature.
 2: Patient ambulatory and capable of all self care, but unable to carry any work activities, up and about more than 50% of waking hours.
 3: Patient capable of only limited self care, confined to bed or chair more than 50% of waking hours.
 4: Patient completely disabled, unable to carry on any self care, totally confined to bed or chair.
 enter value: _____

- equal to 0, next question.
 -equal to 1, next question.
 -equal to 2, next question.
 -equal to 3, INELIGIBLE
 -equal to 4, INELIGIBLE
 -unknown, INELIGIBLE

110. Does the patient intend to use combined androgen blockade (CAB) greater than 30 days from start of therapy?

1. No [next question]
 2. Yes [next question]
 3. Unknown [INELIGIBLE]

120. How long did the patient receive prior adjuvant hormonal therapy?

1. Greater than 12 months [next question]
 2. Less than or equal to 12 months [next question]
 3. Unknown [INELIGIBLE]

130. Will the patient receive bisphosphonates?

1. No [next question]
 2. Yes [next question]
 3. Unknown [INELIGIBLE]

140. Correlative samples should be submitted for analysis as outlined in Sections 10 and 11.

150. Did the patient agree to participate in the scientific laboratory tests that are being done as part of this study?

1. No [next question]
 2. Yes [next question]
 3. Unknown [next question]

160. Have you obtained the patient's consent for his or her blood and tissue to be kept for use in future research to learn about, prevent, treat, or cure cancer?

1. No [next question]
 2. Yes [next question]
 3. Unknown [next question]

170. Have you obtained the patient's consent for his or her blood and tissue to be kept for future research about other health problems (for example: causes of diabetes, Alzheimer's disease, and heart disease)?

1. No [next question]
 2. Yes [next question]
 3. Unknown [next question]

180. Have you obtained the patient's consent for his or her doctor (or someone from the Eastern Cooperative Oncology Group) to contact him or her in the future to ask him or her to take part in more research?

1. No [next question]
 2. Yes [next question]
 3. Unknown [next question]

190. NOTE: The following 3 questions are mandatory for the Quality of Life portion of the study. Please be sure to complete all 3 questions.

200. Enter the name of the contact person who will be responsible for overseeing the administration of the quality of life portion of the study.

enter response:

210. Enter the telephone number for the QOL contact person.

enter response:

Patient's Hospital ID _____

Patient's Social Security Number _____

Method of Payment _____

220. Enter the e-mail address for the QOL contact person.

enter response:

Attending Physician _____

Institution Contact _____

Registrar _____

Country _____

Additional Comments _____

enter response: _____

800. Enter the name of the contact person responsible for sample submission.

810. Enter the telephone number for the contact person responsible for sample submission.

enter response:

I have reviewed the above data and agree the data are accurate and correct.

signature of treating physician _____ date _____
Note: Date must be on or before date of registration

820. Enter the fax number of the contact person responsible for sample submission.

enter response:

830. Enter the e-mail address for the contact person responsible for sample submission.

enter response:

ELIGIBLE...

Demographic Data Required For Patient Registration:

Patient's Sex (m/f) _____ Birthdate (mm/yy) _____

Patient's Race _____ Zip Code _____

EASTERN COOPERATIVE ONCOLOGY GROUP (ECOG)
Authorization (Permission) to Use or Disclose (Release) Identifiable Health Information for Research

Participant's Name: _____

Birth Date: _____

1. What is the purpose of this form?

The Eastern Cooperative Oncology Group (ECOG), is an organization that does research to learn about the causes of cancer, and how to prevent and treat cancer. Researchers would like to use your health information for research. This information may include data that identifies you. Please carefully review the information below. If you agree that researchers can use your personal health information, you must sign and date this form to give them your permission.

2. What personal health information do the researchers want to use?

The researchers want to copy and use the portions of your medical record that they will need for their research. If you enter an ECOG research study, information that will be used and/or released may include the following:

- the history and diagnosis of your disease;
- specific information about the treatments you received, including previous treatment(s) you may have had;
- information about other medical conditions that may affect your treatment;
- medical data, including laboratory test results, tumor measurements, CT scans, MRIs, x-rays, and pathology results;
- information on side effects (adverse events) you may experience, and how these were treated;
- long-term information about your general health status and the status of your disease;
- data that may be related to tissue and/or blood samples that may be collected from you; and
- numbers or codes that will identify you, such as your social security number and medical record number.

(Include other study-specific information as necessary.)

You may request a blank copy of the ECOG data forms from the study doctor or his/her research staff to learn what information will be shared.

3. Why do the researchers want my personal health information?

[Insert Name of Site] will collect your health information and share it with the ECOG Biostatistical Center and the ECOG Operations Center if you enter a cooperative group research study. The ECOG centers will use your information in their cancer research study - E3805, *CHAARTED – ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer*. The purpose of this study is to determine whether giving docetaxel chemotherapy, at the time you are starting hormonal therapy and when the hormonal therapy is no longer working, is more effective than giving it only at the time the hormonal therapy is no longer working.

4. Who will be able to use my personal health information?

(Include the names of collaborators and sponsors who will have access to the data that includes protected health information)

[Insert Name of Site] will use your health information for research. As part of this research, they may give your information to the following groups taking part in the research. **[Insert Name of Site]** may also permit the following groups to come in to review your original records that are kept by **[Insert Name of Site]** so that they can monitor their research study:

- the ECOG Operations Center;
- the ECOG Biostatistical Center;
- the Cancer Trials Support Unit (CTSU) or designees, a research group sponsored by the National Cancer Institute that supports the research of ECOG;
- Public Health agencies and other government agencies (including non-U.S.) as authorized or required by law;
- other people or organizations assisting with ECOG research efforts. This may include drug manufacturers, drug companies that may provide partial support for the study, drug distributors, and/or their designees (Sanofi-Aventis Pharmaceuticals); and
- central laboratories, central review centers, and central reviewers. The central laboratories and review agencies may also give your health information to those groups listed in the bullets above.

5. How will information about me be kept private?

ECOG will keep all patient information private to the extent possible, even though ECOG is not required to follow this federal privacy rule. Only researchers working with ECOG or authorized by ECOG will have access to your information.

ECOG will not release personal health information about you to others except as authorized or required by law. However, once your information is given to other organizations that are not required to follow federal privacy laws, we cannot assure that the information will remain protected.

6. What happens if I do not sign this permission form?

If you do not sign this permission form, you will not be able to take part in the research study for which you are being considered.

7. If I sign this form, will I automatically be entered into the research study?

No, you cannot be entered into any research study without further discussion and separate consent. After discussion, you may decide to take part in the research study. At that time, you will be asked to sign a specific research consent form.

8. What happens if I want to withdraw my permission?

You can change your mind at any time and withdraw your permission to allow your personal health information to be used in the research. If this happens, you must withdraw your permission in writing. Beginning on the date you withdraw your permission, no new personal health information will be used for research. However, researchers may continue to use the health information that was provided before you withdrew your permission.

If you sign this form and enter the research study, but later change your mind and withdraw your permission, you will be removed from the research study at that time.

To withdraw your permission, please contact the person below. He/she will make sure your written request to withdraw your permission is processed correctly.

[Insert Title of Contact Person, Address, Phone and Fax Number]

9. How long will this permission last?

If you agree by signing this form that researchers can use your personal health information, this permission has no expiration date. However, as stated above, you can change your mind and withdraw your permission at any time.

10. What are my rights regarding access to my personal health information?

You have the right to refuse to sign this permission form. You have the right to review and/or copy records of your personal health information kept by **[Insert Name of Site]**. You do not have the right to review and/or copy records kept by ECOG or other researchers associated with the research study.

Signatures

I agree that my personal health information may be used for the research purposes described in this form.

Signature of Patient
or Patient's Legal Representative: _____ Date: _____

Printed Name of Legal Representative (if any): _____

Representative's Authority to Act for Patient: _____

Signature of Person Obtaining Permission: _____ Date: _____

Printed Name of Person Obtaining Permission: _____



Eastern Cooperative Oncology Group

Coordinating Center

Frontier Science
900 Commonwealth Avenue Boston, MA 02215
(617) 632-3610 Fax: (617) 632-2990
Randomization: (617) 632-2022

Group Chair: Robert L. Comis, M.D.
Group Statistician: Robert Gray, Ph.D.

Jean MacDonald, Director of Research Operations
Mary Steele, Director of Group Administration

TO: ECOG Clinical Research Associates and Investigators with Patients for E3805

FROM: ECOG Coordinating Center

DATE: February 7, 2007

SUBJECT: E3805, Addendum #2 and Update #2

This addendum has been reviewed and approved by the Central IRB.

Enclosed is Addendum #2 and Update #2 for E3805 – *CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer.*

A revised cover sheet, indicating the date of the revisions, and replacement pages, incorporating the changes to the protocol, have been provided. Please remove the outdated pages and replace them with the enclosed cover and current pages. We recommend that each institution maintain a file containing the original protocol and all subsequent revisions.

Revised forms are attached.

This addendum and update contain administrative changes and, per OHRP guidelines, expedited IRB review may be appropriate. However, please consult your local IRB's policy, since their requirements may differ and require full board review. This addendum must be submitted and reviewed by your IRB within 90 days of receipt of this notice, unless your local IRB has different written SOPs, which must be available at future ECOG audits.

Addendum #2 includes the following changes:

1. Cover page: The study statistician was changed to Yu-Hui Chen, M.P.H., M.S.
2. Contact Information Page: Anne Foster's email address has been changed from annefoste@iupui.edu to anefoste@iupui.edu.
3. Schema: Under the schema diagram, the definition of "Extensive Disease" was changed to the following:

"Extensive Disease:

 - Visceral metastases (extranodal)
AND/OR
 - Bone Metastases
 - o At least 4 or more bone lesions
 - o One of which must be outside of the vertebral column or pelvis."

4. Section 3.1 (page 5): The first paragraph of this section was revised as follows: "Patients must have histologically or cytologically confirmed prostate cancer. Patients may have begun hormonal therapy but it must not have commenced more than 90 days prior to randomization." The line titled "disease stage" was deleted.
5. Section 3.2 (page 5): This section was revised to read as follows:

"Patients must have extensive disease, defined as:

 - Visceral metastases (extranodal)
AND/OR
 - Bone Metastases
 - o At least 4 or more bone lesions
 - o One of which must be outside of the vertebral column or pelvis.

NOTE: Radiological studies identifying measurable or non-measurable disease must be obtained according to the following criteria:

 - If androgen deprivation therapy has not commenced:
 - Scans must be obtained within 6 weeks of randomization.
 - If androgen deprivation has commenced prior to randomization:
 - Scans done within 6 weeks prior to the start of androgen deprivation therapy may be used to identify measurable or non-measurable disease.
 - If all required imaging had not been completed prior to starting androgen deprivation therapy, then any or all additional/necessary scans may be obtained after starting androgen deprivation but prior to randomization.
6. Section 5.1.3 (page 12): In the second bullet, "(3 months)" was added next to "Week 12." In the third bullet, "(6 months)" was added next to "Week 24." In the fourth bullet, "(9 months)" was added next to "Week 36." In the fifth bullet, "(12 months)" was added next to "Week 48." The first sentence after the bullets was revised to read as follows: "Administration of the four part questionnaire (FACT-P, FACIT-F Fatigue Subscale, FACT-Taxane Subscale and the Brief Pain Inventory) will allow us to examine overall QOL, as well as specific treatment-associated and disease-associated changes in QOL."
7. Section 7.1 (page 27): In the fifth column of the table, "Every 12 weeks" was changed to "Every 3 months." In the "Hormone Refractory" column of the table, the 'X' was removed from the "FACT-P..." row. In footnote 6, "every 12 weeks" was changed to "every 3 months." In footnote 7, "week 12" was changed to "3 months (12 weeks)." Footnote 8 was revised to read as follows: "For both Arms A and B, submit at the end of months 6, 9, and 12."
8. Section 7.2 (page 28): The table was revised to read as follows:

Biological material		NUMBER OF TUBES OR SAMPLES		
		Baseline	Week 24	Progression
Original Diagnostic paraffin embedded tumor block ²		X		
Peripheral Blood ⁴	5 mL EDTA lavender top	1 ^{1,3}		
Plasma ⁴	5 mL EDTA lavender top	1 ¹	1	1
Serum ⁴	10 mL red top	1	1	1

Footnote 3 was revised to read as follows: "Whole blood requested at baseline only. If baseline lavender top tube samples are not taken, please draw at next available time point. Please note on the Material Submission Form the cycle, week number, and day that samples were taken."

9. Section 11.0 (page 37): The table in this section was revised to read as follows:

Biological material		NUMBER OF TUBES OR SAMPLES		
		Baseline	Week 24	Progression

Tumor Block Original Diagnostic paraffin embedded ²		X		
Peripheral Blood	5 mL EDTA lavender top tube	1 ^{1,3}		
Plasma	5 mL EDTA lavender top tube	1 ¹	1	1
Serum	10 mL red top tube	1 ¹	1	1

Footnote 3 was revised to read as follows: "Whole blood requested at baseline only. If baseline lavender top tube samples are not taken, please draw at next available time point. Please note on the Material Submission Form the cycle, week number, and day that samples were taken."

10. Section 11.1.2.2 (page 38): The following text was added to this section: "Plasma to be submitted at baseline, week 24, and progression. Whole blood requested at baseline only. If whole blood baseline draw cannot be done prior to treatment, draw at next available opportunity and indicate cycle status (cycle #, week and day) of draw on the material submission form."
11. Appendix I (page 2): In the "Before you begin the study" section, the following was added to the third bullet: "chest x-ray."
12. Appendix I (page 4): In the box to the right of the Arm A box, the first sentence was revised to read as follows: "Evaluate every 3 weeks while receiving chemotherapy and at week 24."
13. Appendix I (page 12): The third paragraph was revised was revised to read as follows: "You will be asked to complete 5 questionnaires: one on your first visit, one at 3 months (week 12), one at 6 months (week 24), one at 9 months (week 36), and one at 12 months (week 48). Each questionnaire has 4 parts. It takes about 15 minutes to fill out each questionnaire." In the last paragraph, "three Quality of Life Questionnaires" was changed to "five Quality of Life Questionnaires."
14. Appendix I (page 13): In the fourth sentence of the first paragraph, the following phrase was deleted: "days 1."

Update #2 includes the following changes:

1. Cover page: Under "Study Participants," the CTSU language was revised to read as follows: "This study is supported by the NCI Cancer Trials Support Group (CTSUS). Institutions not aligned with ECOG will participate through the CTSU mechanism (please see the index page and Appendix IV for details)."
2. Contact information pages: A new box was added detailing the transition to CTSU basic service. Also as a result of this change, the CTSU contact information box was revised in its entirety.
3. Section 10.2.1 (page 36): The room number for Olson Pavilion was changed from Room 8501 to 8421.
4. Section 11.2 (page 38): In the third sentence of the second paragraph, "Dr. Sweeney's laboratory" was changed to "The HOG laboratory." The address for where to ship samples was changed so that they now go to the Hoosier Oncology Group."
5. Appendix II (page 2): In item #6, The room number for Olson Pavilion was changed from Room 8501 to 8421.
6. Appendix IV (pages 1-4): Due to the transition to basic services for all CTSU studies, this appendix was revised in its entirety.

Enclosure

C: CTSU operation office



Eastern Cooperative Oncology Group

Coordinating Center

Frontier Science

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Group Chair: Robert L. Comis, M.D.
Group Statistician: Robert Gray, Ph.D.

Jean MacDonald, Director of Research Operations
Mary Steele, Director of Group Administration

TO: ECOG Clinical Research Associates and Investigators with Patients for E3805

FROM: ECOG Coordinating Center

DATE: July 9, 2008

SUBJECT: E3805, Addendum #3

This addendum has been reviewed and approved by the Central IRB.

Enclosed is Addendum #3 for E3805 – *CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer.*

A revised cover sheet, indicating the date of the revisions, and replacement pages, incorporating the changes to the protocol, have been provided. Please remove the outdated pages and replace them with the enclosed cover and current pages. We recommend that each institution maintain a file containing the original protocol and all subsequent revisions.

A revised eligibility worksheet and revised forms are attached.

This addendum contains changes to the eligibility section and consent form. Full IRB approval is **required**, unless your local IRB has different, written standard operating procedures (SOPs) for handling protocol addenda. It is the decision of the local IRB whether to re-consent subjects. This addendum must be submitted and reviewed by your IRB within 90 days of receipt of this notice, unless your local IRB has different written SOPs. Documentation of this review or adherence to local IRB SOPs will be requested at subsequent ECOG audits

Addendum #3 includes the following changes:

1. Updated version date to 3/13/2008 on the Cover Page and page 1 of Appendix 1.
2. Cover Page: The following was added below "STUDY STATISTICIAN" : "LABORATORY STUDIES CO-CHAIRS: Christopher Sweeney, MBBS NOAH Hahn, M.D."
3. Contact Information Page: The Study Chair contact information was revised to read as follows:
Christopher Sweeney, MBBS
Director of Clinical Trials
Royal Adelaide Hospital Cancer Centre
Professor, University of Adelaide
North Terrace
Adelaide, South Australia 5000
Australia
Phone: + 61 08 8222 2014
Email: chris.sweeney@health.sa.gov.au
4. Schema: The following was added below "Stratify" in the first box: "**Volume of Disease** High vs. Low". Every instance of "week 24" was replaced by "month 6 (week 24)" and "12 weeks" was replaced by "3

months (12 weeks)". The accrual goal was raised to 600. The first 2 sentences below "Accrual goal" was revised to read as follows: "For Arm A patients, PSA will be checked more frequently (once every 3 weeks starting day 1 of cycle 2). Arm A patients will then have PSA checked every 3 months (12 weeks) once off- treatment. If PSA test rises > 50% of lowest level recorded since starting hormonal therapy, progression will need to be confirmed by a repeat PSA test at least 2 weeks later." In the third sentence of footnote 3, "90 days" was changed to "120 days." In the same footnote of the sentence beginning with "**Premedication**" the following was added before "12, 3 and 1 hour": "approximately". The following was added to the end of footnote # 4: "Investigators are encouraged to continue androgen deprivation therapy at time of progression to HRPC. If however, the investigator does not feel it is in the patient's best interest, then androgen deprivation can be discontinued but must be documented in the study chart." The following NOTE was added to the end of the page: "Docetaxel is a commercially marketed product which has been approved by the FDA for use in some cancers.

However its use in patients eligible for this protocol with prostate cancer is not currently approved by the FDA. When used as directed by this protocol, docetaxel is classified as an "unapproved use of an approved agent" and, by definition, considered as an investigational agent. After patient randomization, a supply of docetaxel may be obtained. Investigators must fax a completed "E3805 Clinical Supplies Shipping Form" (Appendix VII) to the ECOG Coordinating Center, ATTN: Drug Team at 617-632-2063."

5. Section 1.1 (page 1): The first paragraph of section 1.1 was revised to read as follows: "Prostate cancer afflicts approximately 230,000 men per year and results in approximately 30,000 deaths (1). Although the PSA (prostate specific antigen) test has been able to diagnose prostate cancer at an earlier stage, it is unclear whether the overall mortality has decreased. The current treatment for hormone naive metastatic prostate cancer is hormone ablation either by LHRH analogue therapy or orchiectomy as monotherapy or in combination with an antiandrogen. Survival varies depending on the extent of disease at commencement of therapy. With the advent of the PSA test many patients are commenced on hormonal therapy at a very early stage and subjected to the long-term effects of androgen ablation including osteoporosis. However, if patients with an asymptomatic rising PSA after definitive local therapy are observed until they develop overt metastatic disease (i.e. evident by imaging techniques), the median time from PSA relapse to clinical progression is approximately 8 years (2). In the pre-PSA era studies relied upon bone scan and CT scans to document the presence of metastatic disease. The median overall survival for men commencing androgen ablation with clinically evident metastatic disease (i.e. not PSA only disease) is about 30 months (3). This information is derived from the meta-analysis of 27 randomized trials with 8,275 men comparing castration (medical or surgery) alone against combined androgen blockade (3). Currently, patients with overt metastases are placed solely on androgen ablation and followed until symptomatic progression. Most men are continued on androgen deprivation indefinitely. Once hormonal therapy is no longer effective, chemotherapy is employed with palliative intent and can prolong overall survival slightly. The purpose of this proposed trial is to determine whether instituting chemotherapy when starting hormonal therapy can delay the time to progression to a clinically meaningful degree without effecting quality of life in men with metastatic (extensive) disease. Patients with metastatic disease are classified into either high volume metastatic disease (ie. metastases involving appendicular skeleton (including ribs and clavicle) with or without axial involvement and/or visceral metastases or low volume metastatic disease. Patients with high volume metastatic disease have a poorer prognosis with a median time to PSA progression of about only 10 months and median time to clinical progression (e.g. worsening bone metastases) of about 14 months. This was seen in the previous Intergroup phase III trial which accrued 1,387 patients in five years of which 75% had extensive disease (4). This was also observed in a second Intergroup trial in the same population (5). In contrast patients with low volume metastatic disease have a 22 month median time to PSA progression with androgen ablation alone and median time to clinical progression of more than three years. The short time to clinical progression also results in men with high volume disease having a median overall survival of only 24 months compared to about four years for patients with minimal disease. Clearly, men with extensive disease are in urgent need of more effective therapies."
6. Section 1.1 (page 2): In the first sentence of the fourth paragraph of section 1.1, "extensive disease" was revised to read "metastatic disease". The second to last sentence of the fifth paragraph was revised to read: "A clinically significant increase in disease control is considered to be an increase in the median overall survival from 24 to 32 months in men with high volume metastatic disease and 48 to 62 months with low volume metastatic prostate cancer."
7. Section 1.3 (page 3): The following was deleted from the end of the first sentence of the page: "which equates to the patients with "extensive" disease".

8. Section 1.4 (page 3): In the last sentence, “sequential serum analyses” was revised to read, “sequential blood analyses”.
9. Section 2.3.2 (page 4): The following was added before “polymorphisms”: “constitutive”.
10. Section 2.3.3 (page 4): The following was added after “carcinogenic proteins”: “in prostate cancer tissue”.
11. Section 3.1 (page 5): In the last sentence of the first paragraph, “90 days” was changed to “120 days”. In the first line of the second paragraph “and” was revised to read “but”.
12. Section 3.2 (page 5): The first three lines were revised to read as follows: “Patients must have metastatic disease and will be stratified by presence of high volume or low volume disease. Low volume disease is defined as any metastatic disease that is not extensive. High volume is defined as:” The bullets below “If androgen deprivation has commenced prior to randomization:” were revised to read as follows:
 - Scans must be obtained within 6 weeks prior to the start of androgen deprivation therapy.
 - If all required imaging had not been completed prior to starting androgen deprivation, then any additional scans must be obtained after starting androgen deprivation but prior to randomization. (It is assumed the scans of patients with high volume disease would not normalize in less than 120 days to the point a patient would go from “high volume” to “low volume.”)
13. Section 3.3 (page 5): Section 3.3 was revised to read as follows: “Patients are not eligible if the PSA has risen greater than 50% from its lowest point between the beginning of androgen deprivation therapy and the date of randomization. “
14. Section 3.3.4 (page 5a): The following was added to the end of section 3.3.4: “ULN:_____ Date of test:_____”.
15. Section 3.4.4 (page 5a): The first line was revised to read as follows: “ALT ≤ 2.5 X upper limit of normal”. The following was added to the end of section 3.4.4: “ULN:_____ Date of test:_____”.
16. Section 3.4.6 (page 6): Section 3.4.6 was revised to read as follows: “PT, INR ≤ 1.5 x ULN (except if on therapeutic anti-coagulation in which case the patient can be enrolled if stable and anti-coagulation levels are appropriate for their condition per good clinical practice)
PT:_____ Date of test:_____
INR:_____ Date of test:_____
ULN:_____ Date of test:_____”
17. Section 3.4.7 (page 6): Section 3.4.7 was revised to read as follows: “PTT ≤ 1.5 x ULN (except if on therapeutic anti-coagulation in which case the patient can be enrolled if stable and anti-coagulation levels are appropriate for their condition per good clinical practice)
PTT:_____ Date of test:_____
ULN:_____ Date of test:_____”
18. Section 3.4.8 (page 6): The following was removed from section 3.4.7 and separated into section 3.4.8: “Patients can enter the study if on therapeutic anti-coagulation.”
NOTE: All values must be obtained within 4 weeks prior to beginning protocol therapy.”
19. Section 3.5 (page 6): At the end of section 3.5 “beginning protocol therapy” was revised to read as follows: “randomization”.
20. Section 3.6 (page 6): In the first paragraph two instances of “ng/dL” were revised to read “ng/mL”. In the last sentence, “at this time” was revised to read “at or after 11/05”.
21. Section 3.7 (page 6): In the parenthesis “NB” was changed to “NOTE”.

22. Section 3.10 (page 6): At the end of section 3.10 “principal investigator” was revised to read “study chair”.
23. Section 3.15 (page 6a): The following was added as section 3.15: “Patients with prior hormone therapy in the metastatic setting are not eligible.”
24. Section 4.0 (page 7): In the second to last sentence “five working days” was revised to read “seven working days”.
25. Section 4.5.6 (page 8): The following section 4.5.6 was added:
 - 4.5.6 Volume of Disease
 - 4.5.6.1 High
 - 4.5.6.2 Low
26. Section 4.6.2 (page 9): The following NOTE was added to section 4.6.2: “NOTE: ECOG requires that all biological samples submitted from patients participating in this trial be entered and tracked via the online ECOG Sample Tracking System (STS). See Section 10.4.”
27. Section 5.1.1.1 (page 10): In the NOTE, the phrase “90 days before beginning protocol therapy” was changed to “120 days before randomization.” In the last paragraph, “a single agent ≤ 7 days” was revised to read “a single agent ≤ 28 days”. In the last sentence of the last paragraph, “90-day” was changed to “120-day”.
28. Section 5.1.1.2 (page 10): The following NOTE was added to section 5.1.1.2: “Docetaxel is a commercially marketed product which has been approved by the FDA for use in some cancers. However its use in patients eligible for this protocol with prostate cancer is not currently approved by the FDA. When used as directed by this protocol, docetaxel is classified as an “unapproved use of an approved agent” and, by definition, considered as an investigational agent. After patient randomization, a supply of docetaxel may be obtained. Investigators must fax a completed “E3805 Clinical Supplies Shipping Form” (Appendix VII) to the ECOG Coordinating Center, ATTN: Drug Team at 617-632-2063.”
29. Section 5.1.1.5 (page 11): At the end of the first sentence of the page “performed” was revised to read “assessed”. The following was added to the end of section 5.1.1.5: “Investigators are encouraged to continue androgen deprivation therapy at time of progression to HRPC. If however, the investigator does not feel it is in the patient’s best interest, then androgen deprivation can be discontinued but must be documented in the study chart.”
30. Section 5.1.2.1 (page 11): In the NOTE, “90 days before beginning protocol therapy” was changed to read as follows: “120 days before randomization.” In the last paragraph, “a single agent ≤ 7 days” was revised to read “a single agent ≤ 28 days”. In the last sentence of the last paragraph, “90-day” was changed to “120-day”.
31. Section 5.1.2.3 (pages 11-12): At the end of the first sentence of the last paragraph “performed” was revised to read “assessed”. The following was added to the end of section 5.1.2.3: “Investigators are encouraged to continue androgen deprivation therapy at time of progression to HRPC. If however, the investigator does not feel it is in the patient’s best interest, then androgen deprivation can be discontinued but must be documented in the study chart.”
32. Section 5.2.1 (page 13): The word “NOTE” was added to the beginning of the first statement below the “Dose modifications for Myelosuppression” chart. The following was added after “held” in the footnote with the “+” beside it: “due to myelosuppression”.
33. Section 5.2.2 (page 14): The first statement under “Dose Modification for Hepatic Dysfunction” was revised to read as follows: “ALT and Bilirubin will be evaluated pre-study and Day 1 (may be evaluated within 24 hours of day 1) of cycles 1-6:”. The second row of the “Action” column was revised to read as follows: “Treat without delay but reduce docetaxel dose by one dose level.”
34. Section 5.2.8 (page 15): The last paragraph was separated into two sentences. The beginning of the second sentence reads as follows: “If baseline was greater than Grade 1...”
35. Section 5.3.6 (page 17): The following NOTE was added to the “Expedited reporting for commercial agents” chart below the sentence “Expedited reporting requirements for adverse events experienced by patients...”: “NOTE: Arm B patients who have received Docetaxel or Androgen Deprivation therapy must follow these reporting requirements.”

36. Section 5.5 (page 18): The following was added below the first sentence: “Patients will come off study if they meet criteria for PSA or clinical progression.”
 37. Section 6.1.1 (page 20): The last sentence of the second paragraph was revised to read as follows: “All baseline evaluations should be performed as detailed in section 3.”
 38. Section 6.1.4 (page 21): A period was added to the end of section 6.1.4.
 39. Section 6.3.1 (page 24): The following was added below the definition of “Serological Partial Response (PR)”: “PSA Stable Disease – Patients who do not meet the criteria for response (CR or PR) or serological progression for at least 3 months (90 days) will be categorized as having stable disease.” The last sentence under “Serological Progression” was revised to read as follows: “If a patient achieves a PSA less than 4ng/mL, progression will only be deemed to have occurred once the level rises above 4 ng/mL and has risen 50% above the nadir level since starting androgen deprivation therapy.”
 40. Section 6.3.2 (page 24): In the first sentence below “Time to Serological Progression” the following was inserted before “50%”: “≥”. In the same paragraph the last sentence was revised to read as follows: “If a patient achieves a PSA less than 4 ng/mL, progression will only be deemed to occur once the level rises above 4 ng/mL and has risen ≥ 50% above nadir level since starting androgen deprivation therapy.”
 41. Section 6.4 (page 25): The following wording was removed throughout section 6.4: “or death from any cause”.
 42. Section 6.4.3 (page 25): The following was added to the end of “Time to Clinical Progression”: “Free Survival”.
 43. Section 6.4.4 (page 25): The following was added to the end of “Time to Serological Progression”: “Free Survival”.
 44. Section 7.1 (page 27): The following was added to the end of #1 under “Therapeutic Parameters”: “(see section 3.2 for guidance if hormone therapy commenced prior to obtaining all scans.)” In the same paragraph, “of starting hormonal therapy.” was revised to read as follows: “prior to randomization”. In #3 above the “Therapeutic Parameters” chart “Section 3.3” was revised to read as follows: “Section 3.4”. Superscript 5 was added to the “Physical Examination, Vital Signs” row under “Day 1, Cycle 1” in the “Therapeutic Parameters” chart . The following row was deleted from the “Therapeutic Parameters” chart under “Both Arms A and B”: “Creatinine Clearance”. The following was added to the end of footnote # 3: “(see section 3.2 for guidance if hormone therapy commenced prior to obtaining all scans)” and also “accepted” was revised to read “required”.
 45. Section 7.1 (page 27a): The following was added to footnote #6:
Follow-up Schedule:
 - Every 6 months if patient is < 2 years from study entry
 - Every 6 months if patient is 2-5 years from study entry
 - Every 12 months if patient is 5-10 years from study entry
 - If patient is > 10 years from study entry, no forms are required
- Added new footnote # 9 that reads as follows: “For Arm A only (i.e. only patients getting docetaxel).”
Added this footnote number to the PSA row of “Day 1, Cycles 2-6”.

46. Section 7.2 (page 28): The following NOTE was added to section 7.2: “NOTE: ECOG requires that all biological samples submitted from patients participating in this trial be entered and tracked via the online ECOG Sample Tracking System (STS). See Section 10.4.” In the third row under “Biological material,” “5mL EDTA lavender top” was revised to read: “10 mL EDTA lavender top”. Superscript 1 was added to the “Serum” row of the “Biological Sample Submission” chart under “baseline”. Superscript 5 was added to the “week 24” box. Throughout the “Biological Sample Submission” chart all instances of “1” (except for footnotes) were replaced by “X”. The following was added to the end of footnote 2: “and Appendix II”. In the third sentence of footnote #3 “Material Submission Form” was revised to read “upon submission,”. The following was added as footnote #5: “If this blood draw is missed, submit at the time of the next scheduled clinical blood draw.”

47. Section 9.0 (pages 33-35): Revised section 9.0 in its entirety. This revised statistical considerations section reflects design changes incorporated in addendum 3, including expansion of study eligibility and adjustment in accrual rate. Extensive and minimal disease throughout this section was replaced by high volume and low volume disease. This study will enroll all patients with extensive disease and minimal disease; eligibility was no longer restricted to patients with extensive disease. In addition, the duration between beginning of prior hormonal therapy and registration was also changed from 90 days to 120 days. Furthermore, an additional stratification factor was added in this addendum, extent of disease: High volume vs. low volume disease. Due to the change in patient population and the adjustment in accrual rate, the sample size, study duration, and interim analyses plan were all revised.
48. Section 10.0 (page 36): The following NOTE was added to section 10.0: "NOTE: ECOG requires that all biological samples submitted from patients participating in this trial be entered and tracked via the online ECOG Sample Tracking System (STS) as outlined in section 10.4. A shipping manifest generated by the system is to accompany all submissions. If STS is unavailable at time of shipment, samples are to be shipped and the laboratory notified. Once STS is available, retroactively log the shipment into STS, using the actual collection and shipping dates."
49. Section 10.4 (page 36a): This new section was added titled "ECOG Sample Tracking System." As a result, the rest of section 10 was re-numbered.
50. Section 11.0 (page 37): The following NOTE was deleted from section 11.0: "**NOTE:** Kits are available. They will include the EDTA Lavender top tube, Red top tube, and shipping materials. Upon institutional IRB approval, complete Kit Order Form (Appendix VI) and fax to Kristina Kirkpatrick at (317) 921-2053. Kits will NOT be shipped overnight courier unless expressly stated on the form that it is needed within 24 hours de to patient enrollment." The NOTE beginning "An E3805 Material Submission Form #2377..." was replaced by the following: "NOTE: ECOG requires that all biological samples submitted from patients participating in this trial be entered and tracked via the online ECOG Sample Tracking System (STS) as outlined in section 10.4. A shipping manifest generated by the system is to accompany all submissions. If STS is unavailable at time of shipment, samples are to be shipped with a completed E3805 Material Submission Form (#2741) and the laboratory notified. Once STS is available, retroactively log the shipment into STS, using the actual collection and shipping dates." All instance of "1" (except for footnotes) in the "Submission Summary" chart were replaced by "X". In the third row under "Biological material," "5mL EDTA lavender top" was revised to read: "10 mL EDTA lavender top". Superscript 4 was added to the "week 24" box. In footnote #3 of the "submission Summary" chart, "on Material Submission Form" was revised to read: "upon submission". The following was added as footnote 4: "If missed, draw at the time of next scheduled clinical draw."
51. Section 11.1.1 (page 37): The second sentence was revised to read as follows: "If tumor block is not available slides may be submitted as outlined in section 10.1.2."
52. Section 11.1.2.1 (page 38): The following was added to the end of section 11.1.2.1: "until shipped".
53. Section 11.1.2.2.2 (page 38): At the beginning of the second paragraph, "Draw approximately 5 mL" was revised to read: "Draw approximately 10 mL". In the second sentence of the same paragraph, "1.0 ml aliquots" was revised to read "1.5 ml aliquots".
54. Section 11.2 (page 38): The last sentence of paragraph one was revised to read as follows: "The STS generated shipping manifest or appropriate Material Submission Form (#2741) must accompany the samples." In the fourth sentence of paragraph two, "The" was replaced by the following: "If STS is not available at time of shipment, the." In the last sentence of the second paragraph "Raja Shanmugam at rshanmug@iupui.edu and" was replaced by the following: "Kristina Kirkpatrick at kriskirk@iupui.edu or call her at (317) 921-2050."
55. Section 11.3 (pages 39-41): Section 11.3 was revised in it's entirety.
56. Section 11.4 (page 42): In the last sentence of the page "Cancer Pavilion" was revised to read: "university".
57. Section 11.5.1 (page 43): The following was added after "tumor blocks": "or appropriate slides".
58. Section 11.8 (page 43): The following was removed from the bottom of the page:

Please refer to the E3805 Forms Packet for copies of all forms. The E3805 Forms Packet may be downloaded by accessing the ECOG World Wide Web Home Page (<http://www.ecog.org>).

Forms must be submitted to the ECOG Coordinating Center, FSTRF, 900 Commonwealth Avenue, Boston, MA 02215 (ATTN: DATA).

This study will be monitored by the CTEP Data Update System (CDUS) version 3.0. Cumulative CDUS data will be submitted quarterly from the ECOG Coordinating Center to CTEP by electronic means.

59. Appendix I (page 1): The second paragraph was revised to read as follows: “ You are being asked to take part in this study because you have prostate cancer that has spread beyond the prostate and can be seen on scans.” The first paragraph under “WHY IS THIS STUDY BEING DONE?” was revised to read as follows:

“The purpose of this study is to evaluate a new way of treating patients with prostate cancer. In Arm A, you will receive docetaxel chemotherapy both at the time you are starting hormonal therapy and later when/if the hormonal therapy is no longer working per your doctor decision. In Arm B, patients only get chemotherapy when/if the hormonal therapy is no longer working per your doctor decision. We are doing the study to see if there is a benefit to giving chemotherapy at the beginning. Hormonal therapy refers to drugs or surgical procedures such as an orchiectomy (removal of testicles) which lower your testosterone. This puts prostate cancer into remission in most patients as the testosterone is like a fuel for the cancer. It is the standard to give this as the only treatment for your cancer. Normally chemotherapy is reserved for when a patient’s cancer starts to grow again despite having a low testosterone level. In this study you will either get docetaxel when you start your hormonal therapy or when/if your cancer grows with a low testosterone level. If you have chemotherapy when you start hormones for the first time, you may get chemotherapy when/if your cancer grows back with a low testosterone level.”

In the parenthesis of the second to last sentence on the page “or within 90 days” was revised to read as follows: “or within 120 days”.

60. Appendix I (page 2): Under “HOW MANY PEOPLE WILL TAKE PART IN THIS STUDY?”, “568 people” was revised to “600 people”.
61. Appendix I (page 3): Under “WHEN YOU ARE FINISHED TAKING CHEMOTHERAPY”, “12 weeks” was revised to read “3 months”. The “Cycle 1-6” chart was revised to read as follows:

Cycle 1-6	
Day	What you do
Just before chemotherapy	Take dexamethasone (pre-medication to prevent reactions from chemotherapy) 12 hours, 3 hours and 1 hour prior to infusion
Day 1	Check into doctor’s office, get blood tests and if suitable, get chemotherapy

62. Appendix I (page 4): Throughout the “Study Plan” chart “12 weeks” was revised to read “every 3 months (12 weeks)”. “week 24” was revised to read “month 6 (week 24).”
63. Appendix I (page 6): The following was removed from the 11th bullet under “Likely” for Docetaxel: “and, rarely, liver damage or failure.”
64. Appendix I (page 7): The following bullet was added under “rare, but serious” for Docetaxel: “Liver damage or failure”.
65. Appendix I (page 12): In the sentence which read, “If you decide to take part in this study, the only thing you will be asked to do is fill out the three questionnaires”, “three” was replaced with “five”.
66. Appendix I (page 13): The box under “SCIENTIFIC STUDIES” was revised to read as follows:

I agree to participate in the scientific laboratory research studies that are being done as part of this clinical trial.

Yes No

--

The paragraph under “About Using Tissue for Research” was revised to read as follows: “If you participate in the laboratory study associated with this protocol, you will have some blood and the biopsy material from your original diagnostic biopsy sent to central laboratories for analysis. The samples of blood and tissue are referred to as specimens.” In the last paragraph of the page three instances of “blood and tissue” were revised to read “specimens”.

67. Appendix I (page 14): Through out the page every instance of “blood and tissue” or “blood” was revised to read “specimens”.
68. Appendix I (page 15): The boxes under “Making Your Choice” were revised to read as follows:

My specimens may be kept for use in research to learn about, prevent, treat, or cure cancer.
Yes No
My specimens may be kept for use in research about other health problems (for example, causes of diabetes, Alzheimer’s disease, and heart disease).
Yes No
Someone from this institution may contact me in the future to ask me to take part in more research.
Yes No

69. Appendix IV (page 2): The following NOTE was added below “**SPECIAL MATERIALS OR SUBSTUDIES**”: “NOTE: ECOG requires that all biological samples submitted from patients participating in this trial be entered and tracked via the online ECOG Sample Tracking System (STS). See Section 10.4.”

Enclosure

C: CTSU



Eastern Cooperative Oncology Group

Coordinating Center

Frontier Science
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Randomization: (617) 632-2022

Group Chair: Robert L. Comis, M.D.
Group Statistician: Robert Gray, Ph.D.

Jean MacDonald, Director of Research Operations
Mary Steele, Director of Group Administration

TO: ECOG Clinical Research Associates and Investigators with Patients for E3805

FROM: ECOG Coordinating Center

DATE: **July 23, 2008**

SUBJECT: E3805 Update # 3

Enclosed is E3805, Update #3, *CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer.*

A revised cover sheet (indicating the date of the changes) and replacement pages (incorporating the changes to the protocol) are provided. Please remove the outdated pages and replace them with the appropriate pages enclosed. We recommend that each institution maintain a file containing the original protocol and all subsequent modifications.

This update contains administrative changes only and, per OHRP guidelines, expedited IRB review may be appropriate. However, please consult your local IRB policy since their requirements may differ and may require full board review. This addendum must be reviewed by your IRB within 90 days of receipt of this notice.

Update #3 includes the following changes:

1. Updated version date and the NCI update date to July 23, 2008 on the Cover Page and page 1 of Appendix I.
2. Contact Information page: The study chair liaison's name was updated to read "Anne E Younger RN CCRP"
3. Section 5.3.2 (page 16): In the "Step 5" NOTE the "NCI Medical Help Desk" was replaced by the following: "NCI AdEERS Help Desk".
4. Section 5.3.6 (page 17): In order to correct an error, the following NOTE was removed: "**NOTE:** Arm B patients who have received Docetaxel or Androgen Deprivation therapy must follow these reporting requirements."
5. Section 7.1 (page 27a): Under "Follow-up Schedule:" in footnote #6, the first bullet was revised to read "Every 3 months if patient is < 2 years from study entry".

Enclosure



Eastern Cooperative Oncology Group

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Group Chair: Robert L. Comis, M.D.
Group Statistician: Robert Gray, Ph.D.

Jean MacDonald, Director of Research Operations
Mary Steele, Director of Group Administration

TO: ECOG Clinical Research Associates and Investigators with Patients for E3805

FROM: ECOG Coordinating Center

DATE: **October 1, 2008**

SUBJECT: E3805 Update # 4

Enclosed is E3805, Update #4, *CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer.*

A revised cover sheet (indicating the date of the changes) and replacement pages (incorporating the changes to the protocol) are provided. Please remove the outdated pages and replace them with the appropriate pages enclosed. We recommend that each institution maintain a file containing the original protocol and all subsequent modifications.

This update contains administrative changes only and, per OHRP guidelines, expedited IRB review may be appropriate. However, please consult your local IRB policy since their requirements may differ and may require full board review. This update must be reviewed by your IRB within 90 days of receipt of this notice.

Update #4 includes the following changes:

1. On the cover page, updated version date and the NCI update date to October 1, 2008. Updated the version date to 10/01/08 on page 1 of Appendix I.
2. Cover Page: The following was added below the "QOL CO-CHAIR" line: "SWOG CO-CHAIR: Jorge A. Garcia, M.D." The following was added under "ECOG Entire Group": "SWOG".
3. Contact Information page: The following was added to the contact information page:

SWOG CO-CHAIR
Jorge A. Garcia, M.D.
Cleveland Clinic
Taussig Cancer Center
9500 Euclid Ave.
R35
Cleveland, OH 44195
Tel: (216) 444-7774
Fax: (216) 444-9474
Email: garciaj4@ccf.org

Enclosure



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Group Chair: Robert L. Comis, M.D.
Group Statistician: Robert Gray, Ph.D.

Jean MacDonald, Director of Research Operations
Mary Steele, Director of Group Administration

TO: ECOG Clinical Research Associates and Investigators with Patients for E3805

FROM: ECOG Coordinating Center

DATE: **April 23, 2009**

SUBJECT: E3805 Update # 5

Enclosed is E3805, Update #5, *CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer.*

A revised cover sheet (indicating the date of the changes) and replacement pages (incorporating the changes to the protocol) are provided. Please remove the outdated pages and replace them with the appropriate pages enclosed. We recommend that each institution maintain a file containing the original protocol and all subsequent modifications.

This update contains administrative changes only and, per OHRP guidelines, expedited IRB review may be appropriate. However, please consult your local IRB policy since their requirements may differ and may require full board review. This update must be reviewed by your IRB within 90 days of receipt of this notice.

Update # 5 includes the following change:

1. On the cover page, updated version date and the NCI update date to April 23, 2009. Updated the version date to April 23, 2009 on page 1 of Appendix I.
2. Cover Page: The following was added below "STUDY PARTICIPANTS": "United States Sites ONLY".

Enclosure



Eastern Cooperative Oncology Group

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Randomization: (617) 632-2022

Group Chair: Robert L. Comis, M.D.
Group Statistician: Robert Gray, Ph.D.

Jean MacDonald, Director of Research Operations
Mary Steele, Director of Group Administration

TO: ECOG Clinical Research Associates and Investigators with Patients for E3805

FROM: ECOG Coordinating Center

DATE: **August 25, 2009**

SUBJECT: E3805 Update # 6

Enclosed is E3805, Update #6, *CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer*.

A revised cover sheet (indicating the date of the changes) and replacement pages (incorporating the changes to the protocol) are provided. Please remove the outdated pages and replace them with the appropriate pages enclosed. We recommend that each institution maintain a file containing the original protocol and all subsequent modifications.

This update contains administrative changes only and, per OHRP guidelines, expedited IRB review may be appropriate. However, please consult your local IRB policy since their requirements may differ and may require full board review. This update must be reviewed by your IRB within 90 days of receipt of this notice.

A revised eligibility worksheet is attached.

Update # 6 includes the following change:

1. On the cover page, updated version date and the NCI update date to August 25, 2009. Updated the version date to August 25, 2009 on page 1 of Appendix I.
2. Cover Page: The list of "STUDY PARTICIPANTS" was revised to read as follows:
"ECOG: United States Institutions ONLY with the following exception: GECOPERU
SWOG: United States Institutions ONLY
CTSU: United States Institutions ONLY"
3. Contact Information page: The "STUDY CHAIR" email address was revised to read as follows:
"chsweene@gmail.com."
4. Section 7.2 (page 28): In the "Biological Sample Submissions" table the following was removed from the first box below "Biological material": "Original diagnostic".
5. Section 10.1.2.1 (page 36): The following was removed from the first bullet below "Correlative Studies": "from the original diagnostic biopsy".
6. Section 11.0 (page 37): The first box below "Biological material" of the "Submission Summary" table was revised to read as follows: "Paraffin embedded tumor block²".
7. Appendix II (page 3): The following was removed from the first sentence of bullet one under #4: "from the original diagnostic biopsy".

Enclosure



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Group Statistician: Robert Gray, Ph.D.

Jean MacDonald, Director of Research Operations
Mary Steele, Director of Group Administration

TO: ECOG Clinical Research Associates and Investigators with Patients for E3805

FROM: ECOG Coordinating Center

DATE: February 3, 2010

SUBJECT: E3805, Addendum #4

This addendum has been reviewed and approved by the Central IRB.

Enclosed is Addendum #4 for E3805 – *CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer.*

A revised cover sheet, indicating the date of the revisions, and replacement pages, incorporating the changes to the protocol, have been provided. Please remove the outdated pages and replace them with the enclosed cover and current pages. We recommend that each institution maintain a file containing the original protocol and all subsequent revisions.

A memo explaining the determination of serological (PSA) progression is attached.

This addendum contains administrative changes only and, per OHRP guidelines, expedited IRB review may be appropriate. However, please consult your local IRB's policy, since their requirements may differ and require full board review. This addendum must be submitted and reviewed by your IRB within 90 days of receipt of this notice, unless your local IRB has different written SOPs, which must be available at future ECOG audits.

Addendum #4 includes the following changes:

1. Updated version date to 12/31/2009 on the Cover Page and page 1 of Appendix 1.
2. Contact Information Page: The Study Chair contact information was revised to read as follows:

Christopher Sweeney, MBBS
Medical Oncology
Dana Farber Cancer Institute
44 Binney Street, Rm 1230
Boston, MA 02115
Phone: (617) 632-4524
Fax: (617) 632-2165
E-mail: Christopher.Sweeney@dfci.harvard.edu

3. Schema: The following was added after "whether it will be" in footnote # 1: "given or".
4. Section 3.6 (page 6): In the last sentence of Section 3.6 "11/05" was revised in two places to read as follows "11/04".
5. Section 5.1.1 (page 10): In the NOTE under section 5.1.1.1, "within 5 working days" was revised to read "within 7 working days".

6. Section 6.3.1 (page 24): The last paragraph of section 6.3.1 was revised to read as follows:
“**Serological Progression (PD)** is observed when the PSA demonstrates an increase that is more than 50% of nadir, taking as reference the lowest recorded PSA level since starting androgen deprivation therapy (ADT). Two consecutive increases must be documented with each measurement obtained at least two weeks apart. On occasions, there may be an intermediate fluctuant value. In accordance with the recommendations of the Prostate Cancer Clinical Trials Working Group (51), this will **not** restart the evaluation period so long as the intermediate value was not below the previous nadir. The date of first recorded increase (not defeated by a subsequent drop in PSA level to create a new nadir) will be deemed the *date of progression*.
If a patient achieves a PSA that is less than 4 ng/ml:
- The confirming increase must reach a value that is more than 4.0 ng/ml (the unconfirmed and additional intermediate increases may be a value that is less than 4.0 ng/ml, but greater than 50% of nadir since starting ADT).
 - If the patient's nadir is less than 2 ng/ml, the first time the PSA reaches a value greater than 50% of nadir since starting ADT **and** is at least 2 ng/ml will be the date of first progression. However, this progression must be confirmed by a PSA value that is more than 4.0 ng/ml with no intervening declines below 2 ng/ml.”
7. Section 6.3.2 (page 24a): The section entitled “**Time to Serological Progression**” was revised to read as follows: “**Time to Serological Progression** – The period measured from the initiation of therapy until at least a $\geq 50\%$ increase in PSA level, taking as reference the lowest PSA level recorded since start of treatment. The date of the first recorded increase will be deemed the date of progression. If a patient achieves a PSA that is less than 4 ng/ml:
The confirming increase must reach a value that is more than 4.0 ng/ml (the unconfirmed and additional intermediate increases may be a value that is less than 4.0 ng/ml, but greater than 50% of nadir since starting ADT).
If the patient's nadir is less than 2 ng/ml, the first time the PSA reaches a value greater than 50% of nadir since starting ADT **and** is at least 2 ng/ml will be the date of first progression. However, this progression must be confirmed by a PSA value that is more than 4.0 ng/ml with no intervening declines below 2 ng/ml.” As a result of this change, the entire “**Time to Serological Progression**” section was moved to the new page 24a.
8. Section 7.2 (page 28): Under “Biological material” in the “Biological Sample Submissions” table “Paraffin embedded tumor block²” was revised to read as follows: “Prostate Cancer Pathology²”. The following was added to footnote #2 of the same table: “Preferably tissue from the prostatectomy specimen, but if not available please send biopsy specimen from either prostate or metastatic site”.
9. Section 10.2.1 (page 36): In the first sentence below “Submission Schedule”, “original diagnostic block” was revised to read as follows: “prostate cancer pathology”.
10. Section 11.0 (page 37): Under “Biological material” in the “Submission Summary” table “Paraffin embedded tumor block²” was revised to read as follows: “Prostate Cancer Pathology²”. The following was added to footnote #2 of the same table: “Preferably tissue from the prostatectomy specimen, but if not available please send biopsy specimen from either prostate or metastatic site”.
11. Section 11.5.1 (page 43): In the first sentence below “Sample Submission”, “original diagnostic tumor blocks” was revised to read as follows: “prostate cancer pathology”.
12. Section 14 (page 47): The following was added as reference 51: “Scher et al *J Clin Oncol* 26: 1148-119; 2008.”

Enclosure

C: CTSU



Eastern Cooperative Oncology Group

Coordinating Center

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Randomization: (617) 632-2022

Group Chair: Robert L. Comis, M.D.
Group Statistician: Robert Gray, Ph.D.

Jean MacDonald, Director of Research Operations
Mary Steele, Director of Group Administration

TO: ECOG Clinical Research Associates and Investigators with Patients for E3805

FROM: ECOG Coordinating Center

DATE: **June 15, 2011**

SUBJECT: E3805 Update # 7

Enclosed is E3805, Update # 7, *CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer.*

A revised cover sheet (indicating the date of the changes) and replacement pages (incorporating the changes to the protocol) are provided. Please remove the outdated pages and replace them with the appropriate pages enclosed. We recommend that each institution maintain a file containing the original protocol and all subsequent modifications.

This update contains administrative changes only and, per OHRP guidelines, expedited IRB review may be appropriate. However, please consult your local IRB policy since their requirements may differ and may require full board review. This update must be reviewed by your IRB within 90 days of receipt of this notice.

A revised eligibility worksheet is attached.

Update # 7 includes the following change:

1. On the cover page, updated version date and the NCI update date to June 15, 2011. Updated the version date to June 15, 2011 on page 1 of Appendix I.
2. Cover Page: Removed the following from the "STUDY PARTICIPANTS" list: "with the following exception: GECOPERU" as it is no longer relevant.



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Group Chair: Robert L. Comis, M.D.
Group Statistician: Robert Gray, Ph.D.

Jean MacDonald, Director of Research Operations
Mary Steele, Director of Group Administration

TO: ECOG Clinical Research Associates and Investigators with Patients for E3805

FROM: ECOG Coordinating Center

DATE: **September 1, 2011**

SUBJECT: E3805 Update # 8

Enclosed is E3805, Update # 8, *CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer.*

Note: The changes made in this amendment are due to the CTEP CTCAE version 3.0 to CTCAE version 4.0 conversion. Specifically, all AE reporting and toxicity grading in version 3.0 will end on September 30, 2011 and version 4.0 will be utilized beginning October 1, 2011.

A revised cover sheet (indicating the date of the changes) and replacement pages (incorporating the changes to the protocol) are provided. Please remove the outdated pages and replace them with the appropriate pages enclosed. We recommend that each institution maintain a file containing the original protocol and all subsequent modifications.

This update contains administrative changes only and, per OHRP guidelines, expedited IRB review may be appropriate. However, please consult your local IRB policy since their requirements may differ and may require full board review. This update must be reviewed by your IRB within 90 days of receipt of this notice.

Update # 8 includes the following changes:

1. Cover page: Updated version date and NCI update date.
2. Section 5.2, page 13, 2nd paragraph: Replaced sentence due to CTC v4.0 conversion.
3. Section 5.3.2, page 15, "Step 1" and "Step 2": Replaced text due to CTC v4.0 conversion.
4. Appendix I, page 1: updated version date.



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Group Chair: Robert L. Comis, M.D.
Group Statistician: Robert Gray, Ph.D.

Jean MacDonald, Director of Research Operations
Mary Steele, Director of Group Administration

TO: ECOG Clinical Research Associates and Investigators with Patients for E3805

FROM: ECOG Coordinating Center

DATE: December 7, 2011

SUBJECT: E3805, Addendum #5

This addendum has been reviewed and approved by the Central IRB.

Enclosed is Addendum #5 for E3805 – *CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer.*

NOTE: This addendum increases accrual from 600 to 780 patients.

A revised cover sheet, indicating the date of the revisions, and replacement pages, incorporating the changes to the protocol, have been provided. Please remove the outdated pages and replace them with the enclosed cover and current pages. We recommend that each institution maintain a file containing the original protocol and all subsequent revisions.

A revised eligibility worksheet is attached.

This addendum contains administrative changes and an accrual increase, per OHRP guidelines, expedited IRB review may be appropriate. However, please consult your local IRB's policy, since their requirements may differ and require full board review. This addendum must be submitted and reviewed by your IRB within 90 days of receipt of this notice, unless your local IRB has different written SOPs, which must be available at future ECOG audits.

Addendum #5 includes the following changes:

1. Updated version date on the Cover Page and page 1 of Appendix 1.
2. Schema: 1st box, below "Stratify", revised "Bisphosphonate" to read, "FDA Approved Drugs for Delaying Skeletal Related Events". Revised the accrual goal to read, "780". Replaced footnote #2 with more detailed wording.
3. Section 3.3 (page 5): Replaced "greater than 50%" with "and met criteria for progression as defined in section 6.3.1".
4. Section 3.9 (page 6): Added the following to section 3.9 for clarification purposes: "Patients cannot simultaneously enroll on CALGB 90202 due to the potential to confound endpoints of both studies and impact stratification."
5. Section 4.5.5 (page 8): Revised the header for section 4.5.5.
6. Section 5.1.1.1 (page 10): Below NOTE, paragraph 2, sentence 2, added "as sole therapy". End of same paragraph, added new sentence regarding the use of ketoconazole.
7. Section 5.1.1.5 (page 10): 2nd paragraph, replaced "bisphosphonates" with "drugs to prevent skeletal related events".

8. Section 5.1.2.1 (page 11): Below NOTE, paragraph 2, sentence 2, added “as sole therapy”. End of ame paragraph, added new sentence regarding the use of ketoconazole.
9. Section 5.1.2.3 (page 11): 2nd paragraph, replaced “bisphosphonates” with “drugs to prevent skeletal related events”.
10. Section 5.1.3 (page 12): 1st bullet, revised wording to maintain consistency.
11. Section 5.2.1 (page 13): 1st sentence, below the NOTE, added “A need for”. End of page 13, added new paragraph regarding the use of growth factors.
12. Section 5.2.3 (page 14): 3rd sentence, added “Grade $\frac{3}{4}$ stomatitis” after “second”. End of paragraph, added new sentence regarding a “third Grade $\frac{3}{4}$ stomatitis event”.
13. Section 5.2.8 (page 15): Deleted the last sentence and added the following: “and patients treated with a one dose level reduction”.
14. Section 5.3.7 and 5.3.8 (pages 17-18): Replaced existing sections 5.3.7 and 5.3.8 with new reporting guidelines. These guidelines are now entirely contained within 5.3.7, therefore 5.3.8 has been removed. As a result of this change a new page 17a was added.
15. Section 5.4.2 (page 18): 1st sentence, added “or other agents such as RANK ligand inhibitors” after “bisphosphonates”.
16. Section 6.3.1 (page 24): 3rd bullet, added 2 new sentences regarding progression and testosterone level.
17. Section 6.3.2 (page 24-24a): End of paragraph, added 2 new sentences regarding progression and testosterone level. Page 24a, added a new 3rd bullet.
18. Section 6.4.4 (page 25): Added 2 new sentences regarding progression and testosterone level.
19. Section 7.2 (page 28): Replaced section 7.2 with a new updated version.
20. Section 8.2 (page 31): Revised the title to be more specific.
21. Section 8.2.1 (page 31): 1st sentence, replaced “agonists” with “analogues”.
22. Section 9.0- 9.7 (pages 33-35a): Sections 9.0 through 9.7 were revised in its entirety in accordance with the increase of accrual.
23. Section 10.0 (pages 36-36d): Revised section 10 in its entirety due the reformatting of template language. As a result of this change new pages 36a- 36d were added.
24. Section 11-14 (37-44): Revised section 11 in its entirety due the reformatting of template language. As a result of this change, sections 12 through 14 were moved up to different pages.
25. Appendix I (page 2): Below “HOW MANY PEOPLE WILL TAKE PART IN THIS STUDY?”, “600 people” was revised to read: “780 people”.
26. Appendix I (pages 13-15b): Pages 13-15b were updated with new laboratory research template language. As a result of this change new pages 15a and 15b were added.
27. Appendix VI (page 1): Updated appendix VI with new template language.

Enclosure

C: CTSU

TO: ECOG-ACRIN Clinical Research Associates and Investigators with Patients for E3805

FROM: ECOG-ACRIN Operations Office – Boston

DATE: June 4, 2014

SUBJECT: E3805, Addendum 6

Enclosed is Addendum 6 for E3805, *CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer*.

Due to new NCI requirements, the entire protocol has been reformatted. Please note that this reformatting includes the removal of the Informed Consent document from the protocol, as well as a complete pagination revision. The Informed Consent document will continue to be maintained as an independent document on the ECOG website. Please replace your current copy of the protocol with this updated version.

Because this protocol is permanently closed to accrual, this addendum may be handled in one of the following two ways, depending on the status of subjects under the protection of your local IRB. Documentation of this review or adherence to local IRB written SOPs will be requested at subsequent ECOG-ACRIN audits.

1. If all subjects under the protection of the local IRB have discontinued active treatment with androgen deprivation or docetaxel, your local IRB will determine whether or not you must submit this addendum for IRB review.
2. If one or more subjects under the protection of the local IRB continue on active treatment with androgen deprivation or docetaxel, you must submit this addendum for IRB review.

Expedited IRB review of this addendum may be utilized even if there are subjects on active treatment with androgen deprivation or docetaxel. However, please consult your local IRB policy since its requirements may differ and require full board review. It is the decision of the local IRB whether or not subjects are to be re-consented. This addendum must be submitted and reviewed by your IRB within 90 days of receipt of this notice, unless your local IRB has different written SOPs, which must be available at future ECOG-ACRIN audits.

Addendum 6 includes the following changes to the protocol:

	Section	Change
1.	Cover Page	Updated the Version Date.
2.	Cover Page	Logo updated to ECOG-ACRIN. Removed “Eastern Cooperative Oncology Group” from above study title.
3.	Cover Page	Removed ECOG-ACRIN from list of Participating Organization since lead protocol organization. Removed “CTSU” and “SWOG” and Inserted new group names of all participating groups through the CTSU mechanism.

4.	Global change throughout protocol	Changed references to “ECOG” to “ECOG-ACRIN,” references to “ECOG Coordinating Center” to “ECOG-ACRIN Coordinating Office,” and references to “ECOG PCO-RL” to “ECOG-ACRIN PCORL.” Page headers updated from “Eastern Cooperative Oncology Group” to “ECOG-ACRIN Cancer Research Group.” Removed “ECOG” from all form references for administrative simplification.
5.	Schema	Volume of Disease footnote: Corrected spelling of Metastases. In last bullet point regarding bone metastases, clarified that for bone lesions one must be outside of the vertebral column AND pelvis.
6.	Section 3.1.2	In first 2 bullet points, corrected spelling of Metastases. In last bullet point prior to NOTE, clarified that for bone lesions, one must be outside of the vertebral column AND pelvis.
7.	Section 4	Removed CTSU Help Desk hours due to varying schedule. Please refer to the CTSU website for the latest hours.
8.	Section 5.3	Updated entire section to reflect change from AdEERS to CTEP-AERS system.
9.	Section 5.3.2	Steps to determine if an adverse event is to be reported in an expedited manner, step 1 paragraph: Corrected reference to CTCAE version to version 4.0 of CTCAE.
10.	Section 8.1.1	Removed IND number for Docetaxel, as agent is exempt from IND status.
11.	Section 9.1	First paragraph, fourth paragraph: Clarified that for bone lesions, one must be outside of the vertebral column AND pelvis.
12.	Appendix I	Per NCI guidelines, removed consent document from protocol and separated into stand alone consent document.
13.	Appendix IV	Updated phone number for PMB. Revised appendix to reflect change from AdEERS to CTEP-AERS system.

Addendum 6 includes the following changes to the Informed Consent Document:

	Section	Change
1.	Cover Page	Updated the Version Date.
2.	Global change	Updated “ECOG” to “ECOG-ACRIN” and “Eastern Cooperative Oncology Group (ECOG)” to “ECOG-ACRIN Cancer Research Group.” Page headers updated from “Eastern Cooperative Oncology Group” to “ECOG-ACRIN Cancer Research Group.”
3.	WILL MY MEDICAL INFORMATION BE KEPT PRIVATE?	Added new paragraph to end of section with information on this trial posting on www.clinicaltrials.gov
4.	WHAT ARE THE COSTS OF TAKING PART IN THIS STUDY?	Updated website link on clinical trial insurance coverage.

Enclosure

TO: ECOG-ACRIN Clinical Research Associates and Investigators with Patients for E3805

FROM: ECOG-ACRIN Operations Office

ACTIVATION DATE: April 8, 2015

SUBJECT: E3805, Addendum 7

This addendum has been reviewed and approved by the Central IRB.

Enclosed is Addendum 7 for E3805, *CHAARTED: ChemoHormonal Therapy Versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer*.

Addendum 7 will activate on April 8, 2015.

Please replace your current copy of the protocol and Informed Consent document with these updated versions. We recommend that each institution maintain a file containing the original protocol, Informed Consent, and all subsequent revisions/versions.

Because this protocol is permanently closed to accrual, this addendum may be handled in one of the following two ways, depending on the status of subjects under the protection of your local IRB. Documentation of this review or adherence to local IRB written SOPs will be requested at subsequent ECOG-ACRIN audits.

1. If all subjects under the protection of the local IRB have discontinued active treatment with study drugs, your local IRB will determine whether or not you must submit this addendum for IRB review.
2. If one or more subjects under the protection of the local IRB continue on active treatment with study drugs, you must submit this addendum for IRB review.

Expedited IRB review of this addendum may be utilized even if there are subjects on active treatment with study drugs. However, please consult your local IRB policy since its requirements may differ and require full board review. It is the decision of the local IRB whether or not subjects are to be re-consented. This addendum must be submitted and reviewed by your IRB within 90 days of receipt of this notice, unless your local IRB has different written SOPs, which must be available at future ECOG-ACRIN audits.

Addendum 7 includes the following changes to the protocol:

	Section	Change
1.	Cover Page	Updated the version date.
2.	Global Change	Updated the lab name from Hoosier Oncology Group to Hoosier Cancer Research Network.
3.	Global Change	Updated the protocol to reflect the ECOG-ACRIN CBPF moving to the MD Anderson Cancer Center.
4.	Section 10.3.3	Updated the Hoosier Oncology Group laboratory address.

Addendum 7 includes the following changes to the Informed Consent:

	Section	Change
1.	Cover Page	Updated the version date.

Enclosure

CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease in Prostate Cancer

Rev. 7/08

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Rev. 10/08

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Version Date: December 23, 2014

NCI Update Date: June 15, 2011

Rev. 6/14

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United States Institutions ONLY

ALLIANCE / Alliance for Clinical Trials in Oncology

NRG / NRG Oncology Foundation, Inc

SWOG / SWOG.

This study is supported by the NCI Cancer Trials Support Unit (CTSU). Institutions not aligned with ECOG-ACRIN will participate through the CTSU mechanism (please see the index page and [Appendix IV](#) for details)

ACTIVATION DATE

July 28, 2006

Addendum # 1 – Incorporated Prior to Activation

Update #1 – Incorporated Prior to Activation

Addendum # 2 – 2/07

Update #2 – 2/07

Addendum #3 – 7/08

Update #3 – 7/08

Update #4 – 10/08

Update #5 – 4/09

Update #6 – 8/09

Addendum #4 – 2/10

Update #7 – 6/11

Update #8 – 9/11

Addendum #5 – 12/11

Addendum #6 – 6/14

Addendum #7 - 4/15

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2/10

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Rev. 10/08

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Rev. 2/07

Rev. 2/07

This study is supported by the NCI Cancer Trials Support Unit (CTSU).

Institutions not aligned with ECOG-ACRIN will participate through the CTSU mechanism as outlined below and detailed in the CTSU logistical appendix.

- The **study protocol and all related forms and documents** must be downloaded from the protocol-specific Web page of the CTSU Member Web site located at <https://members.ctsu.org>
- Send completed **site registration documents** to the CTSU Regulatory Office. Refer to the CTSU logistical appendix for specific instructions and documents to be submitted.
- **Patient enrollments** will be conducted by the CTSU. Refer to the CTSU logistical appendix for specific instructions and forms to be submitted.
- Data management will be performed by the ECOG-ACRIN. **Case report forms** (with the exception of patient enrollment forms), **clinical reports, and transmittals** must be sent to ECOG-ACRIN unless otherwise directed by the protocol. Do not send study data or case report forms to the CTSU Data Operations.
- **Data query and delinquency reports** will be sent directly to the enrolling site by ECOG-ACRIN (via postal mail). Please send query responses and delinquent data to ECOG-ACRIN and do not copy the CTSU Data Operations. ECOG-ACRIN accepts either mail or fax for responses. A cover memo indicating the study and data manager (if known) should accompany response. Each site should have a designated CTSU Administrator and Data Administrator and must keep their CTEP AMS account contact information current. This will ensure timely communication between the clinical site and the ECOG-ACRIN data center.

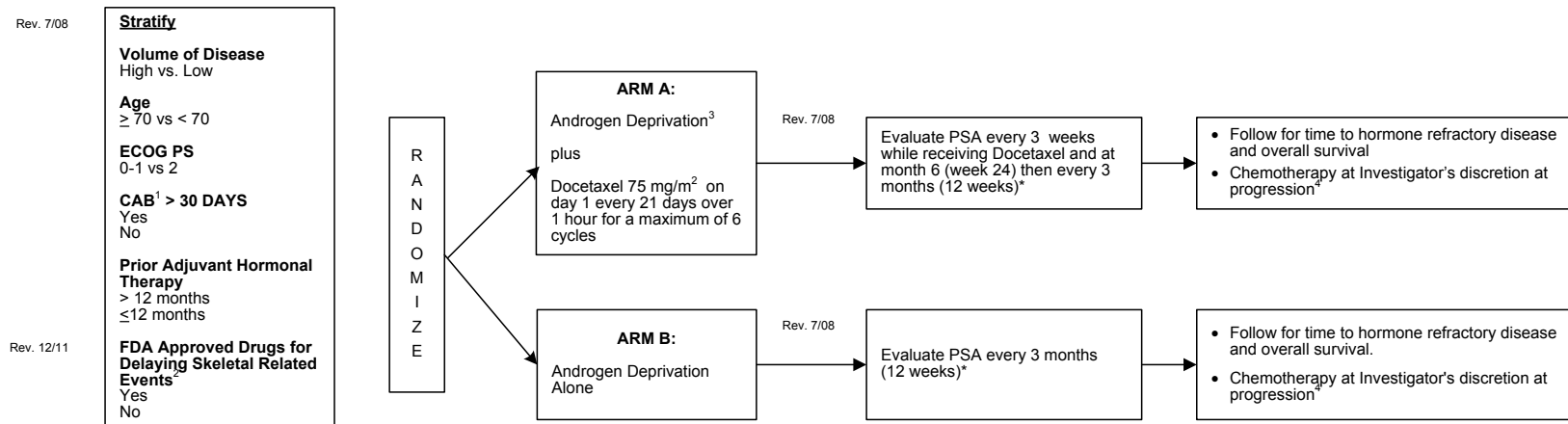
CANCER TRIALS SUPPORT UNIT (CTSU) ADDRESS AND CONTACT INFORMATION

Rev. 2/07

To submit site registration documents:	For patient enrollments:	Submit study data directly to the Lead Cooperative Group unless otherwise specified in the protocol:
CTSU Regulatory Office 1818 Market Street, Suite 1100 Philadelphia, PA 19103 Phone - 1-888-823-5923 Fax – 215-569-0206	CTSU Patient Registration Voice Mail – 1-888-462-3009 Fax – 1-888-691-8039 Hours: 8:00 AM – 8:00 PM Eastern Time, Monday – Friday (excluding holidays) [For CTSU patient enrollments that must be completed within approximately one hour, or other extenuating circumstances, call 301-704-2376. Please use the 1-888-462-3009 number for ALL other CTSU patient enrollments.]	ECOG-ACRIN Operations Office – Boston, FSTRF, 900 Commonwealth Avenue, Boston, MA 02215 (ATTN: DATA). Phone # 617-632-3610 Fax # 617-632-2990 Data should be sent via postal mail (preferred), however fax is accepted. Do not submit study data or forms to CTSU Data Operations. Do not copy the CTSU on data submissions.
For patient eligibility or treatment-related questions: Contact the Study PI of the Coordinating Group.		
For questions unrelated to patient eligibility, treatment, or data submission contact the CTSU Help Desk by phone or e-mail: CTSU General Information Line – 1-888-823-5923, or ctsucontact@westat.com. All calls and correspondence will be triaged to the appropriate CTSU representative.		
The CTSU Public Web site is located at: www.ctsu.org The CTSU Registered Member Web site is located at https://members.ctsu.org		

CTSU logistical information is located in [Appendix IV](#).

Schema



1 cycle = 21 days

Rev. 7/08, 12/11 Accrual goal: 780

Rev. 7/08 * For Arm A patients, PSA will be checked more frequently (once every 3 weeks starting day 1 of cycle 2). Arm A patients will then have PSA checked every 3 months (12 weeks) once off-treatment. If PSA test rises > 50% of lowest level recorded since starting hormonal therapy, progression will need to be confirmed by a repeat PSA test at least 2 weeks later.

Volume of Disease:

- Visceral metastases (extranodal) AND/OR

Rev. 2/07

- Bone Metastases

Rev. 5/14

- At least 4 or more bone lesions

Rev. 5/14

- One of which must be outside of the vertebral column AND pelvis.

Rev. 2/10

1. **CAB:** Combined Androgen Blockade. Investigators are not obliged to use CAB but must declare whether it will be given or continued beyond testosterone surge with LHRH agonists (i.e. 30 days) when starting LHRH agonist.

Rev. 12/11

2. FDA approved drug for delaying skeletal related events: Yes vs No

FDA approved for delaying skeletal related events: Doses and schedule of drugs shown to delay skeletal related events (eg zoledronic acid or denosumab every 3 to 4 week). Patients can not simultaneously enroll on CALGB90202 due to potential to confound endpoints of both studies and impacting stratification. (Please see Section 5.4.2 for Supportive Care Guidelines)

Rev. 7/08

3. **Hormonal Therapy:** Combined Androgen Blockade or monotherapy as surgical or medical castration at investigator's discretion. Investigators are reminded to use combined androgen blockade for four weeks if starting LHRH agonist therapy in men with conditions that could be made worse by an initial rise in testosterone (e.g.: impending spinal cord compression.) Hormonal therapy can be commenced up to 120 days prior to randomization.

Rev. 7/08

Premedication is required to decrease or prevent acute anaphylactoid reactions to docetaxel, and to decrease the severity or delay the onset of late-occurring fluid retention problems. Dexamethasone 8 mg po approximately 12, 3 and 1 hour prior to docetaxel infusion. Use of diphenhydramine is optional.

Docetaxel: 75 mg/m² of docetaxel will be given on day 1 every 21 days.

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4. Type of therapy at time of hormone refractory disease is not being dictated but investigators are encouraged to treat with docetaxel every three weeks on BOTH ARMS so the effect of early versus delayed chemotherapy on overall survival can be assessed. If the investigator deems that the patient is not a suitable candidate for docetaxel chemotherapy, the reason for this decision will be recorded. All therapy for hormone refractory disease will be recorded, including anti-androgen withdrawal, second-line hormonal therapy and chemotherapy. If a patient has more than one therapy, then all therapies will be recorded. Patients will be followed until development of HRPD and for overall survival. Investigators are encouraged to continue androgen deprivation therapy at time of progression to HRPD. If however, the investigator does not feel it is in the patient's best interest, then androgen deprivation can be discontinued but must be documented in study chart.

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NOTE: Docetaxel is a commercially marketed product which has been approved by the FDA for use in some cancers. However its use in patients eligible for this protocol with prostate cancer is not currently approved by the FDA. When used as directed by this protocol, docetaxel is classified as an "unapproved use of an approved agent" and, by definition, considered as an investigational agent. After patient randomization, a supply of docetaxel may be obtained. Investigators must fax a completed "E3805 Clinical Supplies Shipping Form" (Appendix VII) to the ECOG Coordinating Center, ATTN: Drug Team at 617-632-2063.

1. Introduction

Rev. 7/08

1.1 Background

Prostate cancer afflicts approximately 230,000 men per year and results in approximately 30,000 deaths (1). Although the PSA (prostate specific antigen) test has been able to diagnose prostate cancer at an earlier stage, it is unclear whether the overall mortality has decreased. The current treatment for hormone naive metastatic prostate cancer is hormone ablation either by LHRH analogue therapy or orchiectomy as monotherapy or in combination with an antiandrogen. Survival varies depending on the extent of disease at commencement of therapy. With the advent of the PSA test many patients are commenced on hormonal therapy at a very early stage and subjected to the long-term effects of androgen ablation including osteoporosis. However, if patients with an asymptomatic rising PSA after definitive local therapy are observed until they develop overt metastatic disease (i.e. evident by imaging techniques), the median time from PSA relapse to clinical progression is approximately 8 years (2). In the pre-PSA era studies relied upon bone scan and CT scans to document the presence of metastatic disease. The median overall survival for men commencing androgen ablation with clinically evident metastatic disease (i.e. not PSA only disease) is about 30 months (3). This information is derived from the meta-analysis of 27 randomized trials with 8,275 men comparing castration (medical or surgery) alone against combined androgen blockade (3). Currently, patients with overt metastases are placed solely on androgen ablation and followed until symptomatic progression. Most men are continued on androgen deprivation indefinitely. Once hormonal therapy is no longer effective, chemotherapy is employed with palliative intent and can prolong overall survival slightly. The purpose of this proposed trial is to determine whether instituting chemotherapy when starting hormonal therapy can delay the time to progression to a clinically meaningful degree without effecting quality of life in men with metastatic (extensive) disease. Patients with metastatic disease are classified into either high volume disease (i.e., metastases involving appendicular skeleton (including ribs and clavicle) with or without axial involvement and/or visceral metastases or low volume metastatic disease. Patients with high volume metastatic disease have a poorer prognosis with a median time to PSA progression of about only 10 months and median time to clinical progression (e.g. worsening bone metastases) of about 14 months. This was seen in the previous Intergroup phase III trial which accrued 1,387 patients in five years of which 75% had extensive disease (4). This was also observed in a second Intergroup trial in the same population (5). In contrast patients with low volume metastatic disease have a 22 month median time to PSA progression with androgen ablation alone and median time to clinical progression of more than three years. The short time to clinical progression also results in men with high volume disease having a median overall survival of only 24 months compared to about four years for patients with minimal disease. Clearly, men with extensive disease are in urgent need of more effective therapies.

Until recently no therapy for hormone refractory prostate cancer (HRPC) had been shown to confer a survival advantage although numerous approaches improved the quality of life of these patients. These include radio-pharmaceuticals (strontium, samarium) (6), suramin (7), and low dose corticosteroids (8). Mitoxantrone combined with a low dose corticosteroid was

the standard treatment in the 1990's based on two randomized studies that compared this combination to the corticosteroid alone (9, 10). In both studies, there was a palliative benefit as evidenced by a decrease in pain or opiate requirements but no survival benefit for patients who received mitoxantrone.

Recently, two large phase III trials showed a survival benefit for men with hormone refractory prostate cancer treated with a docetaxel-based regimen (11, 12). One is the SWOG 9916 trial of docetaxel plus estramustine versus mitoxantrone plus prednisone. Docetaxel was dosed at 60 mg/m² every 3 weeks plus 5 days of oral estramustine. The lower docetaxel dose was to minimize toxicities seen at 70 mg/m². Patients were escalated from 60 mg/m² to 70 mg/m² if no grade 3 or 4 toxicities occurred in the first cycle. The other is a multinational trial (TAX327) of single agent docetaxel given either weekly (30 mg/m²) or every three weeks (75 mg/m²) versus mitoxantrone plus prednisone. In the latter study prednisone 5 mg twice per day was dosed continuously. In both studies every three week docetaxel was more effective in terms of serological response and overall survival. The mitoxantrone plus prednisone arm in the SWOG study had a PSA response of 27% versus 50% with every three week docetaxel plus estramustine (p<0.001) and the corresponding result in the TAX327 trial for the mitoxantrone plus prednisone arm was 32% versus 45% in the every 3 week docetaxel arm (P<0.001). The overall survival also favored docetaxel based therapy with the median survival for mitoxantrone plus prednisone versus every three week docetaxel being 16 versus 18 months in the SWOG study (p=0.008) and 16.5 versus 18.9 months (p=0.009) in TAX327. Docetaxel every three weeks without estramustine, is therefore the optimal regimen. The weekly regimen did not provide any significant benefits.

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This trial aims to evaluate whether early use of active chemotherapy (docetaxel) in men with metastatic disease in combination with hormonal therapy can delay the time to development of progression/hormone refractory disease and ultimately increase overall survival. Given the fact patients are going to be starting hormonal therapy (androgen ablation) daily prednisone will not be given with the docetaxel in this protocol and will thus minimize the toxicity of chronic low dose steroid exposure. This is expected to make the regimen better tolerated for the 18 weeks it is planned to be administered and will remove this as a confounding variable.

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A theory in support of why early chemotherapy may be more beneficial when starting hormonal therapy is that chemotherapy may be more effective on small volume disease that has not been eradicated by androgen deprivation. This has the potential to delay the time to progression and hence prolong disease control. An argument against combining chemotherapy with hormonal therapy is that the androgen deprivation takes the cells out of cycle and may cause the chemotherapy to be less effective. A randomized phase III trial is required to answer this question. A quality of life analysis will be incorporated to determine whether the postulated clinically meaningful increase in disease control is not associated with an overall decrement in quality of life. A clinically significant increase in disease control is considered to be an increase in the median overall survival from 24 to 32 months in men with high volume metastatic disease and 48 to 62 months with low volume metastatic prostate cancer. This is considered to be of a magnitude large enough that would alter the current standard of care if quality of life was not impaired.

1.2 Rationale for selected approach and trial design

In summary, docetaxel based chemotherapy has shown significant activity in hormone refractory prostate cancer. The therapy to be taken forward to the phase III trial of patients in patients STARTING hormonal therapy has been chosen based on the results of the phase III trials in the hormone refractory setting. A randomized phase III trial is required to definitively answer the question of whether early chemotherapy plus androgen ablation can significantly increase the duration of cancer control versus androgen ablation alone.

1.3 Supporting Preliminary Data

To date, only one trial has been completed and analyzed addressing the question of chemohormonal therapy as primary therapy for metastatic hormone sensitive prostate cancer. The MDACC group, lead by Dr. Millikan compared androgen deprivation to androgen deprivation with ketoconazole, adriamycin, vinblastine and estramustine (KAVE). It is reported that the median time to PSA progression in patients with high volume metastases was 10 months in the 64 patients with androgen deprivation alone and 19 months in those with chemohormonal therapy. For the 50 patients with low volume metastases, there was no improvement in time to PSA progression as it was 22 months in both arms with about 24 patients per arm. At the time of this analysis there was no improvement in overall survival in either group (Personal Communication: Randall Millikan).

From this study we at least know that two relatively ineffective chemotherapy agents (adriamycin and vinblastine) prolonged time to PSA progression in patients with high volume metastatic burden. Docetaxel holds even more promise based on the data showing improvement over mitoxantrone plus prednisone in HRPC. Another theoretical advantage for using docetaxel is that it can abrogate BCL-2's anti-apoptotic properties (13) which has been shown to be involved in the emergence of androgen insensitive clones (14-18). It has also been reported that retreatment after a "drug holiday" with docetaxel, if no progression on docetaxel, can result in significant responses (19). Thus, retreatment with docetaxel at the time of progression has the possibility of compounding its own clinical benefit by adding to the total time of disease control when re-instituted when a patient is hormone refractory.

1.4 Rationale for Correlative Studies

In addition to asking the question of whether chemotherapy at the time of starting hormonal therapy prolongs survival, this clinical trial provides a unique opportunity to collect specimens (tumor specimens and blood) that will be annotated with clinical information. The intent of the genetic studies will be to identify genes in the patients' germline DNA and/or tumor that predict response and toxicity. The sequential blood analyses will allow for proteins to be studied longitudinally and thus further facilitate potential biomarker discovery.

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2. Objectives

2.1 Primary Objective

- 2.1.1 To evaluate the ability of early chemotherapy to improve overall survival in men commencing androgen deprivation for metastatic prostate cancer.

2.2 Secondary Objectives

- 2.2.1 To determine whether early chemotherapy can increase the time to clinical progression (radiographic or symptomatic deterioration due to disease) over hormonal therapy alone.
- 2.2.2 To determine whether early chemotherapy can increase the time to development of hormone refractory disease over hormonal therapy alone.
- 2.2.3 To determine whether early chemotherapy can increase time to serological progression over hormonal therapy alone.
- 2.2.4 To determine rates of biochemical response at 6 months and 12 months in the chemohormonal arm versus the hormonal therapy alone arm.
- 2.2.5 To determine the frequency of adverse events and the tolerability of chemotherapy combined with hormonal therapy versus hormonal therapy alone.
- 2.2.6 To determine whether the postulated clinically meaningful increase in disease control is associated with an alteration in overall quality of life using the Functional Assessment of Cancer Therapy–Prostate questionnaire (FACT-P self-administered).
- 2.2.7 To determine the ability of PSA changes to be a surrogate for clinical benefit from therapy and overall survival.

2.3 Tertiary Objectives

- 2.3.1 To determine if there are proteins differentially translated from the genome in hormone sensitive prostate cancer, prostate cancer that has responded to hormonal therapy and hormone refractory prostate cancer.
- 2.3.2 To determine the frequency of constitutive polymorphisms of enzymes involved in steroid metabolism and other carcinogenic processes.
- 2.3.3 To determine if the amount and frequency of certain carcinogenic proteins in prostate cancer tissue, such as CXCR4 and Manganese Superoxide Dismutase, can be correlated with a poor prognosis.

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3. Selection of Patients

Each of the criteria in the following section must be met in order for a patient to be considered eligible for this study. Use the spaces provided to confirm a patient's eligibility. For each patient, this section should be photocopied, completed and maintained in the patient's chart.

In calculating days of tests and measurements, the day a test or measurement is done is considered Day 0. Therefore, if a test is done on a Monday, the Monday four weeks later would be considered Day 28.

ECOG-ACRIN Patient No. _____

Patient's Initials (L, F, M) _____

NOTE: All questions regarding eligibility should be directed to the ECOG-ACRIN Operations Office – Boston at (617) 632-3610.

NOTE: Institutions may use the eligibility checklist as source documentation if it has been reviewed, signed, and dated prior to randomization by the treating physician.

3.1 Eligibility Criteria

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Rev. 7/08

- _____ 3.1.1 Patients must have histologically or cytologically confirmed prostate cancer. Patients may have begun hormonal therapy but it must not have commenced more than 120 days prior to randomization.

Rev. 7/08

Rarely pathology is not available but if clinical situation confirms prostate cancer (such as response to androgen ablation) pathology is not required and patient can be enrolled after discussed with study chair.

Rev. 2/07, 7/08

- _____ 3.1.2 Patients must have metastatic disease and will be stratified by presence of high volume or or low volume disease. Low volume disease is defined as any metastatic disease that is not extensive. High volume disease is defined as:

Rev. 6/14

- Visceral Metastases (extranodal)
- AND/OR

Rev. 6/14

- Bone Metastases
 - At least 4 or more bone lesions
 - One of which must be outside of the vertebral column AND pelvis.

Rev. 6/14

NOTE: Radiological studies identifying measurable or non-measurable disease must be obtained according to the following criteria:

- If androgen deprivation therapy has not commenced:
 - Scans must be obtained within 6 weeks of randomization
- If androgen deprivation has commenced prior to randomization:

- Rev. 7/08
- Scans must be obtained within 6 weeks prior to the start of androgen deprivation therapy.
- Rev. 7/08
- If all required imaging had not been completed prior to starting androgen deprivation, then any additional scans must be obtained after starting androgen deprivation but prior to randomization. (It is assumed the scans of patients with high volume disease would not normalize in less than 120 days to the point a patient would go from "high volume" to "low volume.")
- Rev. 7/08
12/11
- _____ 3.1.3 Patients are not eligible if the PSA has risen and met criteria for progression as defined in Section [6.3.1](#) from its lowest point between the beginning of androgen deprivation therapy and the date of randomization.
- _____ 3.1.4 Patients must have adequate organ function within 4 weeks prior to randomization and evidenced by:
- _____ 3.1.4.1 Absolute Neutrophil Count $\geq 1500/\text{mm}^3$
ANC: _____ Date of test: _____
- _____ 3.1.4.2 Platelet count $\geq 100,000/\text{mm}^3$
Platelet count: _____ Date of test: _____
- _____ 3.1.4.3 Total bilirubin $\leq$ ULN
Total bilirubin: _____ Date of test: _____
- Rev. 7/08
Rev. 7/08
- _____ 3.1.4.4 ALT $\leq 2.5 \times$ upper limit of normal
ALT: _____ Date of test: _____
ULN: _____ Date of test: _____
- Rev. 7/08
- _____ 3.1.4.5 Creatinine clearance of $\geq 30 \text{ mL/min}$. Creatinine clearance (CrCl) should be calculated at screening using the Cockcroft-Gault formula:
Creatinine clearance for males (mL/min) = $(140 - \text{age})(\text{body weight in kg})/[72 \times (\text{serum creatinine in mg/dl})]$
Creatinine clearance: _____ Date of test: _____
- Rev. 7/08
- _____ 3.1.4.6 PT, INR $\leq 1.5 \times$ ULN (except if on therapeutic anti-coagulation in which case the patient can be enrolled if stable and anti-coagulation levels are appropriate for their condition per good clinical practice).
PT: _____ Date of test: _____
INR: _____ Date of test: _____
ULN: _____ Date of test: _____

- Rev. 7/08 _____ 3.1.4.7 PTT $\leq$ 1.5 x ULN (except if on therapeutic anti-coagulation in which case the patient can be enrolled if stable and anti-coagulation levels are appropriate for their condition per good clinical practice).
PTT: _____ Date of test: _____
ULN: _____ Date of test: _____
- Rev. 7/08 _____ 3.1.4.8 Patients can enter the study if on therapeutic anti-coagulation.
NOTE: All values must be obtained within 4 weeks prior to beginning protocol therapy.
- Rev. 7/08 _____ 3.1.5 If a patient has had major surgery, the patient must be longer than 4 weeks post major surgery and recovered from all toxicity prior to randomization.
- Rev. 7/08 _____ 3.1.6 Patients must have discontinued hormonal therapy in the adjuvant and/or neoadjuvant setting 12 months prior to beginning protocol therapy, AND must not have exceeded 24 months of therapy AND have shown to have no evidence of disease (PSA < 0.1 ng/dL after prostatectomy plus hormonal therapy and < 0.5 ng/dL and not have doubled above nadir after radiation therapy plus hormonal therapy) at least 12 months after completing adjuvant or neoadjuvant hormonal therapy. Patients with prior chemotherapy in the adjuvant or neoadjuvant setting are ineligible. The last depot injection must have expired by the time the 24 month mark was reached.
(e.g., patient completed 24 months of adjuvant therapy with 8 three month depot LHRH agonists injections – last dose given 8/03 – 24 months completed 11/03. Patient is eligible if no evidence of disease at or after 11/04 and commenced hormonal therapy for metastatic disease on or after 11/04).
- Rev. 2/10
- Rev. 7/08 _____ 3.1.7 Patients must have ECOG performance status of 0- 2. (NOTE: Patients with PS 2 are only eligible if the decline in PS is due to metastatic prostate cancer).
- _____ 3.1.8 Patients must be at least 18 years of age.
- Rev. 12/11 _____ 3.1.9 Patients must not have participated in another clinical trial within 30 days prior to randomization. Patients may participate in non-therapeutic trials. Patients can not simultaneously enroll on CALGB 90202 due to the potential to confound endpoints of both studies and impact the stratification.
- _____ 3.1.10 Patients must not have prior history of malignancy in the past 5 years with the exception of basal cell and squamous cell carcinoma of the skin. Other malignancies that are considered to have a low potential to progress (eg: grade 2, T1A TCC) may be enrolled if approved by study chair.
- Rev. 7/08 _____ 3.1.11 Peripheral neuropathy must be $\leq$ grade 1.

- _____ 3.1.12 Patients with a history of severe hypersensitivity reaction to Docetaxel® or other drugs formulated with polysorbate 80 must be excluded.
- _____ 3.1.13 No active cardiac disease defined as active angina, symptomatic congestive heart failure, or myocardial infarction within previous six months.
- _____ 3.1.14 Patients may be enrolled if they have had prior palliative radiation therapy. However, this has to have been commenced within 30 days of starting androgen deprivation.
- _____ 3.1.15 Patients with prior hormone therapy in the metastatic setting are not eligible.

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4. Randomization Procedures

Submitting Regulatory Documents

Before an ECOG-ACRIN Institution may enter patients, protocol specific regulatory documents must be submitted to the CTSU Regulatory Office at the following address:

CTSU Regulatory Office
Coalition of National Cancer Cooperative Groups
1818 Market Street, Suite 1100
Philadelphia, PA 19103
FAX: (215) 569-0206

Required Protocol Specific Regulatory Documents

1. CTSU Regulatory Transmittal Form.
2. Copy of IRB Informed Consent Document.

NOTE: Any deletion or substantive modification of information concerning risks or alternative procedures contained in the sample informed consent document must be justified in writing by the investigator and approved by the IRB.

3. A. CTSU IRB Certification Form.

Or

- B. HHS 310 Form.

Or

- C. IRB Approval Letter

NOTE: The above submissions must include the following details:

- Indicate all sites approved for the protocol under an assurance number.
- OHRP assurance number of reviewing IRB
- Full protocol title and number
- Version Date
- Type of review (full board vs. expedited)
- Date of review.
- Signature of IRB official

The CTSU encourages you to link to the following RSS2.0 webpage so that more information on RSS2.0 as well as the submission forms can be accessed http://www.ctsu.org/rss2_page.asp. If you have questions regarding regulatory document submission, please telephone the CTSU Help Desk at 1-888-823-5923 or E-mail CTSUContact@westat.com.

Patients must not start protocol treatment prior to randomization.

Treatment should start within seven working days after randomization.

NOTE: Please refer to [Appendix IV](#) for CTSU registration guidelines.

Institutions may register eligible patients to this study via the ECOG webpage 24 hours a day, 7 days a week, using the Web-based Patient Registration Program

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(<https://webreg.ecog.org>). If you need assistance or have questions, please telephone the Central Randomization Desk at the ECOG-ACRIN Operations Office – Boston at (617) 632-2022, Monday through Friday 9:00am – 5:00pm Eastern Time. Please note that a password is required to use this program. The following information will be requested:

4.1 Protocol Number

4.2 Investigator Identification

4.2.1 Institution and affiliate name

4.2.2 Investigator's name

4.3 Patient Identification

4.3.1 Patient's initials and chart number

4.3.2 Patient's Social Security number

4.3.3 Patient demographics

4.3.3.1 Sex

4.3.3.2 Birth date (mm/yyyy)

4.3.3.3 Race

4.3.3.4 Ethnicity

4.3.3.5 Nine-digit ZIP code

4.3.3.6 Method of payment

4.4 Eligibility Verification

Patients must meet all of the eligibility requirements listed in Section [3](#). An eligibility checklist has been appended to the protocol. A confirmation of registration will be forwarded by the ECOG-ACRIN Operations Office – Boston.

4.5 Stratification Factors

4.5.1 Age

4.5.1.1 ≥ 70

4.5.1.2 < 70

4.5.2 ECOG Performance Status

4.5.2.1 PS 0-1

4.5.2.2 PS 2

4.5.3 CAB > 30 Days

4.5.3.1 Yes

4.5.3.2 No

4.5.4 Prior Adjuvant Hormonal Therapy

4.5.4.1 > 12 months

4.5.4.2 ≤ 12 months

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4.5.5 FDA Approved Drugs for Delaying Skeletal Related Events

4.5.5.1 Yes

4.5.5.2 No

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4.5.6 Volume of Disease

4.5.6.1 High

4.5.6.2 Low

4.6 Additional Requirements

4.6.1 Patients must provide a signed and dated, written informed consent form.

4.6.2 Correlative samples should be submitted for analysis as outlined in Sections [10](#) and [11](#).

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NOTE: ECOG-ACRIN requires that all biological samples submitted from patients participating in this trial be entered and tracked via the online ECOG-ACRIN Sample Tracking System (STS). See Section [10.4](#).

4.7 Instructions for Patients who Do Not Start Assigned Protocol Treatment

If a patient does not receive any assigned protocol treatment, baseline and follow-up data will still be collected and must be submitted according to the instructions in the E3805 Forms Packet. Document the reason for not starting protocol treatment on the Off Treatment Form. Also report the date and type of the first non-protocol treatment that the patient receives.

5. Treatment Plan

5.1 Administration Schedule

5.1.1 Arm A: Androgen Deprivation plus Docetaxel

5.1.1.1 Androgen Deprivation Therapy

Options for androgen-deprivation (i.e., hormone ablation) therapy include the following:

- LHRH agonist or antagonist therapy
- Surgical castration

NOTE: Patients can be randomized to the study within 7 working days before beginning docetaxel and may begin androgen-deprivation therapy up to 120 days before randomization.

Use of combined androgen blockade (medical or surgical castration combined with antiandrogen treatment) is to be instituted at the investigator's discretion.

Antiandrogens (e.g., bicalutamide or flutamide) may be used in addition to androgen-deprivation therapy, but may not be used as the sole hormonal therapy. Anti-androgens combined with 5-alpha reductase inhibitors are also not allowed as sole therapy. An investigator may administer this class of drug as a single agent ≤ 28 days before medical castration to cover the testosterone surge associated with LHRH agonists. *The 120-day window of androgen-deprivation or hormonal therapy starts with commencement of either the antiandrogen or LHRH therapy.* An agent such as ketoconazole may be used in addition if urgent control of cancer is required but must be discontinued prior to randomization.

5.1.1.2 Docetaxel

75 mg/m² of docetaxel will be given on day 1 every 21 days over one hour for up to 6 cycles (1 cycle = 21 days).

NOTE: Docetaxel is a commercially marketed product which has been approved by the FDA for use in some cancers. However its use in patients eligible for this protocol with prostate cancer is not currently approved by the FDA. When used as directed by this protocol, docetaxel is classified as an "unapproved use of an approved agent" and, by definition, considered as an investigational agent. After patient randomization, a supply of docetaxel may be obtained. Investigators must fax a completed "E3805 Clinical Supplies Shipping Form" ([Appendix VII](#)) to the ECOG-ACRIN Operations

Office – Boston, ATTN: Drug Team at 617-632-2063.

5.1.1.3 Premedication for Docetaxel

Premedication is required to decrease or prevent acute anaphylactoid reactions to docetaxel, and to decrease the severity or delay the onset of late-occurring fluid retention problems. Dexamethasone 8 mg po approximately 12, 3 and 1 hour prior to docetaxel infusion. Use of diphenhydramine is optional.

5.1.1.4 Calcium Carbonate

Patients will receive concomitant treatment with calcium carbonate at a dose of at least 500 mg orally per day every evening (any product with amount stated on the bottle and verified by investigator, e.g., TumsTM). Calcium is best absorbed when taken with meals.

5.1.1.5 Vitamin D

Patients will receive concomitant treatment with vitamin D by administration of any multivitamin containing at least 400 IU of vitamin D (any product with amount stated on the bottle and verified by investigator).

For guidance on use of drugs to prevent skeletal related events, please see Section [5.4.2](#) (Supportive Care).

At the time of hormone refractory disease, investigators are encouraged to treat with docetaxel therapy every three weeks ON BOTH ARMS so the effect of early versus delayed chemotherapy on overall survival can be assessed. If the investigator deems that the patient is not a suitable candidate for docetaxel chemotherapy, the reason for this decision will be recorded. All therapy for hormone refractory disease will be recorded including anti-androgen withdrawal, second-line hormonal therapy and chemotherapy. If a patient has more than one therapy, then all therapies will be recorded. Patients will be followed until development of HRPC and for overall survival. Investigators are encouraged to continue androgen deprivation therapy at time of progression to HRPC. If however, the investigator does not feel it is in the patient's best interest, then androgen deprivation can be discontinued but must be documented in the study chart.

5.1.2 Arm B: Androgen Deprivation Alone

5.1.2.1 Androgen Deprivation Therapy

Options for androgen-deprivation (i.e., hormone ablation) therapy include the following:

- LHRH agonist or antagonist therapy
- Surgical castration

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NOTE: Patients may begin androgen-deprivation therapy up to 120 days before randomization.

Use of combined androgen blockade (medical or surgical castration combined with antiandrogen treatment) is to be instituted at the investigator's discretion.

Antiandrogens (e.g., bicalutamide or flutamide) may be used in addition to androgen-deprivation therapy, but may not be used as the sole hormonal therapy. Anti-androgens combined with 5-alpha reductase inhibitors are also not allowed as sole therapy. An investigator may administer this class of drug as a single agent $\leq$ 28 days before medical castration to cover the testosterone surge associated with LHRH agonists. *The 120-day window of androgen-deprivation or hormonal therapy starts with commencement of either the antiandrogen or LHRH therapy.* An agent such as ketoconazole may be used in addition if urgent control of cancer is required but must be discontinued prior to randomization.

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5.1.2.2 Calcium Carbonate

Patients will receive concomitant treatment with calcium carbonate at a dose of at least 500 mg orally per day every evening (any product with amount stated on the bottle and verified by investigator, e.g., TumsTM). Calcium is best absorbed when taken with meals.

5.1.2.3 Vitamin D

Patients will receive concomitant treatment with vitamin D by administration of any multivitamin containing at least 400 IU of vitamin D (any product with amount stated on the bottle and verified by investigator).

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For guidance on use of drugs to prevent skeletal related events, please see Section [5.4.2](#) (Supportive Care).

At the time of hormone refractory disease, investigators are encouraged to treat with docetaxel therapy every three weeks ON BOTH ARMS so the effect of early versus delayed chemotherapy on overall survival can be assessed. If the investigator deems that the patient is not a suitable candidate for docetaxel chemotherapy, the reason for this decision will be recorded. All therapy for hormone refractory disease will be recorded including anti-androgen withdrawal, second-line hormonal therapy and chemotherapy. If a patient has more than one therapy, then all therapies will be recorded. Patients will be followed until development of HRPC and for overall survival. Investigators are encouraged to continue androgen deprivation therapy at time of progression to HRPC. If however, the investigator does not feel it is in the patient's

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best interest, then androgen deprivation can be discontinued but must be documented in the study chart.

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5.1.3

For Arms A and B: Quality of Life (QOL)

Quality of Life will be measured at the following timepoints:

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- Baseline (within 1 week prior to docetaxel therapy on Arm A and within 2 weeks of randomization)
- Week 12 (3 months)
- Week 24 (6 months)
- Week 36 (9 months)
- Week 48 (12 months)

Administration of the four part questionnaire (FACT-P, FACIT-F Fatigue Subscale, FACT-Taxane Subscale and the Brief Pain inventory) will allow us to examine overall QOL, as well as specific treatment-associated and disease-associated changes in QOL. The questionnaire should be administered to patients by staff during visits. Differences between pre-treatment and subsequent scores will be analyzed using appropriate statistical tests.

- 5.1.3.1 The questionnaire must be administered at the timepoints listed above unless the patient refuses. The patient should be instructed to respond to the questionnaire in terms of his experience during the time frame specified on each questionnaire.
- 5.1.3.2 The patient should be asked to read the instructions at the beginning of each questionnaire and complete all of the items. It is permissible to assist the patient with the completion of the questionnaire as long as the staff person does not influence the patient's responses.
- 5.1.3.3 The questionnaire must be reviewed by the protocol nurse or research coordinator as soon as the patient completes them to ensure all items were marked appropriately. If more than one answer was marked, the patient should be asked to choose the answer which best reflects how he is feeling. If a question was not answered, the patient should be asked if he would like to answer it. The patient should always have the option to refuse. If the patient refuses, it should be indicated on the questionnaire that he declined to answer the item.
- 5.1.3.4 Completed questionnaires must be returned to the ECOG-ACRIN Operations Office – Boston. If a questionnaire cannot be completed by the patient, the reason should be noted on the Assessment Compliance Form and the form should be returned to the ECOG-ACRIN Operations Office – Boston.
- 5.1.3.5 If a patient misses an appointment on the scheduled date, the questionnaire may be completed by telephone on the

appointed date or at the time the appointment is rescheduled. If the missed date is on a treatments date, the QOL assessment will be done when the patient comes for the rescheduled treatment.

- 5.1.3.6 If a patient cannot complete the questionnaire because he is too sick, this should be documented on the Assessment Compliance Form.

5.2 Dose Modifications

There will be no dose modifications for androgen deprivation. Intermittent hormonal therapy is not allowed.

For docetaxel, no more than two dose modifications should be allowed for any patient. If a patient requires a third reduction of docetaxel, they will be followed every 12 weeks and treated with androgen deprivation alone. Dose adjustments are to be made according to the system showing the greatest degree of toxicity. **All toxicities should be graded according to the NCI Common Terminology Criteria for Adverse Events (CTCAE) version 4.0. The CTCAE version 4.0 is identified and located on the CTEP website at <http://ctep.cancer.gov>. All appropriate treatment areas should have access to a copy of the CTCAE version 4.0.**

Dose adjustments for toxicity should be made according to the guidelines that follow. If a dose is reduced due to toxicity the dose will not be re-escalated back to starting level. Treatment may be delayed no more than three weeks to allow recovery from toxicity. If treatment must be delayed longer than three weeks from scheduled day of dosing, patient will be treated with androgen deprivation alone.

Dose Level	Docetaxel (mg/m ²)
Level 0	75 mg/m ²
Level - 1	65 mg/m ²
Level - 2	55 mg/m ²

5.2.1 Dose Modifications for Myelosuppression

Dose modifications to be made based on granulocyte and/or platelet count drawn prior to planned treatment (can be done day prior to planned dose):

Docetaxel	Granulocytes/mm ³ Day 1 of treatment	Platelet/mm ³ Day 1 of treatment
No change	> 1,500 ^ψ	or > 100,000
Reduce one dose level #	1,000 to 1,499	or 75,000 to 99,999
Delay dosing one week ^{* +}	<1000	or < 75,000

NOTE: If a dose reduction is made, maintain the lower dose for all subsequent cycles

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- # A need for three dose reductions will require that the patient discontinue docetaxel and continue on androgen ablation alone.
- * If planned day 1 dose must be delayed for three consecutive weeks, discontinue docetaxel and continue on androgen ablation alone.
- + If a dose is held due to myelosuppression, the patient will be retreated with a one level dose reduction once granulocyte count has recovered to at least 1,500/mm³ and platelet count has recovered to at least 100,000/mm³.
- ψ Management of prolonged grade 4 neutropenia and neutropenic fever- In order to maximize dose intensity, patients with afebrile Grade 4 neutropenia ≥ 7 days or Grade ≥ 3 neutropenia associated with fever (one reading of oral temperature > 38.5°C, or three readings of oral temperature >38.0°C in a 24-hour period) should be retreated after recovery to an absolute neutrophil count to 1,500 granulocytes/mm³ but with a one level dose reduction. The fever must have resolved and if an infection is identified, it must be adequately treated and have clinically resolved before restarting therapy. If prior bacteremia, blood cultures must be negative on recheck. Patient can continue with chemotherapy dosing while on antibiotics.

Use of growth factors is not required as the dose and scheduled dose not meet ASCO guidelines. If however, the investigator considers it in patients best interest growth factors can be used per investigator discretion.

5.2.2 Dose Modification for Hepatic Dysfunction

ALT and Bilirubin will be evaluated pre-study and Day 1 (may be evaluated within 24 hours of day 1) of cycles 1-6:

Patients who develop abnormal liver function tests for any reason while on the study will have the following dose reductions:

Dose Modifications for Abnormal Liver Function

Bilirubin		ALT/ SGPT	Action
> ULN	or	> 5 x ULN	Wait ≤ 3 weeks. If recovered*, reduce docetaxel dose by one dose level. If not, discontinue docetaxel.
≤ ULN	and	> 3 x ULN	Treat without delay but reduce docetaxel dose by one dose level

* Recovery is < 3X ULN for ALT/SGPT and WNL for bilirubin. Dose modifications are based on SGPT alone due to confounding effect of bone disease on SGOT levels.

5.2.3 Dose Modification for Stomatitis

If stomatitis ≥ grade 2 is present on day 1 of any cycle, docetaxel should be held until stomatitis has resolved. If Grade 3/4 stomatitis occurs at any time, the dose of docetaxel will be reduced one dose level for all subsequent doses. If a second Grade 3/4 stomatitis event is incurred, docetaxel will be reduced one more dose level. If a third Grade 3/4 stomatitis event occurs, the patient will be taken off study.

- 5.2.4 Dose Modification for Peripheral Neuropathy
- If $\geq$ Grade 3, the patient will discontinue docetaxel.
- If Grade 2, the Docetaxel will be held and the patient should be retreated upon recovery to a $\leq$ Grade 1 toxicity with a dose reduction of Docetaxel by one level.
- If Grade 2 or greater neurotoxicity persists for more than 3 weeks, the patient will discontinue docetaxel.
- 5.2.5 Hypersensitivity Reactions for Docetaxel
- See [Appendix V](#) for treatment of hypersensitivity reactions. Treatment should be discontinued for Grade 4 hypersensitivity reactions. There are no dose reductions for hypersensitivity reactions.
- Grade 4 Hypersensitivity is defined as a reaction that is life threatening and requires pressor and/or ventilator support or shock associated with acidemia and impairing vital organ function due to tissue hypoperfusion.
- Patients with two episodes of Grade 3 Hypersensitivity reactions or one Grade 4 Hypersensitivity reaction are to discontinue docetaxel.
- 5.2.6 Diarrhea
- If patients experience $>$ grade 2 diarrhea and concurrent grade 3 or 4 neutropenia, hold Docetaxel until $ANC > 1000/mm^3$ and diarrhea $\leq$ grade 2.
- If patients experience significant diarrhea (>3 loose stools/24hrs over baseline), they should be treated prophylactically in subsequent cycles with loperamide or diphenoxylate. If patient experiences $>$ grade 2 diarrhea despite prophylaxis, Docetaxel should be reduced one dose level. If patients experience $>$ grade 2 diarrhea despite prophylaxis AND dose reduction, they should discontinue docetaxel.
- 5.2.7 Delay of Therapy:
- If docetaxel has to be delayed for more than 3 weeks from planned day of dosing because of any toxicity, then the patient is to be treated with androgen deprivation alone.
- 5.2.8 Other Toxic Effects Thought to be related to docetaxel:
- If toxicities $\leq$ Grade 2, manage the subject symptomatically if possible, and retreat without dose reduction.
- If toxicities $\geq$ Grade 3 and clinically significant (not mentioned above), docetaxel should be withheld (except for anemia as patients can be transfused) until resolution to $\leq$ Grade 1 or baseline and patients treated with a one dose level reduction.

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5.3 Adverse Event Reporting Requirements

5.3.1 Purpose

Adverse event data collection and reporting, which are required as part of every clinical trial, are done to ensure the safety of patients enrolled in the studies as well as those who will enroll in future studies using similar agents. Adverse events are reported in a routine manner at scheduled times during a trial (please refer to the E3805 Forms Packet for the list of forms with directions for routine adverse event reporting). Additionally, certain adverse events must be reported in an expedited manner for more timely monitoring of patient safety and care. The following sections provide information about expedited reporting.

5.3.2 Determination of reporting requirements

Reporting requirements may include the following considerations: 1) whether the patient has received an investigational or commercial agent; 2) the characteristics of the adverse event including the grade (severity), the relationship to the study therapy (attribution), and the prior experience (expectedness) of the adverse event; 3) the phase (1, 2, or 3) of the trial; and 4) whether or not hospitalization or prolongation of hospitalization was associated with the event.

Commercial agents are those agents not provided under an IND but obtained instead from a commercial source. The NCI, rather than a commercial distributor, may on some occasions distribute commercial agents for a trial.

Steps to determine if an adverse event is to be reported in an expedited manner:

Step 1: *Identify the type of event using the NCI Common Terminology Criteria for Adverse Events (CTCAE) Version 4.0.* The CTCAE provides descriptive terminology and a grading scale for each adverse event listed. A copy of the CTCAE can be downloaded from the CTEP home page (<http://ctep.cancer.gov>). Additionally, if assistance is needed, the NCI has an Index to the CTCAE that provides help for classifying and locating terms. All appropriate treatment locations should have access to a copy of the CTCAE.

Step 2: *Grade the event using the NCI CTCAE.*

Step 3: *Determine whether the adverse event is related to the protocol therapy (investigational or commercial).* Attribution categories are as follows: Unrelated, Unlikely, Possible, Probable, and Definite.

Step 4: *Determine the prior experience of the adverse event.* *Expected* events are those that have been previously identified as resulting from administration of the agent. An adverse event is considered *unexpected*, for expedited reporting purposes only, when either the type of event or the severity of the event is **NOT** listed in:

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- **Arm A and B** – the drug package insert or protocol

Step 5: Review Section [5.3.6](#) for E3805 and/or ECOG-ACRIN specific requirements for expedited reporting of specific adverse events that require special monitoring.

NOTE: For general questions regarding expedited reporting requirements, please contact the AEMD Help Desk at aemd@tech-res.com or 301-897-7497.

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5.3.3 Reporting methods

- **Arm A and B** – This study requires that expedited adverse event reporting use CTEP's Adverse Event Reporting System (CTEP-AERS). CTEP's guidelines for CTEP-AERS can be found at <http://ctep.cancer.gov>. For questions regarding the use of the CTEP-AERS application, please contact the NCI Technical Help Desk at ncictephelp@ctep.nci.nih.gov or by phone at 1-888-283-7457.

A CTEP-AERS report must be submitted to ECOG-ACRIN and the appropriate regulatory agencies by one of the following methods:

- Electronically submit the report via the CTEP-AERS Web-based application located at <http://ctep.cancer.gov>
or
- Fax the completed NCI Adverse Event Expedited Report – Single Agent or Multiple Agents paper template located at <http://ctep.cancer.gov> to ECOG-ACRIN (617-632-2990), Attention: AE and the FDA (800-332-0178).

NOTE: Paper copies of CTEP-AERS reports will only be accepted if the CTEP-AERS system is down. Once the system is restored, a report submitted on a paper template must be entered into the CTEP-AERS system by the original submitter of the report at the site.

Any supporting or follow up documentation must be faxed to ECOG-ACRIN (617-632-2990), Attention: AE. In addition, supporting or follow up documentation must be faxed to the FDA (800-332-0178).

5.3.4 When to report an event in an expedited manner

When an adverse event requires expedited reporting, submit a full CTEP-AERS report within the timeframes outlined in Section [5.3.6](#).

NOTE: Adverse events that meet the reporting requirements in Section [5.3.6](#) and occur within 30 days of the last dose of protocol treatment must be reported on an expedited adverse event report form (using CTEP-AERS). For any adverse events that occur more than 30 days after the last dose of treatment, only those that have an attribution of possibly, probably, or definitely AND meet the reporting requirements in Section [5.3.6](#) must be reported on an expedited adverse event report form (using CTEP-AERS).

5.3.5 Other recipients of adverse event reports

ECOG-ACRIN will forward CTEP-AERS reports to the appropriate regulatory agencies and pharmaceutical company, if applicable.

Adverse events determined to be reportable must also be reported by the institution, according to the local policy and procedures, to the Institutional Review Board responsible for oversight of the patient.

5.3.6 Expedited reporting for commercial agents

Commercial reporting requirements are provided below. The commercial agent used in arms A and B of this study is Docetaxel.

Expedited reporting requirements for adverse events experienced by patients on arm(s) with commercial agents only – Arms A and B

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Attribution	Grade 4		Grade 5 ^a		ECOG-ACRIN and Protocol-Specific Requirements
	Unexpected	Expected	Unexpected	Expected	See footnote (b) for special requirements.
Unrelated or Unlikely			7 calendar days	7 calendar days	
Possible, Probable, Definite	7 calendar days		7 calendar days	7 calendar days	

7 Calendar Days: Indicates a full CTEP-AERS report is to be submitted within 7 calendar days of learning of the event.

a This includes all deaths within 30 days of the last dose of treatment regardless of attribution.

NOTE: Any death that occurs > 30 days after the last dose of treatment and is attributed possibly, probably, or definitely to the treatment must be reported within 7 calendar days of learning of the event.

b Protocol-specific expedited reporting requirements: The adverse events listed below also require expedited reporting for this trial:

Serious Events: Any event following treatment that results in persistent or significant disabilities/incapacities, congenital anomalies, or birth defects must be reported via CTEP-AERS within 7 calendar days of learning of the event. For instructions on how to specifically report these events via CTEP-AERS, please contact the NCI Medical Help Desk at 301-897-7497.

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5.3.7 Reporting secondary Primary Cancers

All cases of second primary cancers (defined as a cancer caused by any treatment for a previous malignancy) including acute myeloid leukemia (AML) and myelodysplastic syndrome (MDS), that occur in patients on NCI-sponsored trials during or following their chemotherapy for cancer must be reported to ECOG-ACRIN. Report the following information following a diagnosis of a secondary primary cancer:

- **For secondary primary cancers that are NOT related to any anti-cancer treatment (including the treatment on this protocol):**

1. Submit a completed Second Primary Form within 30 days to ECOG-ACRIN at
ECOG-ACRIN Operations Office – Boston
FSTRF
900 Commonwealth Avenue
Boston, MA 02215
 2. Submit a copy of the pathology report to ECOG-ACRIN confirming the diagnosis.
 3. If the patient has been diagnosed with AML/MDS, submit a copy of the cytogenetics report (if available) to ECOG-ACRIN.
 - **For secondary primary cancers that ARE related to any anti-cancer treatment (including the treatment on this protocol):**
 1. Submit a completed Second Primary Form within 30 days to ECOG-ACRIN at
ECOG-ACRIN Operations Office – Boston
FSTRF
900 Commonwealth Avenue
Boston, MA 02215
 2. Report the diagnosis via CTEP-AERS at <http://ctep.cancer.gov>
Report under 'Secondary Malignancy – Other'
 3. Submit a copy of the pathology report to ECOG-ACRIN and NCI/CTEP confirming the diagnosis.
 4. If the patient has been diagnosed with AML/MDS, submit a copy of the cytogenetics report (if available) to ECOG-ACRIN and NCI/CTEP.
- NOTE:** The Second Primary Form and the CTEP-AERS report should not be used to report recurrence or development of metastatic disease.
- NOTE:** If a patient has been enrolled in more than one NCI-sponsored study, the Second Primary Form must be submitted for the most recent trial. ECOG-ACRIN must be provided with a copy of the form and the associated pathology report and cytogenetics report (if available) even if ECOG-ACRIN was not the patient's most recent trial.
- NOTE:** Once data regarding survival and remission status are no longer required by the protocol, no follow-up data should be submitted via CTEP-AERS or by the Second Primary Form.

5.4 Supportive Care

- 5.4.1 All supportive measures consistent with optimal patient care will be given throughout the study.
- 5.4.2 It is unknown whether bisphosphonates or other agents such as RANK ligand inhibitors at doses shown to prevent skeletal

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complication in hormone refractory disease is beneficial when a patient is just starting hormonal therapy. If a patient is to be treated with high dose bisphosphonates (not osteoporosis prevention dosing) then the investigator must declare this at the beginning to ensure patients treated in this manner are equally stratified to both arms. Once a patient is stratified, bisphosphonates at doses to prevent skeletal related events will be prohibited for patients who were not stratified by use of these agents. These patients can be treated with this class of drug at these doses once a skeletal related event occurs or once a patient develops hormone refractory disease. Osteoporosis prevention doses can be used at the investigator's discretion.

- 5.4.3 Medications to treat treatment-related side effects, such as nausea, hot flashes, diarrhea, are to be given at the investigator's discretion.

5.5 Duration of Therapy

Chemotherapy will be given for a maximum of six (6) cycles to patients on Arm A. Patients will come off study if they meet criteria for PSA or clinical progression.

Patients will receive protocol therapy unless:

- 5.5.1 Extraordinary Medical Circumstances: If at any time the constraints of this protocol are detrimental to the patient's health, protocol treatment should be discontinued. In this event submit forms according to the instructions in the E3805 Forms Packet.

- 5.5.2 Patient withdraws consent.

- 5.5.3 Unacceptable toxicity

At the time of hormone refractory disease investigators are encouraged to treat with docetaxel every three weeks ON BOTH ARMS so the effect of early versus delayed chemotherapy on overall survival can be performed. If the investigator deems that the patient is not a suitable candidate for docetaxel chemotherapy, the reason for this decision will be recorded. All therapy for hormone refractory disease will be recorded including anti-androgen withdrawal, second-line hormonal therapy and chemotherapy. If a patient has more than one therapy then all therapies will be recorded. Patients will be followed until development of HRPc and for overall survival.

5.6 Duration of Follow-up

For this protocol, all patients, including those who discontinue chemotherapy early, will be followed until development of hormone refractory disease and for survival for 10 years from the date of registration. All patients must also be followed through completion of all protocol therapy.

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6. Measurement of Effect

6.1 Solid Tumor Response Criteria (RECIST)

6.1.1 Malignant Disease Evaluation

To assess objective response, it is necessary to estimate the overall tumor burden at baseline to which subsequent measurements will be compared. Measurable disease is defined by the presence of at least one measurable lesion.

All measurements should be recorded in metric notation by use of a ruler or calipers. The same method of assessment and the same technique should be used to characterize each identified lesion at baseline and during follow-up. All baseline evaluations should be performed as detailed in Section [3](#).

The term evaluable in reference to measurability will not be used because it does not provide additional meaning or accuracy.

At baseline, tumor lesions will be characterized as either measurable or non-measurable.

6.1.1.1 Measurable

Lesions that can be accurately measured in at least one dimension (longest diameter to be recorded) as > 20 mm (2.0 cm) with conventional techniques or as > 10 mm (1.0 cm) with **spiral** CT scan.

If the measurable disease is restricted to a solitary lesion, its neoplastic nature should be confirmed by cytology/histology.

6.1.1.2 Non-Measurable

All other lesions, including small lesions [longest diameter < 20 mm (2.0 cm) with conventional techniques or < 10 mm (1.0 cm) with **spiral** CT scan] and truly non-measurable lesions.

Lesions considered to be truly non-measurable include the following: bone lesions, leptomeningeal disease, ascites, pleural/pericardial effusion, inflammatory breast disease, lymphangitis cutis/pulmonis, abdominal masses that are not confirmed and followed by imaging techniques, and cystic lesions.

Masses growing in a previously radiated field (e.g. prostatic fossa mass) are measurable if documented to have increased in size since XRT.

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6.1.2 Definitions of Response

6.1.2.1 Target Lesions

All measurable lesions up to a maximum of five lesions per organ and 10 lesions in total, representative of all involved organs. Target lesions should be selected on the basis of their size (those with the longest diameters) and their suitability for accurate repeated measurements.

The sum of the longest diameters of all target lesions will be calculated at baseline and reported as the *baseline sum longest diameter*. The *sum longest diameter* will be used to characterize the objective tumor response. For lesions measurable in 2 or 3 dimensions, always report the longest diameter at the time of each assessment.

6.1.2.2 Complete Response (CR)

The disappearance of all target lesions. To be assigned a status of complete response, changes in tumor measurements must be confirmed by repeat assessments performed **no less than four weeks** after the criteria for response are first met.

6.1.2.3 Partial Response (PR)

At least a 30% decrease in the sum of the longest diameters of target lesions, taking as reference the *baseline sum longest diameter*. To be assigned a status of partial response, changes in tumor measurements must be confirmed by repeat assessments performed **no less than four weeks** after the criteria for response are first met.

6.1.2.4 Progressive Disease (PD)

At least a 20% increase in the sum of the longest diameters of target lesions, taking as reference the *smallest sum longest diameter* recorded since the baseline measurements, or the appearance of one or more new lesion(s).

6.1.2.5 Stable Disease (SD)

Neither sufficient shrinkage to qualify for partial response nor sufficient increase to qualify for progressive disease. To be assigned a status of stable disease, measurements must have met the stable disease criteria at least once after study entry at a minimum interval of at least 8 weeks.

6.1.3 Nontarget Lesions

All other lesions or sites of disease. Measurements of these lesions are not required, but the presence or absence of each should be noted throughout follow-up.

6.1.3.1 Complete Response (CR)

The disappearance of all nontarget lesions and normalization of tumor marker levels, if applicable. To be assigned a status of complete response, changes in tumor measurements must be confirmed by repeat assessments performed **no less than four weeks** after the criteria for response are first met.

6.1.3.2 Incomplete Response/Stable Disease (SD)

The persistence of one or more nontarget lesion(s) and/or the maintenance of tumor marker levels above the normal limits. To be assigned a status of stable disease, measurements must have met the stable disease criteria at least once after study entry at a minimum interval of at least 8 weeks.

6.1.3.3 Progressive Disease (PD)

The appearance of one or more new lesion(s) and/or unequivocal progression of existing nontarget lesions.

6.1.4 Symptomatic Deterioration

Patients with a global deterioration of health status requiring discontinuation of treatment without objective evidence of disease progression at that time should be classified as having symptomatic deterioration.

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6.2 Evaluation of Patient's Best Overall Response

The best overall response is the best response recorded from registration until disease progression/recurrence, taking as reference for progressive disease the smallest measurements recorded since registration. The table below provides overall responses for all possible combinations of tumor responses in target and nontarget lesions, with or without new lesions.

To be assigned a status of complete or partial response, changes in tumor measurements must be confirmed by repeat assessments performed **no less than four weeks** after the criteria for response are first met.

To be assigned a status of stable disease, measurements must have met the stable disease criteria at least once after study entry at a minimum interval of at least 8 weeks.

Overall Response for all Possible Combinations of Tumor Response

NOTE: For this protocol, measurable disease response is not being evaluated. However, time to radiographic progression is being evaluated and the following table will apply.

Target Lesions	Nontarget Lesions	New Lesions	Overall Response
CR	CR	No	CR
CR	Incomplete response/SD	No	PR
PR	Non-PD	No	PR
SD	Non-PD	No	SD
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

CR = complete response; PR = partial response; SD = stable disease; PD = progressive disease

6.2.1 First Documentation of Response

The time between initiation of therapy and first documentation of PR or CR.

6.2.2 Confirmation of Response

To be assigned a status of complete or partial response, changes in tumor measurements must be confirmed by repeat assessments performed **no less than four weeks** after the criteria for response are first met.

6.2.3 Duration of Response

Duration of overall response – the period measured from the time that measurement criteria are met for complete or partial response (whichever status is recorded first) until the first date that recurrent or progressive disease is objectively documented, taking as reference the smallest measurements recorded since treatment started.

6.2.3.1 Duration of Overall Complete Response

The period measured from the time measurement criteria are met for complete response until the first date that recurrent disease is objectively documented.

6.2.3.2 Duration of Stable Disease

A measurement from registration until the criteria for disease progression is met, taking as reference the smallest measurements recorded since registration. To be assigned a status of stable disease, measurements must have met the stable disease criteria at least once after study entry at a minimum interval or at least 8 weeks.

6.2.4 Methods of Measurement

Imaging based evaluation is preferred to evaluation by clinical examination. The same imaging modality must be used throughout the study to measure disease.

6.2.4.1 CT and MRI

CT and magnetic resonance imaging (MRI) are the best currently available and most reproducible methods for measuring target lesions. Conventional CT and MRI should be performed with contiguous cuts of 10 mm or less in slice thickness. Spiral CT should be performed by use of a 5 mm contiguous reconstruction algorithm. This specification applies to tumors of the chest, abdomen, and pelvis, while head and neck tumors, and those of the extremities require specific procedures.

6.2.4.2 Chest X-Ray

Lesions on chest x-ray are acceptable as measurable lesions when they are clearly defined and surrounded by an aerated lung. However, CT is preferable.

6.2.4.3 Clinical Examination

Clinically detected lesions will only be considered measurable when they are superficial (e.g., skin nodules and palpable lymph nodes). For skin lesions, documentation by color photography, including a ruler to estimate size of the lesion, is recommended. Photographs should be retained at the institution.

6.2.4.4 Cytology and Histology

Cytologic and histologic techniques can be used to differentiate between complete and partial response in rare cases (e.g., after treatment to differentiate residual benign lesions and residual malignant lesions in germ cell tumors). Cytologic confirmation of the neoplastic nature of any effusion that appears or worsens during treatment is required when the measurable tumor has met response or stable disease criteria.

6.2.4.5 Endoscopy and Laparoscopy

Endoscopy and laparoscopy have not been fully or widely validated, so their use should be limited to validation studies in specialized institutions, and to confirming complete histopathologic response when biopsy specimens have been obtained.

6.2.4.6 Ultrasound

Ultrasound may be used only as an alternative to clinical measurements for superficial palpable lymph nodes, subcutaneous lesions, and thyroid nodules, and for confirming complete disappearance of superficial lesions usually assessed by clinical examination.

6.3 Serological Response

6.3.1 Response According to PSA Levels

PSA levels will be assessed before hormone therapy, at time of randomization, then every 12 weeks thereafter. Patients PSA will be assigned a response according to the following criteria:

Complete Serological Response – PSA level less than 0.2 ng/mL measured for 2 consecutive measurements at least 4 weeks apart.

Serological Partial Response (PR) – decline of PSA value, referenced to the pre-androgen therapy level, by greater than or equal to 50% for 2 consecutive measurements at least 4 weeks apart.

PSA Stable Disease – Patients who do not meet the criteria for response (CR or PR) or serological progression for at least 3 months (90 days) will be categorized as having stable disease.

Serological Progression (PD) is observed when the PSA demonstrates an increase that is more than 50% of nadir, taking as reference the lowest recorded PSA level since starting androgen deprivation therapy (ADT). Two consecutive increases must be documented with each measurement obtained at least two weeks apart. On occasions, there may be an intermediate fluctuant value. In accordance with the recommendations of the Prostate Cancer Clinical Trials Working Group (51), this will **not** restart the evaluation period so long as the intermediate value was not below the previous nadir.

The date of first recorded increase (not defeated by a subsequent drop in PSA level to create a new nadir) will be deemed the *date of progression*.

If a patient achieves a PSA that is less than 4 ng/ml:

- The confirming increase must reach a value that is more than 4.0 ng/ml (the unconfirmed and additional intermediate increases may be a value that is less than 4.0 ng/ml, but greater than 50% of nadir since starting ADT).
- If the patient's nadir is less than 2 ng/ml, the first time the PSA reaches a value greater than 50% of nadir since starting ADT **and** is at least 2 ng/ml will be the date of first progression. However, this progression must be confirmed by a PSA value that is more than 4.0 ng/ml with no intervening declines below 2 ng/ml.
- Progression will be declared if patient has documented to have a testosterone less than 50 ng/dL. In cases where a testosterone level has not been obtained and patient has not had an orchiectomy (as detailed in the study calendar), PD will be deemed if patient had a documented rise in PSA on LHRH analogue therapy.

6.3.2 Definitions of Serological Response

The following definitions of serological responses apply in this study:

First Documentation of Response – The period measured from the initiation of therapy until first documentation of partial or complete response.

Confirmation of Response – To be assigned a status of complete or partial response, changes in PSA level must be confirmed by repeat assessments performed *no less than 4 weeks* after the criteria for response are first met.

Duration of Response – The period measured from the time that measurement criteria are met for complete or partial response (whichever status is recorded first) until the first date that progressive disease is objectively documented, taking as reference the lowest PSA level recorded since androgen deprivation therapy started. Progression will be declared if patient has documented to have a testosterone less than 50 ng/dL. In cases where a testosterone level has not been obtained and patient has not had an orchiectomy (as detailed in the study calendar), PD will be deemed if patient had a documented rise in PSA on LHRH analogue therapy.

Duration of Complete Response – The period measured from the time that measurement criteria are met for complete response until the first date that progressive disease is objectively documented.

Time to Serological Progression – The period measured from the initiation of therapy until at least a $\geq 50\%$ increase in PSA level, taking as reference the lowest PSA level recorded since start of treatment. The date of the first recorded increase will be deemed the date of progression. If a patient achieves a PSA that is less than 4 ng/ml:

- The confirming increase must reach a value that is more than 4.0 ng/ml (the unconfirmed and additional intermediate increases may be a value that is less than 4.0 ng/ml, but greater than 50% of nadir since starting ADT).
- If the patient's nadir is less than 2 ng/ml, the first time the PSA reaches a value greater than 50% of nadir since starting ADT **and** is at least 2ng/mL will be the date of first progression. However, this progression must be confirmed by a PSA value that is more than 4.0 ng/ml with no intervening declines below 2 ng/ml.
- Progression will be declared if patient has documented to have a testosterone less than 50 ng/dL. In cases where a testosterone level has not been obtained and patient has not had an orchiectomy (as detailed in the study calendar), PD will be deemed if patient had a documented rise in PSA on LHRH analogue therapy.

Rev. 7/08	6.4	<u>Endpoint Definitions</u>
	6.4.1	Overall Survival Survival will be measured as the period from randomization until death due to any cause.
	6.4.2	Time to Hormone Refractory Disease: The interval between randomization and the date of documented clinical or serological progression with testosterone less than 50 ng/dL.
Rev. 7/08	6.4.3	Time to Clinical Progression Free Survival: The interval between randomization and the date of documented clinical progression. Clinical progression is defined as increasing symptomatic bone metastases, progression per RECIST criteria or clinical deterioration due to cancer per investigator's opinion.
Rev. 7/08	6.4.4	Time to Serological Progression Free Survival: The interval between randomization and the date of documented PSA progression. Progression will be declared if patient has documented to have a testosterone less than 50 ng/dL. In cases where a testosterone level has not been obtained and patient has not had an orchiectomy (as detailed in the study calendar), PD will be deemed if patient had a documented rise in PSA on LHRH analogue therapy.
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6.5 Quality of Life (FACT-P)

As an additional secondary endpoint, Quality of Life, including benefit and adverse effects of the treatment and the disease, will be measured in all patients.

- 6.5.1 Overall Quality of Life
- The Functional Assessment of Cancer Therapy-Prostate (FACT- P) is a self-report measure of both general and disease-specific quality of life (40). The FACT-P (version 4) contains 39 likert items distributed over 5 subscales: Physical (7 items), Social (7 items), Emotional (6 items), and Functional (7 items) well-being, and the additional concerns related to Prostate Cancer Scale (12 items). The FACT-P is scored by summing responses to the items in each subscale. The subscale scores are then combined to create a total QOL score. A trial outcome index (TOI) can also be calculated by summing the Physical and Functional Well-being Scales and the Prostate Cancer Scale. The FACT-P requires only a sixth-grade reading level and can be completed in 8 – 10 minutes (41).
- The FACT-P scales have been widely used to evaluate quality of life in clinical trials for prostate cancer treatments. Internal consistency of the FACT-P (version 4) subscales ranges from .85 to .89. Concurrent, construct and discriminant validity has been documented (41). Sensitivity to change in performance status and PSA score over two months has been demonstrated for the Physical Well-being, Functional Well-being, Prostate Cancer Scales, the Trial Outcome Index, and the total FACT-P (version 4) scale (41).
- The FACT-P instrument has been validated in samples of patients with extensive disease prostate cancer (41), and used to assess

quality of life in studies of patients with extensive disease prostate cancer (42). However, treatment trials in this patient population have not always demonstrated a significant change in global measures of QOL, such as the FACT-P and the PROSQOLI (43,44). It has been hypothesized that for patients with extensive disease prostate cancer who are undergoing treatment, QOL is the product of both the alleviation of disease-related symptoms and the introduction of treatment-related symptoms (44). Thus, a thoughtful evaluation of the impact of the study treatment on QOL in this sample will require that in addition to overall QOL, disease-related symptomatology and treatment-associated symptomology symptoms must be assessed, as well.

6.5.2 Treatment-associated and Disease-associated Symptoms

The FACT-Taxane instrument is a sub-scale developed to supplement QOL assessment by the FACT scales. It is a 16-item scale that assesses patients' experience of the symptoms most commonly associated with taxane treatment, and has been validated for this purpose (40). The FACT-Taxane subscale has been demonstrated to have excellent internal consistency reliability, validity and responsiveness to change over time (40). Significant decreases in the FACT-Taxane scale have been demonstrated to coincide with the emergence of neurotoxicity in patients undergoing taxane treatment and the relationship of the FACT-taxane score to duration of taxane treatment is linear (40).

Approximately 90% of patients undergoing taxane treatment experience anemia (45). A clinical symptom associated with anemia, fatigue is also a prevalent correlate of extensive disease prostate cancer (44,46). Thus, fatigue, a negative correlate of QOL in cancer patients in general, as a correlate of both extensive disease prostate cancer and the treatment in Arm A, can be expected to play a particularly important role in the evaluation of QOL in this study.

Fatigue will be evaluated using the 13-item Fatigue Subscale of the FACIT-F, which was developed to assess the physical experience of fatigue in cancer patients (47,48). Summed with the four core subscales of the FACT-P (Physical, Social, Emotional and Functional) it creates the Functional Assessment of Cancer Therapy-Fatigue (FACIT-F) score, a measure of both the physical experience and the functional consequences of fatigue (47,48). The Fatigue Subscale and the FACIT-F exhibit strong convergent and discriminant validity, internal consistency, test-retest reliability, and sensitivity to group differences in performance status (47,48).

Pain is the dominant disease-associated symptom experienced by men with metastatic prostate cancer. Therefore, reduction in pain has been the most significant palliative endpoint in treatment trials in this population. Pain will be assessed using the Brief Pain Inventory (BPI) Short Form (49). The BPI is a rapidly and easily completed self-report instrument, designed to measure pain intensity, interference with daily functioning, and relief. Reliability and validity of the BPI are well-established (49, 50).

7. Study Parameters

7.1 Therapeutic Parameters

1. Prestudy scans and x-rays used to assess all measurable or non-measurable sites of disease must be done within 6 weeks prior to randomization. (See Section [3.1.2](#) for guidance if hormonal therapy commenced prior to obtaining all scans).
2. Prestudy CBC (with differential and platelet count) should be done ≤ **4 weeks** before randomization.
3. All required prestudy chemistries, as outlined in Section [3.1.4](#), should be done ≤ **4 weeks** before randomization – unless specifically required on Day 1 as per protocol.

Visit/ Assessment	Baseline	Day 1, Cycle 1	Day 1, Cycles 2-6	Follow-up; Every 3 months ⁶	Hormone Refractory
Arm A					
Physical Examination, Vital Signs	X	X ^{2, 5}	X ⁵	X	
CBC with diff, Platelets ¹	X	X ²	X ¹		
Creatinine clearance, PTT, PT, INR	X	X ²			
Total Bilirubin, ALT	X	X ²	X ¹		
Arm B					
Physical Examination, Vital Signs	X	X ²		X	
CBC with diff, platelets ¹	X				
Creatinine clearance, PTT, PT, INR	X				
Both Arms A and B					
Medical History, ECOG PS	X				
CT Abdomen/Pelvis	X				X
Chest X-ray/CT chest ³	X				X
Bone Scan	X				X
[Deleted Addendum #3]					
FACT-P, FACT-Taxane, FACIT-F, and the BPI	X		X ⁷	X ⁸	
PSA	X		X ⁹	X	X
Testosterone					X ⁴
Biological Sample Submissions	Refer to Section 7.2				

- 1 CBCs (with differential and platelet count which includes WBC, ANC, Platelets, Hgb, and Hct) and ALT and total bilirubin which are required for protocol therapy on days 1 of cycles 2-6, must be done day prior to or day of treatment with docetaxel. If holidays or weather prevent treatment on day 1 as planned, day 1 will be the next feasible day of treatment and should be no more than a 7 day delay.
- 2 Does not need to be repeated if within 7 days of screening PE.
- 3 All patients must at least have a CXR prior to randomization to assess status of disease. CT chest at investigator's discretion. CXR within 6 weeks of starting hormonal therapy is required. (See Section [3.1.2](#) for guidance if hormonal therapy commenced prior to obtaining all scans).
- 4 To document castrate level at progression if patient has not had a bilateral orchiectomy.
- 5 Physical examinations can be done up to 72 hours prior to dosing of docetaxel.
- 6 Follow up every 3 months will continue beyond the time of HRPD for overall survival and will require

physical examination and PSA measurements and recording of all subsequent therapy.

Follow-up Schedule:

- | | | |
|-----------|---|--|
| Rev. 7/08 | | <ul style="list-style-type: none">• Every 3 months if patient is < 2 years from study entry• Every 6 months if patient is 2-5 years from study entry• Every 12 months if patient is 5-10 years from study entry• If patient is > 10 years from study entry, no forms are required |
| Rev. 2/07 | 7 | For both Arms A and B, submit at 3 months (12 weeks). |
| Rev. 2/07 | 8 | For both Arms A and B, submit at the end of months 6, 9, and 12. |
| Rev. 7/08 | 9 | For Arm A only (i.e., only patients getting docetaxel). |

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7.2 Biological Sample Submissions

Samples for the correlative studies should be submitted as outlined in Sections [10](#) and [11](#). Collection and submission of samples is to be limited to those patients who have given written consent to participate in the correlative studies or banking or both.

NOTE: ECOG-ACRIN requires that all biological samples submitted from patients participating in this trial be entered and tracked via the online ECOG-ACRIN Sample Tracking System (STS). See Section [10.4](#).

Samples		Baseline	Week 24 ²	Progression	Submit to:
Prostate Cancer Pathology		X			ECOG-ACRIN CBPF (Section 10.2)
Peripheral Blood	6 mL EDTA lavender top tube	Baseline, prior to trt preferred ^{1,2}			HCRN ³ (Section 10.3)
Plasma	6 mL EDTA lavender top tube x 2	X ¹	X	X	
Serum	10 mL red top tube	X ¹	X	X	

1. Baseline draws are to be collected AFTER randomization, prior to start of treatment. No samples are to be submitted prior to randomization.
2. If the indicated draws were missed, draw at the next scheduled clinical draw and note in STS, upon submission, the cycle, week number, and day that samples were taken.
3. Kits are available to order and will include all supplies needed for the collection and the shipping of the samples. It is important to only use the collection kits and shippers that the Hoosier Cancer Research Network provides. To order kits, fax Kit Order Form ([Appendix VI](#)) to Kristina Kirkpatrick at (317) 921-2053.

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8. Drug Formulation and Procurement

8.1 Docetaxel

Please refer to the FDA approved package insert for additional information.

8.1.1 Other Names

Taxotere, RP 56976, NSC #628503.

8.1.2 Classification

Antimicrotubule agent.

8.1.3 Mode of Action

Docetaxel, a semisynthetic analog of taxol, promotes the assembly of tubulin and inhibits microtubule depolymerization. Bundles of microtubules accumulate and interfere with cell division.

8.1.4 Storage and Stability

Store intact vials between 2° and 25°C (36° and 77°F). Retain in the original package to protect from bright light. The final dilution (in either 0.9% sodium chloride or 5% Dextrose solution) is stable for 4 hours if stored between 2° and 25°C (36° and 77°F).

8.1.5 Dose Specifics

75 mg/m² of docetaxel will be given on day 1 every 21 days for a maximum of six (6) cycles.

8.1.6 Preparation

Just prior to use, allow the docetaxel vial to reach room temperature for 5 minutes. Add the entire contents of the ethanol diluent vial and mix by gently rotating the vial for 45 seconds. Allow to stand for 5 minutes at room temperature, and check that the solution is homogeneous and clear (persistent foam is normal). The resulting solution contains 10 mg/mL of docetaxel. Please note that the solution contains 15% overfill. Dosing amounts should be based in the concentration per extractable volume, not the total volume of the vial. The desired dose is diluted in D5W or NS. The volume of the infusion should be adjusted in order to have a final docetaxel concentration of between 0.3 mg/mL and 0.74 mg/mL. Non-PVC-containing intravenous infusion bags and administration sets should be used to avoid patient exposure to the plasticizer DEHP.

8.1.7 Administration

Docetaxel will be administered to patients as an IV infusion over approximately one hour.

8.1.8 Incompatibilities

Intravenous bags and administration sets containing DEHP (di-[2-ethylexyl] phthalate). No further information available.

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8.1.9 Availability

Docetaxel is a commercially marketed product which has been approved by the FDA for: the treatment of locally advanced or metastatic breast cancer that returns after any prior chemotherapy; locally advanced or metastatic non-small cell lung cancer that recurs after prior platinum-based chemotherapy; and in combination with prednisone for the treatment of advanced metastatic prostate cancer. However its use in patients eligible for this protocol with prostate cancer is not currently approved by the FDA. When used as directed by this protocol, docetaxel is classified as an "unapproved use of an approved agent" and, by definition, considered as an investigational agent. However, while it is not an indication currently approved by the FDA, the use of docetaxel in this protocol is exempt from the requirements of an IND as described under Title 21 CFR 312.2(b).

NOTE: During the course of this study, if the FDA approves docetaxel for use in patients with prostate cancer, then the patient may have to pay for the amount of drug needed to complete the study.

Docetaxel (Taxotere[®]) is available and distributed free of charge from Sanofi-Aventis Pharmaceuticals. The vials of docetaxel will be packaged as 80 mg/2 mL vials. The initial shipment will be 6 vials, enough for 3 cycles. Vials of diluent will also be included.

After patient randomization, a supply of docetaxel may be obtained. Investigators must fax a signed and completed "E3805 Clinical Supplies Shipping Form" ([Appendix VII](#)) to the ECOG-ACRIN Operations Office – Boston, ATTN: Drug Team at 617-632-2063. Do not send drug requests directly to Sanofi-Aventis Pharmaceuticals. Allow 4-5 business days for receipt of the drug. See below for re-ordering instructions.

During the Study

There will be no weekend or holiday delivery of drugs. All deliveries will be made by express carrier.

Receipt of drug shipment: Complete the "Receipt From" section of the shipping form and fax to Debra Dawson at 908-243-7745.

Reorders: Each institution is responsible for reordering study drug. Complete and fax the E3805 Clinical Supplies Shipping Form ([Appendix VII](#)) to the ECOG-ACRIN Operations Office – Boston at 617-632-2063. Please allow 4-5 business days for processing reorders.

Drug Inventory Records: The Investigator or a responsible party, designated by the investigator, must maintain careful record of the inventory and dispensation of all drugs received. Please use the NCI Drug Accountability Record Form or equivalent. (See the NCI Investigators Handbook for Procedures for Drug Accountability and Storage.) The unused docetaxel must be accounted for.

At the Completion of the Study:

At the completion of the study, all used and unused drugs may be destroyed at the site, in accordance with the applicable regulations, guidelines and institution's procedures. Please maintain appropriate records of the disposal, including dates, quantities and method of destruction.

8.1.10 Side Effects

1. Cardiac: arrhythmias, pericardial effusions.
2. Hematologic: dose-related neutropenia, leukopenia, thrombocytopenia, anemia, hypoglycemia, hypernatemia.
3. Gastrointestinal: nausea and vomiting, diarrhea, oral mucositis, pancreatitis, esophagitis.
4. Neurologic: reversible dyesthesias or paresthesias, peripheral neuropathy, mild or moderate lethargy or somnolence, headache, seizures.
5. Hypersensitivity: hypersensitivity (local or general skin rash, flushing, pruritus, drug-fever, chills and rigors, low back pain), severe anaphylactoid reactions (flushing with hypo- or hypertension, with or without dyspnea).
6. Dermatologic: alopecia, desquamation following localized pruriginous maculopapular eruption, skin erythema with edema, extravasation reaction (erythema, swelling, tenderness, pustules), reversible peripheral phlebitis, nail changes.
7. Hepatic: increased transaminase, alkaline phosphatase, bilirubin; hepatic failure; hepatic drug reaction.
8. Pulmonary: dyspnea with restrictive pulmonary syndrome, pleural effusions.
9. Other: asthenia, dysgeusia, anorexia, conjunctivitis, arthralgia, muscle aches, myopathy, peripheral edema, fluid retention syndrome, ascites.

8.1.11 Nursing Implications

1. Monitor CBC and platelet count prior to drug administration.
2. Symptom management of expected nausea, vomiting, and mucositis.
3. Advise patients of possible hair loss.
4. Monitor for signs and symptoms of hypersensitivity reactions. Insure that recommended premedications are given.
5. Monitor liver function tests.
6. Evaluate site regularly for signs of infiltration.
7. Monitor for symptoms of peripheral neuropathy.
8. Monitor for signs of fluid retention and cutaneous reactions.

8.1.12 References

Investigator's Brochure: Docetaxel. Rhone-Poulenc Rorer, June 14, 1995.

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Bissett D, *et al.* Phase I study of Taxotere (RP56976) given as a 24-hour infusion. Proc Am Assoc Cancer Res 33:526, 1992.

Bruno R, *et al.* Clinical pharmacology of Taxotere (RP56976) given as a 1-2 hour infusion every 2-3 weeks. Proc Am Assoc Cancer Res 33:261, 1992.

DeValeriola D, *et al.* Phase I pharmacokinetic study of Taxotere (RP56976) administered as a weekly infusion. Proc Am Assoc Cancer Res 33:1563, 1992.

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8.2 LHRH analogue (such as leuprolide, goserelin, lcape degarelix)

Please refer to the FDA approved package insert for additional information.

8.2.1 Description:

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LHRH analogues are long-acting analogs of the native LHRH peptide and are effective at reducing serum testosterone. Analog approved by the FDA can be used in this study.

8.2.2 Supply:

Commercial available.

8.2.3 Administration:

LHRH analogs are administered with a variety of techniques such as subcutaneously, intramuscularly, or insertion. The manufacturer's instructions should be followed.

8.2.4 Toxicity:

The most common side effect is vasomotor hot flashes; edema, gynecomastia, bone pain, thrombosis, and gastrointestinal disturbances. Other side effects include impotence and loss of libido, weight gain, depression, dizziness, anemia, skin redness, pain at the injection site, unusual taste in the mouth, increased thirst and urination, and rarely allergic generalized rash and difficulty breathing.

8.3 Antiandrogens (flutamide and bicalutamide)

Please refer to the FDA approved package insert for additional information.

8.3.1 Availability:

Commercial available.

8.3.2 Administration:

The drugs are administered orally as prescribed.

8.3.3 Toxicity (flutamide):

Diarrhea and anemia. A high percentage of patients treated with flutamide alone developed gynecomastia within 2 to 8 months. There have been post marketing reports of hospitalization, and rarely, death due to liver failure in patients taking flutamide. Other side effects include elevated transaminase levels, jaundice, hepatic encephalopathy, and death related to acute hepatic failure, which is reversible after prompt discontinuation in some patients; impotence and loss of libido, fatigue, and rarely photosensitivity.

8.3.4 Toxicities (bicalutamide):

Breast tenderness, breast swelling, and hot flashes. When bicalutamide 50 mg was given in combination with an LHRH analog, the LHRH analog adverse event profile predominated with a high incidence of hot flashes (53%), and relatively low incidences of gynecomastia (4.7%) and breast pain (3.2%). Other side effects include impotence and loss of libido, fatigue, and rarely photosensitivity.

9. Statistical Considerations

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9.1 Introduction

This is the second revision of the statistical considerations section. The first changes reflect design changes incorporated in addendum 3. Addendum 3 expanded study eligibility to all patients with both low and high volume metastatic disease; eligibility was no longer restricted to patients with high volume disease. High volume disease is defined as visceral metastases and/or bone metastases (at least 4 or more bone lesions, and one of which must be outside of the vertebral column AND pelvis). The duration between beginning of prior hormonal therapy and registration was also changed from 90 days to 120 days. Furthermore, an additional stratification factor was added in this addendum, extent of disease: high vs. low volume disease.

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The second revision is to increase the number of patients to be accrued from 600 patients to 780 patients. This is based on data from S9346 demonstrating that the overall survival of men with prostate cancer starting androgen deprivation therapy (ADT) in the modern era is longer than the historical data which was used to develop the assumptions of addendum 3. In addition, the estimated proportion of patients with high volume disease is also changed based on the preliminary data released from the ECOG-ACRIN Data Monitoring Committee (DMC) meeting in September, 2011. As such, the projections which are guiding the statistical design need to be updated and require a trial of 780 patients.

9.2 Endpoints

The primary clinical endpoint is the difference across treatments in overall survival. Patients will be stratified based on age, ECOG PS, use of combined androgen blockade and FDA approved drugs for delaying skeletal related events, time of prior adjuvant hormonal therapy, and extent of disease; no subgroup analyses are planned. Primary endpoint will be studied for all intent-to-treat patients.

Secondary endpoints include PSA response, change of PSA over time, time to hormone refractory disease, time to clinical progression and time to PSA progression. All patients who receive treatment, regardless of eligibility, will be evaluated for toxicity.

9.3 Sample Size and Accrual

Based on prior studies and the preliminary data from SWOG study S9346, there will be approximately 50% of patients with high volume disease and 50% of patients with low volume disease in this population. Median survivals in arm B were expected to be 24 months and 48 months for patients with high and low volume disease, respectively. Under these assumptions above, the study required 600 patients to be accrued and randomized equally to each of the two arms (300 patients per arm). With the projected accrual rate of 8 patients per month, the accrual goal was to be reached in 75 months with a follow up period of 2.5 years.

Data from S9346 in the modern era indicates that median overall survival of men treated with ADT and having low volume disease is approximately 67 months.

The median overall survival for men with high volume disease is 33 months. This is probably due to active agents for castrate resistant prostate cancer and stage migration. The study is designed to demonstrate a 25% reduction in the survival hazard rate, which corresponds to an improvement in median overall survival from 33 months to 44 months in patients with high volume disease and from 67 months to 89.3 months in patients with low volume disease. After the activation of addendum 3, E3805 was enrolling approximately 12 patients per month and exceeds the anticipated accrual rate of 8 patients per month detailed in addendum 3. In addition, based on the preliminary data released from the DMC meeting in September, 2011, about 70% of patients have high volume disease on this study. With the assumptions above, the study will need 780 patients to maintain adequate power, and the accrual will be completed in 5.5 years with additional 1.5 years for follow-up.

9.4 Primary Objective

The study will have 80% power to detect a 33.3% improvement in median overall survival across treatments, using a log-rank test stratified on age, ECOG PS, use of combined androgen blockade and FDA approved drugs for delaying skeletal related events, time of prior adjuvant hormonal therapy, and extent of disease with a one-sided 0.025 significance level. The primary comparison will be an intent-to-treat analysis including all randomized patients.

Interim analyses will be performed for all semi-annual ECOG-ACRIN DMC meetings beginning when approximately 25% of the planned full information has occurred, about 35.5 months, continuing every 6 months until either criterion for early stopping are met or full information is reached. Twenty-five percent of the full information under the alternative hypothesis is equal to 100 failures. Full information will be achieved when 399 patients have died. The final analysis is scheduled for this time point. This study will be monitored for early stopping in favor of the null hypothesis using repeated confidence interval methodology similar to that described by Jennison and Turnbull, 1990 (Statistical Science 5:299-317). At each interim analysis the nominal $(1-2 \times \alpha)$ confidence interval on the overall survival hazard ratio comparing arm B to arm A will be computed, where α is the nominal one-sided significance level of the use function boundary at the information fraction for the particular analysis time. If the upper bound of the repeated confidence interval is smaller than 1.33, then the data monitoring committee may consider closing an arm of the trial or stopping the study early for overall lack of treatment differences. Under the accrual and failure rate assumptions above, the following table gives the expected interim analysis times, information times, and the corresponding truncated O'Brien Fleming boundaries (K. Lan and D. DeMets, 1983, Biometrika 70(3):659-663). Information from interim analyses will be reviewed by the DMC, who may decide to stop the study early if efficacy or futility has been demonstrated. Because of delays in initiation of accrual and delays in data submission and processing, it is likely that the actual analysis times will be 6-12 months later.

Repeated Analyses	Real Time (Months)	Information Time	Number of Failures		P-value	Upper Boundary	Number of Patients
			Under Ho	Under H1			
1	35.5	0.25	108.85	99.87	0.0005	3.29	426
2	41.5	0.33	143.88	132.01	0.0005	3.29	498
3	47.5	0.42	182.44	167.39	0.0005	3.29	570
4	53.5	0.52	224.17	205.68	0.0010	3.08	642
5	59.5	0.62	268.74	246.57	0.0035	2.69	714
6	65.5	0.73	315.82	289.77	0.0070	2.46	780
7	71.5	0.83	360.92	331.15	0.0109	2.29	780
8	77.5	0.92	401.27	368.17	0.0148	2.18	780
9	83	1.00	434.54	398.69	0.0180	2.10	780

9.5 Randomization and Stratification

Randomization will be performed with the following stratifications: 1) age: ≥ 70 vs. < 70 ; 2) ECOG PS: 0-1 vs. 2; 3) intended use of combined androgen blockade greater than 30 days from start of therapy; 4) prior adjuvant hormonal therapy > 12 months vs. ≤ 12 months; 5) use of FDA approved drugs for delaying skeletal related events; and 6) extent of disease: high volume vs. low volume disease.

The method of permuted blocks will be used for subject randomization. No per-site treatment-allocation balance will be implemented.

9.6 Quality of Life Analysis

The primary quality of life objective is to describe quality of life over time within each arm. Quality of life will be evaluated at baseline (within 1 week prior to therapy), week 12, week 24, week 36 and week 48. The primary comparison of interest will be the comparison between baseline (pre-treatment) and week 12 for scores on the FACT-P instrument. Treatment-associated and disease-associated changes in QOL will also be compared between baseline and week 12 for scores on the FACIT-F Fatigue Subscale, FACT-Taxane Subscale and the Brief Pain inventory. A comparative analysis will also be performed between arms A and B at each time point.

There are two sources of missing assessments in this study. The first source of missing assessments is among patients who are alive but do not complete the assessment, for reasons such as missed appointments, staff forgetting to administer the instrument, or patient refusal. We assume that this missingness is random, and will analyze the remaining questionnaires under this assumption. We expect only about 2% of the assessments to be missing at baseline, 10% by week 12, and about 30% to be missing by 24 weeks. The other type of

missingness is due to patients who will die during the study. If the median survival is 3-5 years and we assume that deaths follow an exponential distribution, we would expect about 3%-6% of the patients to have died by week 12. Because quality of life is likely to be related to survival, this censoring may be non-ignorable. For this reason, the comparison of week 12 to baseline will be considered the primary analysis.

We expect 85% to be alive and have completed baseline and week 12 assessments. A paired t-test will be used to test the null hypothesis that the change in quality of life score is not equal to 0 for each arm. Assuming 330 patients on each arm with measures on both days and a two-sided test with $\alpha=0.05$, we have 90% power to detect a difference of 0.18 standard deviation. This translates into a change of about 2.6 points assuming the standard deviation of the difference is 14.5 points, which was the standard deviation of the difference in QOL scores between baseline and week 8 from patients in E1898. Differences in quality of life between treatment arms will also be of interest. A t-test will be used to test for pairwise differences between treatment arms at week 12. Assuming a two-sided test with $\alpha=0.05$, the study will have about 90% power to detect a difference of 0.25 standard deviation between two treatment arms. This translates into a change of about 3.6 points assuming the common standard deviation is 14.5 points. Methods developed by Schluchter (1992) will be used to jointly model quality of life and survival data adjusting for non-ignorable missing data due to death or patient withdrawal. Similar analyses will be performed on the FACIT-F Fatigue Subscale scores, FACT-Taxane Subscale score and the Brief Pain inventory scores.

9.7 Gender and Ethnicity

Based on the preliminary data from **S9346**, the anticipated accrual in subgroups defined by gender and race is:

Ethnic Category	Gender		
	Females	Males	Total
Hispanic or Latino	0	34	34
Not Hispanic or Latino	0	746	746
Ethnic Category: Total of all subjects	0	780	780
Racial Category			
American Indian or Alaskan Native	0	3	3
Asian	0	12	12
Black or African American	0	87	87
Native Hawaiian or other Pacific Islander	0	0	0
White	0	678	678
Racial Category: Total of all subjects	0	780	780

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10. Sample Submissions

Blood and tissue samples are to be submitted from consenting patients for the laboratory research projects defined in Section [11](#) and for possible use in future research projects.

ECOG-ACRIN requires that all samples submitted on this trial be entered and tracked via the online ECOG-ACRIN Sample Tracking System (STS). An STS shipping manifest from must be generated and shipped with the sample submissions. See Section [10.4](#).

10.1 Submission Summary

No samples are to be submitted prior to patient randomization.

Tissue sample submissions are outlined in Section [10.2](#) are to be submitted from patients who answer “Yes” to “*I agree to participate in the scientific laboratory research studies that are being done as part of this clinical trial*” or “*My specimens may be kept for use in research to learn about, prevent, treat, or cure cancer.*”

Blood samples submissions are outlined in Section [10.3](#) and are to be submitted from patients who answer “Yes” to “*I agree to participate in the scientific laboratory research studies that are being done as part of this clinical trial.*” Instructions for ordering the collection/shipping kits are outlined in Section [10.3](#). Note that international sites outside the United States and Canada are not required to submit the blood samples because of the costs and problems associated with international shipping.

Samples		Baseline	Week 24 ¹	Progression	Submit to:
Prostate Cancer Pathology		X			ECOG-ACRIN CBPF (Section 10.2)
Peripheral Blood	6 mL EDTA lavender top tube	Baseline, prior to trt preferred ¹			HCRN (Section 10.3)
Plasma	6 mL EDTA lavender top tube x 2	X	X	X	
Serum	10 mL red top tube	X	X	X	

1. If the indicated draws were missed, draw at the next scheduled clinical draw and note in STS, upon submission, the cycle, week number, and day that samples were taken.

10.2 Pathology Submissions

Tumor tissue from the prostatectomy specimen is preferred, but if not available please send biopsy specimen from either prostate or metastatic site. Submit from patients who have given written consent for the use of their samples for the laboratory research studies or for possible use in future research.

When a patient is registered to receive protocol therapy, the submitting pathologist and clinical research associate should refer to [Appendix II](#) (Pathology Submission Guidelines), for submission requirements.

10.2.1 The materials to be submitted are:

Forms:

- A copy of the surgical pathology report, if available.
- STS generated shipping manifest
- If STS is down, please complete an ECOG-ACRIN Generic Specimen Submission Form (#2981). Please identify the clinical status of the submitted material (i.e., pretreatment as opposed to remission and relapse).

Biological Material Submission

Indicate in STS whether the tumor tissue materials are from the prostatectomy (preferred) or prostate biopsy or metastatic site.

- One representative tumor block.

NOTE: If a tumor block is unavailable, one tissue core punch, 10 unstained positively charged slides (3-4 microns), and one (1) H & E slide may be substituted.

10.2.2 Shipping Procedures

Submit at ambient temperature within one month following randomization to:

ECOG-ACRIN Central Biorepository and Pathology Facility
MD Anderson Cancer Center
Department of Pathology, Unit 085
Tissue Qualification Laboratory for ECOG-ACRIN, Room G1.3586
1515 Holcombe Blvd
Houston, TX 77030
Phone: Toll Free 1-844-744-2420 (713-745-4440 Local or International Sites)
Fax: 713-563-6506
Email: eachbpf@mdanderson.org

10.2.3 Central Biorepository and Pathology Facility: Sample processing and Routing

As outlined in Section [11](#), the appropriate materials for the protocol-defined correlative studies will be distributed to investigators for analysis. Residuals will be stored for future research.

10.3 Blood Specimen Submissions

Sample draws are requested as follows:

- Baseline, after randomization prior to treatment: serum, plasma, peripheral blood*
- Week 24*: serum, EDTA plasma
- Note: if this time point is missed, please draw at the next scheduled clinical draw.
- Progression: serum, plasma

*If any draws are delayed, in the comment field of STS indicate the actual cycle, week number, and day that samples were collected.

10.3.1 Kit Requests

To order kits, complete Kit Order Form ([Appendix VI](#)) and fax to Kristina Kirkpatrick at (317) 921-2053.

A minimum of five (5) working days notice of kit needs is required. Kits will NOT be shipped by overnight courier unless expressly stated on the form that the kit is needed within 24 hours due to patient enrollment. The first kit request may be done upon institutional IRB approval.

Collection kits will be provided and will contain everything needed for the sample collections and shipping. **It is important to ONLY use the collection kits and shippers provided by the Hoosier Cancer Research Network.**

10.3.2 Sample Preparation

Complete the pre-printed cryovial labels for the appropriate timepoints (baseline, week 24 or progression) using only a Sharpie marker. Fill in the patient initials, date of collection, and ECOG-ACRIN patient ID number then place on relevant cryovial (this should be done before the cryovials are frozen).

10.3.2.1 Red Top Tube

Collect at baseline, week 24, and progression.

SERUM SAMPLES MUST BE FROZEN WITHIN ONE HOUR OF BLOOD DRAW.

- Draw approximately 10 ml of blood into the red top vacutainer tube.
- Allow to coagulate for 30 minutes,
- Centrifuge at 3500 rpm for 30 minutes.
- Without delay, divide the serum equally into three (3) labeled cryovials in approximately 1.5 ml aliquots.
- Freeze immediately at -70°C until time of shipment. If a -70°C freezer is not available serum samples may be stored at -20°C until shipped. Samples **MUST** be shipped via Fed-Ex priority overnight as soon as possible. **DO NOT ALLOW THE SAMPLES TO THAW.**

10.3.2.2 EDTA Lavender Top Tubes

Whole Blood Processing

Collect at baseline. If baseline is missed, collect at next scheduled draw.

- Draw 6 mL of whole blood into one EDTA lavender top tube.
- Invert gently then divide sample equally into two (2) labeled cryovials.
- Freeze immediately at -70°C until time of shipment. If a -70°C freezer is not available samples may be stored

at -20°C until shipped. Samples MUST be shipped via Fed-Ex priority overnight as soon as possible. DO NOT ALLOW THE SAMPLES TO THAW.

Plasma Processing

Collect at baseline, week 24, and progression.

SAMPLE PROCESSING MUST BE STARTED WITHIN 30 MINUTES OF BLOOD DRAW.

- Draw approximately 6 mL of blood into each of the two EDTA lavender top vacutainer tubes.
- Invert gently.
- Centrifuge them at 3500 rpm for 30 minutes.
- Without delay, divide plasma equally into six (6) labeled cryovials in ~ 1 ml aliquots, and then freeze the samples immediately at -70°C. If a -70°C freezer is not available, plasma samples may be stored at -20°C until shipped. Samples MUST be shipped as soon as possible and sent by overnight courier, DO NOT ALLOW SAMPLES TO THAW.

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10.3.3 Shipping Guidelines

If multiple patients' samples are shipped together, bag each patients' samples separately using provided baggies. Make sure that cryovials are double-bagged for shipment) The STS-generated shipping manifest must accompany the samples. If STS is unavailable, see Section [10.4](#).

Using only the Thermosafe shippers provided by the Hoosier Cancer Research Network, completely fill the shipper with dry ice covering the samples.

Ship Monday through Thursday only (to arrive Tuesday through Friday). Shipments that arrive on Saturdays and holidays will NOT be accepted. The HCRN laboratory must be contacted PRIOR to shipping, email notification to: Kristina Kirkpatrick at kkirkpatrick@hoosiercancer.org or call at (317) 921-2050. Please include the Fed-Ex tracking number in the email.

Ship samples to:

Hoosier Cancer Research Network
Attn: Kristina Kirkpatrick
500 North Meridian Street
Suite 100
Indianapolis, IN 46204

NOTE: If your facility does not have dry-ice, contact Kristina Kirkpatrick to review potential sources.

10.4 ECOG-ACRIN Sample Tracking System

It is **required** that all samples submitted on this trial be entered and tracked using the ECOG-ACRIN Sample Tracking System (STS). The software will allow the use of either 1) an ECOG-ACRIN user-name and password previously assigned (for those already using STS), or 2) a CTSU username and password.

When you are ready to log the collection and/or shipment of the samples required for this study, please access the Sample Tracking System software by clicking <https://webapps.ecog.org/Tst>.

Important: Any case reimbursements associated with specimen submissions may not be captured if specimens are not logged into STS. Additionally, please note that the STS software creates pop-up windows, so you will need to enable pop-ups within your web browser while using the software. A user manual and interactive demo are available by clicking this link: <http://www.ecog.org/general/stsinfo.html>. Please take a moment to familiarize yourself with the software prior to using the system.

An STS generated shipping manifest should be generated and shipped with all sample submissions.

Please direct your questions or comments pertaining to the STS to ecog.tst@jimmy.harvard.edu.

Study Specific Notes

If the STS is unavailable,

- ECOG-ACRIN CBPF
FAX the completed ECOG-ACRIN Generic Specimen Submission Form (#2981) to the ECOG-ACRIN CBPF at 1-844-744-2420. Include the Fed-Ex tracking number in the comment section of the form.
- Hoosier Cancer Research Network (HCRN):
The day the samples are shipped, substitute the STS-generated shipping manifest with a completed E3805 Specimen Submission Form (#2741) AND email Kristina Kirkpatrick at kkirkpatrick@hoosiercancer.org and or call (317) 921-2050. Include the Fed-Ex tracking number in the comment section of the form.

Retroactively enter all specimen collection and shipping information when STS is available.

10.5 Banking

Residual material from the specimens submitted will be retained at the ECOG-ACRIN Central Biorepository for possible use in future ECOG-ACRIN approved studies. Any residual blocks will be available for purposes of individual patient management on specific written request. If future use is denied or withdrawn by the patient, the samples will be removed from consideration for use in any future study.

10.6 Sample Inventory Submission Guidelines

Inventories of all samples submitted will be tracked via the ECOG-ACRIN STS and receipt and usability verified by the receiving laboratory. Inventories of specimens forwarded and utilized for the will be submitted by the laboratory to the ECOG-ACRIN Operations Office – Boston on a monthly basis in an electronic format defined by the ECOG-ACRIN Operations Office – Boston.

10.7 Lab Data Transfer Guidelines

The data collected and generated on the laboratory research studies will be submitted electronically using a secured data transfer to the ECOG-ACRIN Operations Office – Boston by the investigating laboratories on a quarterly basis or per joint agreement between ECOG-ACRIN and the investigator. The quarterly cut-off dates are March 31, June 30, September 30, and December 31. Quarterly data reports are due at the ECOG-ACRIN Operations Office – Boston 1 week after these cut-off dates.

Rev. 12/11 **11. Correlative Studies**

Blood and tissue samples are to be submitted as outlined in Section [10](#) for use in the laboratory research studies described below. As technologies are developed and refined, the samples from E3805 will be used for additional research projects to explore various additional biological parameters, including molecular and genetic, and their relevance to prostate cancer. Any additional projects will only be conducted after receipt of the appropriate reviews and approvals.

11.1 Proteomic Study

Two-dimensional polyacrylamide gel electrophoresis (2D PAGE) has been the principal tool for the separation and simultaneous analysis of multiple proteins (20). This methodology, which is able to resolve numerous proteins in one experiment, provides the highest resolution in protein separation. Although the resolving power of 2D PAGE remains unchallenged, the high sensitivity, speed and reproducibility of mass spectrometry-based approaches have boosted their application in all aspects of protein analysis, including profiling, discovery and identification (21, 22), creating what is now called “proteomics.”

A high throughput proteomic analysis system for prostate cancer using patient sera and plasma will be developed for this study. The methodologies/proteomic platforms to be used will be those that are a part of the NCI supported “Clinical Proteomic Technologies for Cancer program”.

In essence, these technologies develop spectra of individual proteins or peptides which correspond to the mass dependent time-of-flight to the detector. Resulting protein patterns will then be analyzed as clusters or groupings that are similar or not similar. A set of known “training” samples will be used to segregate the data into two types of clusters: those containing samples from patients known to have hormone refractory prostate cancer (n=100) and clusters containing samples from patients with hormone sensitive prostate cancer (n=100). After “training” a computer algorithm, the pattern of an unknown sample will be diagnostically classified by its similarity to the diseased or unaffected clusters found in the training set. Training the computer will be done by means of computer-simulated Darwinian evolution in which the outcome is “survival of the fittest” subset of data in a sample. The algorithm starts by randomly selecting many proteomic patterns within the training data for analysis. Each chosen pattern is tested to see how well it can discriminate affected from unaffected individuals in the training set, by cluster analysis. Successful proteomic patterns are kept and recombined, whereas unsuccessful patterns are discarded. Ultimately, a best pattern emerges after many iterations by the algorithm. This pattern, which best segregates the training sets, is used to diagnostically classify unknown samples.

In summary, the plasma and serum samples that result from the clinical trial will be subjected to different proteomic platforms, and potential biomarkers will be identified by MS/MS or multiplex ELISA platforms. This approach will permit a comprehensive search for changes in polypeptide profiles that are present at the various stages of prostate cancer. The hypothesis to be tested is that there are proteins differentially translated from the genome in hormone sensitive prostate cancer, prostate cancer that has responded to hormonal therapy and hormone

refractory prostate cancer. This analysis will be performed by investigators with appropriate expertise under the direction of Dr. Christopher Sweeney.

The following definitions will be applied:

The “hormone sensitive” patients will be defined as patients who have met eligibility criteria. It is known that 90% of these patients will have a PSA response. Patients who do not respond (per Section [6.3](#)) to androgen ablation will be removed from the analysis.

“Prostate cancer that has responded to hormonal therapy” will be defined in patients with a PSA that has met PSA criteria for PSA response (per Section [6.3](#)) and is maintained at the 6 month mark from start of protocol therapy. In some cases, patients will have been on therapy for up to 12 months (120 day window for androgen ablation prior to enrollment). Patients treated with androgen ablation alone will be analyzed separately and in combination with the patients who received chemotherapy.

Plasma and serum from patients with “hormone refractory prostate cancer” will be obtained from patients who have provided serum and plasma at time of hormone refractory disease (see Section [6.4.2](#) for definition).

To ensure reproducibility the following analyses will be performed:

All samples will be run in duplicate. A run-in pilot study on samples from 70 patients with HRPC has been conducted at Indiana University and the optimized conditions from these experiments will be applied to the samples from this protocol. A normal serum control sample will be added to one spot on each chip to serve as quality control for chip integrity, process integrity, and mass spectrometer function. All samples and controls will be added randomly to the chips. Samples of replicates will be run in separate batches to assess inter-assay variability. Samples from different “disease states” (e.g. hormone sensitive and hormone refractory) will be intermingled to avoid assay batch effects.

Planned Analysis - Training Set/Test-set Approach

Samples from the Hoosier Cancer Research Network (HCRN) trial “A phase III, randomized, double-blind, placebo-controlled trial evaluating the ability of risedronate to prevent skeletal related events in patients with metastatic prostate cancer commencing hormonal therapy” as well as the first 30 patients from E3805 will serve as the training set. Samples for the HCRN study have been collected at the same time points as E3805. Classifiers will be completely specified based on data from this study. There are approximately 60 patients from the HCRN study and will be combined with the first 40 patients from E3805 to develop the “training set” and the second 100 patients from E3805 to form the “test set”. Using the test set of size 100, overall accuracy of the classifier can be estimated to within approximately +/- 10%.

11.1.1 Sample Submission

Plasma and serum aliquots submitted as outlined in Section [10](#) will be used for this study. No additional samples are requested.

11.2 Germline Genetic Analysis

In addition to analyzing the blood borne protein, an analysis of the germline DNA may also provide insight into the causes of prostate cancer and outcomes from treatment. We acknowledge that analyses of genetic polymorphisms will initially be exploratory. However, the potential value of obtaining the DNA and performing the analyses will provide the basis for further directed research. Moreover, this well powered prospective trial provides the platform to serve as a validation set for other findings. Dr Sweeney will oversee this work and will collaborate with investigators with the appropriate expertise in this endeavor (34-38). The same precautions for protecting the anonymity of the patients and obtaining consent to perform the studies will be adhered to as those described for the proteomic analysis. We will obtain ethnicity information on all patients as polymorphisms often differ between ethnic groups. Two examples of the analyses that can be performed are the following:

1. Pharmacogenetic analysis of drug metabolism and transport genes

Cytochrome p450 3A4 and mdr genes metabolize and transport chemotherapeutics such as docetaxel respectively. As such variants of these genes may predict for altered efficacy and toxicity. Preliminary data from a Hoosier Cancer Research Network trial assessing two docetaxel based regimens for hormone refractory prostate cancer found that patients with the more functional ABCG2 (breast cancer resistance protein – BCRP) gene with C/C at position 421 have a lower chance of being alive beyond 15 months (27% of 44 patients) compared with those with the less functional transporter (C/A). In the later group, 4 of 6 patients were alive beyond 15 months ($p=0.05$). It is therefore hypothesized that, the presence of a less functional germline transporter predicts for less efflux of cytotoxic drugs from cancer cells and better survival. It should be recognized this was from an analysis of only 52 patients. Therefore a study such as E3805 with a larger number of patients is needed to confirm or refute the significance of this finding. The C/A variant is found in about 25% of Caucasians. Therefore, approximately 70 patients (25% of 270) treated with docetaxel when commencing hormonal therapy will have this variant. It is hypothesized that this group will have a longer time to hormone refractory disease than those with the C/C genotype (approximately 200 patients) as the latter group will have a more effective drug transporter to remove docetaxel from the tumor. The time to progression of patients with C/A and C/C genotypes treated with androgen ablation alone will be analyzed and compared with their counterparts on the chemohormonal therapy arm. An example of the statistical power that will be available to address pharmacogenetic questions such as the one detailed with the ABCG2 gene is the following. Assume that the C/A genotype occurs in approximately 25% of patients and it confers a better overall survival.

2. Analysis of genetic polymorphisms that may alter the activity of enzymes involved in activation or deactivation of steroid hormones

The rationale for this is the following: The development of prostate cancer is possibly modulated by steroid hormones and many steroids and environmental carcinogens are subject to cytochrome P450 mediated metabolism which can generate reactive metabolites and may promote tumor initiation. Polymorphisms in the enzymes that are specifically involved in the

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metabolism of steroids may therefore be a specific risk factor for this disease. Some studies have suggested that this may in fact be the case (26-33). We will access the DNA of patients enrolled onto this trial and determine the frequency of polymorphisms of enzymes that are possibly involved in the genesis of prostate cancer. This will then be compared with frequency of these polymorphisms that will be identified from a screened population (e.g. samples and analyses can be obtained from a screening study that is being conducted at Indiana University). The analysis in the screening study will employ a case-control design. The frequency of polymorphisms in E3805 will then be compared informally with the results from the screening study. The demographics and ethnicity of patients from both E3805 and the screening study will be known and allow investigators to investigate the generalizability of the data from the two studies. One published example of a polymorphism of interest is an analysis in a Japanese population of the cytochrome p450 1B1 gene. This gene encodes for an enzyme involved in the hydroxylation of steroid hormones. The odds ratio for developing prostate cancer if the genotype T/T at codon 119 was 4.02 compared to the reference genotype of G/G (95% confidence interval: 1.73-9.38 - $P < 0.001$). Validating the frequency and significance of this polymorphism in a study of patients with metastatic disease and in what we expect to be predominately Caucasian population is required. Genes involved in the metabolism of steroid hormones have been reported to impact efficacy of ADT and E3805 has the potential to serve not only as a discovery but also a validation set.

This analysis will be performed with collaborators with appropriate expertise under the direction of Dr. Christopher Sweeney. ival for patients receiving docetaxel. With genotype results from 270 patients receiving docetaxel, there will be approximately 90% power to detect a hazard ratio of 1.8 when comparing overall survival between C/C and C/A groups using a two-sided 0.05 level logrank test and assuming a median overall survival of 32 months in the docetaxel arm and accrual over 45 months with 36 months additional follow-up.

11.2.1 Sample Submission

DNA will be extracted and analyzed from the lavender top tube blood samples submitted as outlined in Section [10](#). No additional samples are requested.

11.3 Tumor Tissue Analysis.

Tumor tissue obtained in a standardized fashion with clinical data from patients with metastatic disease is limited. E3805 provides a unique opportunity to obtain tissue tumor from this unique population. Moreover, technologies have advanced to allow extraction of DNA, RNA, micro-RNA and analyses of other tumor and tumor-stroma components including proteins from formalin fixed paraffin embedded (FFPE) tissue. An example of how this tissue can be studied in the following. Specifically, it has been noted that the expression of nuclear factor kappa B has been reported to be over-expressed in organ confined prostate cancer (39). However, the impact of NFkB activation on prognosis is unknown at this time. CXCR4 and Manganese superoxide dismutase are proteins under the control of nuclear factor kappa B and are associated with metastasis and avoidance of apoptosis respectively. Other proteins of interest

under control of NFκB that will be analyzed include TRAF-1, Bcl-XL, and cIAP2, The amount and frequency of expression of these genes and proteins as well as BCL-2 will be analyzed in the samples using immunohistochemistry and multiplex mRNA platforms from patients on this trial (with metastatic disease) and compared with samples from patients who never developed metastatic disease and will also be correlated with nuclear factor kappa B expression. This analysis will be performed under the direction of Dr. Christopher Sweeney with collaborators with the requisite expertise.

11.3.1 Sample Submission

The prostate cancer pathology or appropriate slides that were submitted as outlined in Section [10](#) will be used for this study. No additional samples are requested.

12. Records to Be Kept

Please refer to the E3805 Forms Packet for the forms submission schedule and copies of all forms. The E3805 Forms Packet may be downloaded by accessing the ECOG World Wide Web Home Page (<http://www.ecog.org>). Forms must be submitted to the ECOG-ACRIN Operations Office – Boston, FSTRF, 900 Commonwealth Avenue, Boston, MA 02215 (ATTN: DATA).

This study will be monitored by the CTEP Data Update System (CDUS) version 3.0. Cumulative CDUS data will be submitted quarterly from the ECOG-ACRIN Operations Office – Boston to CTEP by electronic means.

13. Patient Consent and Peer Judgment

Current FDA, NCI, state, federal and institutional regulations concerning informed consent will be followed.

14. References

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Appendix I

**Informed Consent Template for Cancer Treatment Trials (English Language)
[Deleted in Addendum #6]**

**INFORMED CONSENT INTENTIONALLY REMOVED FROM
PROTOCOL DOCUMENT**

Appendix I was removed from the protocol document in Addendum #6 and is posted as a separate document on the ECOG website. This was removed from the protocol to comply with NCI formatting guidelines.

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Appendix II

Pathology Submission Guidelines

The following items are included in Appendix II:

1. Guidelines for Submission of Pathology Materials
(instructional sheet for Clinical Research Associates [CRAs])
2. Instructional memo to submitting pathologists
3. List of Required Materials for E3805
4. ECOG-ACRIN Generic Specimen Submission Form (#2981)

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Guidelines for Submission of Pathology Materials

The following items should always be included when submitting pathology materials to the ECOG-ACRIN Central Biorepository and Pathology Facility:

Institutional Surgical Pathology Report

Pathology materials (see attached List of Required Material)

ECOG-ACRIN Generic Specimen Submission Form (#2981)

Instructions:

1. Place the Patient ID label provided by the ECOG-ACRIN Operations Office – Boston in Part A of the Pathology Material Submission Form.

If a label is not available, **TYPE or PRINT** the following information in **Part A** of the form:

- Patient's name (last, first)
 - Protocol number
 - Protocol case number (the patient's ECOG-ACRIN sequence number; for intergroup studies, include both the ECOG-ACRIN and other group's sequence numbers)
 - Patient's hospital number
 - Institution
 - Affiliate (if appropriate)
2. Complete blank areas of the pathologist's instructional memo, and forward it, along with the List of Required Material and the Pathology Material Submission Form, to the appropriate pathologist.
 3. The pathologist should return to you the required pathologic samples and surgical pathology reports, along with the completed. If any other reports are required, they should be obtained from the appropriate department at this time.
 4. Keep a copy of the ECOG-ACRIN Generic Specimen Submission Form (#2981) for your records (the original should be sent to the ECOG-ACRIN CBPF).
 5. Double check that ALL required forms, reports, and pathology samples are included in the package to send to the Central Biorepository and Pathology Facility (see appropriate List of Required Material).

Pathology specimens submitted for a patient WILL NOT be processed by the Central Biorepository and Pathology Facility until all necessary items are received.

6. Mail pathology materials to:

ECOG-ACRIN Central Biorepository and Pathology Facility
MD Anderson Cancer Center
Department of Pathology, Unit 085
Tissue Qualification Laboratory for ECOG-ACRIN, Room G1.3586
1515 Holcombe Blvd
Houston, TX 77030

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If you have any questions concerning the above instructions, or if you anticipate any problems in meeting the pathology material submission deadline of one month, contact the Pathology Coordinator at the ECOG-ACRIN Central Biorepository and Pathology Facility at Tel: 1-844-744-2420 or Email: eacbpf@mdanderson.org.

List of Required Material

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Pre-Treatment

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1. ECOG-ACRIN Generic Specimen Submission Form (#2981)
2. Institutional pathology report (***must be included with EVERY pathology submission***).
3. Study-specific On-Study Form in addition to the form sent to the ECOG-ACRIN Operations Office – Boston.
4. Requested pathology materials:

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- One tumor block. If a tumor block is not available, five (5) unstained positively charged slides and one H&E slide may be substituted.

NOTE: Blocks are available for return upon written request for purposes of patient management. However, since blocks are being used for laboratory studies, in some cases the material may be depleted and therefore, the block may not be available for return.

MEMORANDUM

TO: _____
(Submitting Pathologist)

FROM: Stanley Hamilton, M.D., Chair
ECOG-ACRIN Laboratory Science and Pathology Committee

DATE: _____

SUBJECT: Submission of Pathology Materials for E3805

The patient named on the attached ECOG-ACRIN Generic Specimen Submission Form (#2981) has been entered onto an ECOG-ACRIN protocol by _____ (ECOG-ACRIN Investigator). This protocol requires the submission of pathology materials for **laboratory studies**.

Please complete PART B of the Submission Form. Keep a copy for your own records, and return the completed Submission Form, the surgical pathology report(s), the slides and/or blocks, and any other required material (see attached List of Required Material) to the Clinical Research Associate (CRA). The CRA will forward all required pathology material to the ECOG-ACRIN Central Biorepository and Pathology Facility.

Blocks and slides submitted for this study will be retained at the ECOG-ACRIN Central Repository for future studies. Paraffin blocks will be returned for purposes of patient management upon written request.

Please note: Since blocks are being *used for laboratory studies, in some cases the material may be depleted, and, therefore, the block may not be returned.*

If you have any questions regarding this request, please feel free to contact the Central Biorepository and Pathology Facility at Tel: 1-844-744-2420 OR EMAIL: eacbpf@mdanderson.org.

The ECOG-ACRIN CRA at your institution is:

Name: _____

Address: _____

Phone: _____

Thank you.

Institution Instructions: This form is to be completed and submitted with **all specimens** ONLY if the Sample Tracking System (STS) is not available. **Use one form per patient, per time-point.** All specimens shipped to the laboratory must be listed on this form. Enter all dates as MM/DD/YY. Keep a copy for your files. Retroactively log all specimens into STS once the system is available. **Contact the receiving lab to inform them of shipments that will be sent with this form.**

Protocol Number _____ Patient ID _____ Patient Initials Last _____ First _____

Date Shipped _____ Courier _____ Courier Tracking Number _____

Shipped To (Laboratory Name) _____ Date CRA will log into STS _____

FORMS AND REPORTS: Include all forms and reports as directed per protocol, e.g., pathology, cytogenetics, flow cytometry, patient consult, etc.

Required fields for all samples				Additional fields for tissue submissions				Completed by Receiving Lab
Protocol Specified Timepoint:								
Sample Type (fluid or fresh tissue, include collection tube type)	Quantity	Collection Date and Time 24 HR		Surgical or Sample ID	Anatomic Site	Disease Status (e.g., primary, mets, normal)	Stain or Fixative	Lab ID

Fields to be completed if requested per protocol. Refer to the protocol-specific sample submissions for additional fields that may be required.

Leukemia/Myeloma Studies:	Diagnosis	Intended Treatment Trial	Peripheral WBC Count (x1000)	Peripheral Blasts %	Lymphocytes %
Study Drug Information:	Therapy Drug Name	Date Drug Administered	Start Time 24 HR	Stop Time 24HR	
Caloric Intake:	Date of Last Caloric Intake		Time of Last Caloric Intake 24HR		

CRA Name _____ CRA Phone _____ CRA Email _____

Comments _____

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Appendix III

Patient Thank You Letter

We ask that the physician use the template contained in this appendix to prepare a letter thanking the patient for enrolling in this trial. The template is intended as a guide and can be downloaded from the web site at <http://www.ecog.org>. As this is a personal letter, physicians may elect to further tailor the text to their situation.

This small gesture is a part of a broader program being undertaken by ECOG-ACRIN and the NCI to increase awareness of the importance of clinical trials and improve accrual and follow-through. We appreciate your help in this effort.

[PATIENT NAME]

[DATE]

[PATIENT ADDRESS]

Dear [PATIENT SALUTATION],

Thank you for agreeing to take part in this important clinical program. Programs like this offer a chance to get the best care while helping us make better care available for all patients. Many questions remain unanswered in cancer. With the help of people like you who participate in these programs, we will achieve our goal of effectively treating and ultimately curing cancer.

We believe this program will provide you with high quality, thorough care. Your physician and research staff will maintain very close contact with you. This is important to allow your physician to provide you with the best care while learning as much as possible to help you and other patients.

On behalf of **[INSTITUTION]** and ECOG-ACRIN, we thank you again and look forward to helping you.

Sincerely,

[PHYSICIAN NAME]

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Appendix IV

Cancer Trials Support Unit (CTSU) Participation Procedures

REGISTRATION/RANDOMIZATION

Prior to the recruitment of a patient for this study, investigators must be registered members of the CTSU. Each investigator must have an NCI investigator number and must maintain an “active” investigator registration status through the annual submission of a complete investigator registration packet (FDA Form 1572 with original signature, current CV, Supplemental Investigator Data Form with signature, and Financial Disclosure Form with original signature) to the Pharmaceutical Management Branch, CTEP, DCTD, NCI. These forms are available on the CTSU registered member Web site or by calling the PMB at 240-276-6575 Monday through Friday between 8:30 a.m. and 4:30 p.m. Eastern time.

Each CTSU investigator or group of investigators at a clinical site must obtain IRB approval for this protocol and submit IRB approval and supporting documentation to the CTSU Regulatory Office before they can enroll patients. Study centers can check the status of their registration packets by querying the Regulatory Support System (RSS) site registration status page of the CTSU member web site at <http://members.ctsuo.org>

All forms and documents associated with this study can be downloaded from the E3805 Web page on the CTSU registered member Web site (<https://members.ctsuo.org>). Patients can be registered only after pre-treatment evaluation is complete, all eligibility criteria have been met, and the study site is listed as ‘approved’ in the CTSU RSS.

Requirements for E3805 site registration:

- CTSU IRB Certification
- CTSU IRB/Regulatory Approval Transmittal Sheet

Prestudy requirements for patient enrollment on E3805:

- Patient must meet all inclusion criteria, and no exclusion criteria should apply
- Patient has signed and dated all applicable consents and authorization forms
- All baseline laboratory tests and prestudy evaluations performed within the time period specified in the protocol.
- Patient completes baseline QOL forms prior to treatment start.

CTSU Procedures for Patient Enrollment

1. Contact the CTSU Patient Registration Office by calling 1-888-462-3009. Leave a voicemail to alert the CTSU Patient Registrar that an enrollment is forthcoming. For immediate registration needs, e.g. within one hour, call the registrar cell phone at 1-301-704-2376.

2. Complete the following forms:
 - CTSU Patient Enrollment Transmittal Form
 - E3805 Eligibility Checklist
3. Fax these forms to the CTSU Patient Registrar at 1-888-691-8039 between the hours of 8:00 a.m. and 8:00 p.m., Mon-Fri, Eastern Time (excluding holidays). The CTSU registrar will check the investigator and site information to ensure that all regulatory requirements have been met. The registrar will also check that forms are complete and follow-up with the site to resolve any discrepancies.
4. Once investigator eligibility is confirmed and enrollment documents are reviewed for compliance, the CTSU registrar will contact the ECOG-ACRIN. The CTSU registrar will access the ECOG-ACRIN's on-line registration system, to obtain and assignment of a treatment arm and a unique patient ID (to be used on all future forms and correspondence). The CTSU registrar will confirm registration by fax.
 - Protocol treatment should start within five working days after randomization.

DATA SUBMISSION AND RECONCILIATION

1. All case report forms (CRFs) and transmittals associated with this study must be downloaded from the E3805 Web page located on the CTSU registered member Web site (<https://members.ctsu.org>). Sites must use the current form versions and adhere to the instructions and submission schedule outlined in the protocol.
2. Submit all completed CRFs (with the exception of patient enrollment forms), clinical reports, and transmittals directly to the ECOG-ACRIN [refer to contacts table] unless an alternate location is specified in the protocol. Do not send study data to the CTSU. A completed CTSU-ECOG-ACRIN coversheet should accompany all data submissions.
3. The ECOG-ACRIN Operations Office – Boston will mail query notices and delinquency reports directly to the site for reconciliation. Please send query responses and delinquent data to the ECOG-ACRIN Operations Office – Boston and do not copy the CTSU Data Operations. Each site should have a designated CTSU Administrator and Data Administrator and must keep their CTEP AMS account contact information current. This will ensure timely communication between the clinical site and the ECOG-ACRIN Operations Office – Boston.

SPECIAL MATERIALS OR SUBSTUDIES

NOTE: ECOG-ACRIN requires that all biological samples submitted from patients participating in this trial be entered and tracked via the online ECOG-ACRIN Sample Tracking System (STS). See Section [10.4](#).

1. Pathology Submission (Protocol Section [10.0](#))
 - Submission of pathologic material for correlative studies is optional.
 - Collect, prepare, and submit specimens as outlined in the protocol.
 - Do not send specimens, supporting clinical reports, or transmittals to the CTSU.
2. Specimen collection for correlatives (Protocol Section [11.0](#))
 - Participation in the correlative studies is optional and requires patient consent.

- Kits are available for the collection of samples. Complete the Kit Order Form ([Appendix VI](#)) and fax to Kristina Kirkpatrick at (317) 921-2053.
 - Collect, prepare, and submit specimens as outlined in the protocol.
 - Do not send specimens, supporting clinical reports, or transmittals to the CTSU.
3. Quality of Life Substudies (Protocol Section [5.1](#))
- Submit completed forms as outlined in the protocol.

SERIOUS ADVERSE EVENT (AE) REPORTING (Section [5.3](#))

1. CTSU sites must comply with the expectations of their local Institutional Review Board (IRB) regarding documentation and submission of adverse events. Local IRBs must be informed of all reportable serious adverse reactions.
2. CTSU sites will assess and report adverse events according to the guidelines and timelines specified in the protocol. You may navigate to the CTEP's Adverse Event Reporting System (CTEP-AERS) from either the Adverse Events tab of the CTSU member homepage (<https://members.ctsu.org>) or by selecting Adverse Event Reporting Forms from the document center drop down list on the E3805 Web page.
3. Do not send adverse event reports to the CTSU.
4. Secondary AML/MDS/ALL reporting: Report occurrence of secondary AML, MDS, or ALL via the NCI/CTEP AML-MDS Report Form in lieu of CTEP-AERS. Submit the completed form and supporting documentation as outlined in the protocol.

DRUG PROCUREMENT (Section [8.0](#))

Investigational agent: Docetaxel (This drug is a commercially marketed drug, however, when used as directed by this protocol, docetaxel is classified as an "unapproved use of an approved agent" and, by definition, considered as an investigational agent. However, while it is not an indication currently approved by the FDA, the use of docetaxel in this protocol is exempt from the requirements of an IND as described under Title 21 CFR 312.2(b).)

Docetaxel (Taxotere®) is available and distributed free of charge from Sanofi-Aventis Pharmaceuticals. After patient randomization, a supply of docetaxel may be obtained. Investigators must fax a signed and completed "E3805 Clinical Supplies Shipping Form" ([Appendix VI](#)) to the ECOG-ACRIN Operations Office – Boston, ATTN: Drug Team at 617-632-2063. Do not send drug requests directly to Sanofi-Aventis Pharmaceuticals. Allow 4-5 business days for receipt of the drug. See Section [8.0](#) for further drug information and re-orders.

Commercial agents: LHRH agonists (such as leuprolide, goserelin, triptorelin); Antiandrogens (flutamide and bicalutamide)

1. Information on drug formulation, procurement, storage and accountability, administration, and potential toxicities are outlined in Section [8.0](#) of the protocol.
2. You may navigate to the drug forms by selecting Pharmacy Forms from the document center drop down list on the E3805 Web page.

REGULATORY AND MONITORING

Study Audit

To assure compliance with Federal regulatory requirements [CFR 21 parts 50, 54, 56, 312, 314 and HHS 45 CFR 46] and National Cancer Institute (NCI)/ Cancer Therapy Evaluation Program (CTEP) Clinical Trials Monitoring Branch (CTMB) guidelines for the conduct of clinical trials and study data validity, all protocols approved by NCI/CTEP that have patient enrollment through the CTSU are subject to audit.

Responsibility for assignment of the audit will be determined by the site's primary affiliation with a Cooperative Group or CTSU. For Group-aligned sites, the audit of a patient registered through CTSU will become the responsibility of the Group receiving credit for the enrollment. For CTSU Independent Clinical Research Sites (CICRS), the CTSU will coordinate the entire audit process.

For patients enrolled through the CTSU, you may request the accrual be credited to any Group for which you have an affiliation provided that Group has an active clinical trials program for the primary disease type being addressed by the protocol. (e.g., NSABP members may only request credit for protocols pertaining to breast or colorectal cancers). Registrations to protocols for other disease sites may still take place through CTSU without receiving credit for your NSABP activities. Per capita reimbursement will be issued directly from CTSU.

Details on audit evaluation components, site selection, patient case selection, materials to be reviewed, site preparation, on-site procedures for review and assessment, and results reporting and follow-up are available for download from the CTSU Operations Manual located on the CTSU Member Web site.

Health Insurance Portability and Accountability Act of 1996 (HIPAA)

The HIPAA Privacy Rule establishes the conditions under which protected health information may be used or disclosed by covered entities for research purposes. Research is defined in the Privacy Rule referenced in HHS 45 CFR 164.501. Templated language addressing NCI-U.S. HIPAA guidelines are provided in the HIPAA Authorization Form located on the CTSU website.

The HIPAA Privacy Rule does not affect participants from outside the United States. Authorization to release Protected Health Information is NOT required from patients enrolled in clinical trials at non-US sites.

Clinical Data Update System (CDUS) Monitoring

This study will be monitored by the Clinical Data Update System (CDUS) Version 3.0. Cumulative CDUS data will be submitted quarterly to CTEP by electronic means. The sponsoring Group fulfills this reporting obligation by electronically transmitting to CTEP the CDUS data collected from the study-specific case report forms.

CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease Prostate Cancer

Appendix V

Treatment of Hypersensitivity Reactions

MANAGEMENT OF ACUTE HYPERSENSITIVITY

Severity of Symptoms	Treatment Guidelines
Mild symptoms: localized cutaneous reactions such as mild pruritus, flushing, rash	- consider decreasing the rate of infusion until recovery from symptoms, stay at bedside and monitor patient - then, complete docetaxel infusion at the initial planned rate
Moderate symptoms: any symptom that is not listed above (mild symptoms) or below (severe symptoms) such as generalized pruritus, flushing, rash, dyspnea, hypotension with systolic BP > 80 mm Hg	- interrupt docetaxel infusion - give diphenhydramine 50 mg IV with or without dexamethasone 10 mg IV; monitor patient until resolution of symptoms - resume docetaxel infusion after recovery of symptoms; depending on the physician's assessment of the patient, docetaxel infusion should be resumed at a slower rate, then increased incrementally to the initial planned rate, (e.g. <i>infuse at an 8 hour rate for 5 minutes, then at a 4-h rate for 5 minutes, then at a 2-h rate for 5 minutes, then finally, resume at the 1-h infusion rate</i>) - depending on the intensity of the reaction observed, additional oral or IV premedication with an antihistamine should also be given for the next cycle of treatment, and the rate of infusion should be decreased initially and then increased back to the recommended 1-hour infusion, (e.g. <i>infuse at an 8 hour rate for 5 minutes, then at a 4-h rate for 5 minutes, then at a 2-h rate for 5 minutes, and finally, administer at the 1-h infusion rate</i>)
Severe symptoms: any reaction such as bronchospasm, generalized urticaria, systolic BP ≤ 80mm Hg, angioedema	- immediately discontinue docetaxel infusion - give diphenhydramine 50 mg IV with or without dexamethasone 10 mg IV and/or epinephrine as needed; monitor patient until resolution of symptoms - the same treatment guidelines outlined under moderate symptoms (i.e. the third and fourth bullets) should be followed.
Anaphylaxis (NCI grade 4 reaction)	NO FURTHER STUDY DRUG THERAPY

**CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for
Extensive Disease Prostate Cancer**

Appendix VI

E3805 Collection and Shipping Kit Order Form

Upon institutional IRB approval complete and fax this form as indicated below. A minimum of 5 working days notice is required. Kits will NOT be shipped overnight unless clearly stated on this form that it is needed within 24 hours **due to patient enrollment** only.

Provided sample collection kits will contain all items needed including: blood tubes, cryovials, labels and Thermosafe shippers. It is important that you use only the collection kits and Thermosafe shippers provided by the Hoosier Cancer Research Network.

Rev. 1/15

TODAYS DATE: _____ Date of Institutional IRB approval: _____

Date Kits are Needed: _____ (Provide 5 working days notice)

Kits Needed:

TYPE	HOW MANY NEEDED?	Comments
Baseline Collection Kits		
Week 24 Collection Kits		
Time of Progression Kits		
Thermosafe Shippers (2 sent per shipment)		

*Comments: _____

Kits are to be shipped to:

Institution Name: _____

Institution Contact: _____

Phone number for contact: _____

E-mail for contact: _____

Institution Address: _____

FAX Completed form to Kristina Kirkpatrick at (317) 921-2053.

Questions are to be directed to the Kristina Kirkpatrick by email at kriskirk@iupui.edu or by calling (317) 921-2050.

CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease Prostate Cancer

Appendix VII

E3805 Clinical Supplies Shipping Form

To Investigator Only

SECTION 1 – TO BE COMPLETED BY SITE

Requested By:

Mail stop :

Telephone:

Fax :

Protocol No. : E3805

IND/CTX No. : N/A

Ship to the attention of :

Investigator No.:

Address :

Investigator Name:

Telephone:

Fax :

SECTION 2 All line items MUST be completed by the ECOG-ACRIN OPERATIONS OFFICE – BOSTON DRUG TEAM

- a) ☐ Investigator shipments: The regulatory requirements are fulfilled at the clinical study center according to valid SOPs
b) ☐ IND studies: Required documents (e.g. protocols. 1572, CV) have been provided to the Regulatory Representative for IND submission/maintenance ☒ Not applicable
c) ☐ Initial shipment ☐ Reshipment

Requestor Name and Title :

Date :

Signature :

REQUESTED SHIP DATE : TO BE COMPLETED BY SITE

Financial Account No. N/A

ECOG-ACRIN Patient Sequence Numbers:

We are pleased to send you the following Investigational Products: COMPLETED BY SA

Number of Patient kits or quantity of bulk drug product	Treatment Period (if applicable)	Product Name/ Strength/ Form	PR or Lot Number	Use by Date

STORAGE CONDITIONS :

PR Number:

Sanofi-aventis Tracking N° :

Number of containers used for shipment:

Carrier/AWB Number :

☐ Invoice

☐ Code Breaks Number of set(s) :

☐ Release Certificate and Certificate of Conformance :

☐ Additional document:

☐ Certificate(s) of Analysis

☐ Randomization list

Comments :

Packed by :

Checked by :

Actual Shipment Date :

ACKNOWLEDGEMENT OF RECEIPT – TO BE COMPLETED BY SITE

Please acknowledge receipt of the shipment by checking the contents and signing approval below:

Number of containers received:

Contents received intact: Yes ☐ No ☐; If No, please explain: _____

Date received: _____ Signature : _____ Print Name: _____

Please fax acknowledgement of receipt to Aventis Attention: Debra Dawson at fax No. 908-243-7745
(local contact)

CHAARTED: ChemoHormonal Therapy versus Androgen Ablation Randomized Trial for Extensive Disease Prostate Cancer

Version Date: December 23, 2014

This is a clinical trial, a type of research study. Your doctor will explain the clinical trial to you. Clinical trials include only people who choose to take part. Please take your time to make your decision about taking part. You may discuss your decision with your friends and family. You can also discuss it with your health care team. If you have any questions, you can ask your doctor for more explanation.

Rev. 7/08

You are being asked to take part in this study because you have prostate cancer that has spread beyond the prostate and can be seen on scans.

WHY IS THIS STUDY BEING DONE?

Rev. 7/08

The purpose of this study is to evaluate a new way of treating patients with prostate cancer. In Arm A, you will receive docetaxel chemotherapy both at the time you are starting hormonal therapy and later when/if the hormonal therapy is no longer working per your doctor decision. In Arm B, patients only get chemotherapy when/if the hormonal therapy is no longer working per your doctor decision. We are doing the study to see if there is a benefit to giving chemotherapy at the beginning. Hormonal therapy refers to drugs or surgical procedures such as an orchiectomy (removal of testicles) which lower your testosterone. This puts prostate cancer into remission in most patients as the testosterone is like a fuel for the cancer. It is the standard to give this as the only treatment for your cancer. Normally chemotherapy is reserved for when a patient's cancer starts to grow again despite having a low testosterone level. In this study you will either get docetaxel when you start your hormonal therapy or when/if your cancer grows with a low testosterone level. If you have chemotherapy when you start hormones for the first time, you may get chemotherapy when/if your cancer grows back with a low testosterone level.

Rev. 7/08

If your disease worsens while you are on hormone treatments, your doctor will consider your next treatment very carefully. This protocol has asked, but not required, that your doctor use docetaxel (a type of chemotherapy) once your disease worsens, even if you had received docetaxel when you started hormone therapy. Your doctor may also try another hormone treatment before starting you on docetaxel chemotherapy. The purpose of this study is to determine whether receiving docetaxel both when you start hormone therapy (or within 120 days of beginning hormonal therapy) as well as when your disease worsens, is better than only receiving docetaxel when your disease gets worse. This trial will let us know which approach is more effective in treating your disease.

HOW MANY PEOPLE WILL TAKE PART IN THE STUDY?

Rev. 7/08, 12/11 **About 780 people will take part in this study.**

WHAT WILL HAPPEN IF I TAKE PART IN THIS RESEARCH STUDY?

BEFORE YOU BEGIN THE STUDY

You will need to have the following exams, tests or procedures to find out if you can be in the study. These exams, tests or procedures are part of regular cancer care and may be done even if you do not join the study. If you have had some of them recently, they may not need to be repeated. This will be up to your doctor.

- Physical Examination with medical history and vital signs
- Blood tests
- Scans of the body (bone scan, chest scan, chest x-ray, abdomen/pelvis scan)

Rev. 2/07

You will be "randomized" into one of the study groups described below. Randomization means that you are put into a group by chance. A computer program will place you in one of the study groups. Neither you nor your doctor can choose the group you will be in. You will have a *one in two* chance of being placed in the group that receives chemotherapy when starting hormonal therapy (Group A). Group B patients will be treated with hormonal therapy alone. Hormonal therapy refers to lowering of your testosterone and can be done by removal of testicles or injections ("shots"). Chemotherapy is normally given at the time the cancer learns to grow without testosterone. This study is trying to see if people's cancer can be controlled for longer if chemotherapy is given when starting hormonal therapy and what the effect the chemotherapy has on a patient's quality of life.

DURING THE STUDY

If the exams, tests and procedures show that you can be in the study, and you choose to take part, then you will need the following tests and procedures. They are part of regular cancer care.

If you are to receive chemotherapy immediately, you will have the following done every 3 weeks, prior to your infusion:

- Physical Examination, vital signs
- Blood tests

If you are to receive hormonal therapy alone, you will have the following tests every 12 weeks:

- Physical Examination, vital signs
- Blood test

WHEN YOU ARE FINISHED TAKING CHEMOTHERAPY

Rev. 7/08

You will be followed every 3 months for a doctor visit to assess the status of your health and your cancer.

STUDY CHART

Study Chart

You will receive hormonal therapy (treatments to lower testosterone) continuously and if you are randomized to receive chemotherapy when starting hormonal therapy, you will get docetaxel every three weeks for up to 6 doses in this study. This three week period of time is called a cycle. The cycle will be repeated six times. Each cycle is numbered in order. The chart below shows what will happen to you during Cycle 1 and future treatment cycles as explained previously if you are to receive chemotherapy when you start hormonal therapy. The left-hand column shows the day in the cycle and the right-hand column tells you what to do on that day.

Cycle 1-6

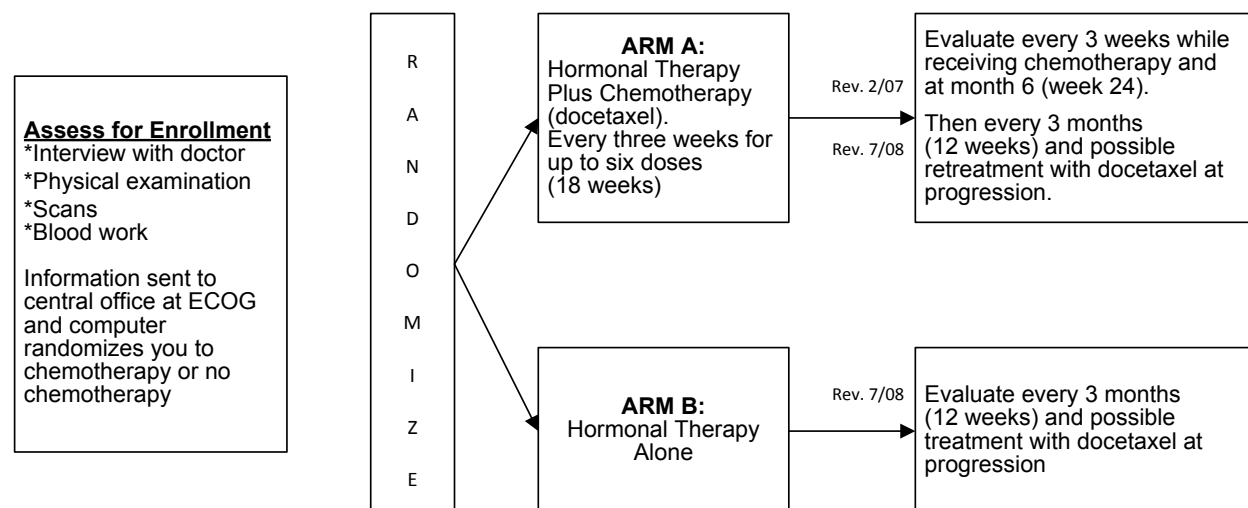
Day What you do

Rev. 7/08

Just before chemotherapy	Take dexamethasone (pre-medication to prevent reactions from chemotherapy) 12 hours, 3 hours and 1 hour prior to infusion
Day 1	Check into doctor's office, get blood tests and if suitable, get chemotherapy

If you are randomized to hormonal therapy alone, you will come to the doctor's office and be followed and receive "hormone" shots if your testicles were not removed. Chemotherapy is to be reserved for when your cancer progresses with a low testosterone.

STUDY PLAN



HOW LONG WILL I BE IN THE STUDY?

We think you will be in the study for several months.

We would like to keep track of your medical condition for 10 years to look at the long-term effects of the study.

CAN I STOP BEING IN THE STUDY?

Yes. You can decide to stop at any time. Tell the doctor if you are thinking about stopping or decide to stop. He or she will tell you how to stop safely.

It is important to tell the doctor if you are thinking about stopping so any risks from the chemotherapy and/or hormonal therapy can be evaluated by your doctor. Another reason to tell your doctor that you are thinking about stopping is to discuss what follow-up care and testing could be most helpful for you.

The doctor may stop you from taking part in this study at any time if he/she believes it is in your best interest; if you do not follow the study rules; or if the study is stopped.

WHAT SIDE EFFECTS OR RISKS CAN I EXPECT FROM BEING IN THE STUDY?

You may have side effects while on the study. Everyone taking part in the study will be watched carefully for any side effects. However, doctors don't know all the side effects that may happen. Side effects may be mild or very serious. Your health care team may give you medicines to help lessen side effects. Many side effects go away soon after you stop taking the chemotherapy and/or hormonal therapy. In some cases, side effects can be serious, long lasting, or may never go away. Death is rare, but possible.

You should talk to your doctor about any side effects that you have while taking part in the study.

Risks and side effects related to the study include those which are:

Docetaxel (Taxotere)

Likely

- Nausea, vomiting or diarrhea
- Lowered white blood cell count (may make you more likely to get an infection)
- Lowered platelets (may make you more likely to bruise or bleed)
- Lowered red blood cell count (may make you feel tired or weak)
- Numbness and pain of the hands and feet
- Hair loss
- Muscle weakness/muscle and joint aches

Rev. 7/08

- Mild to severe allergic reaction at the time the infusion is given
- Nail changes - drying and lines
- Fluid in arms and/or legs
- Changes in liver enzymes
- Fever
- Skin rash or dry skin
- Loss of appetite
- Taste changes
- Fatigue (feeling tired)

Less Likely

- Shortness of breath
- Sores in the mouth or throat
- Itching
- Headaches
- Fluid around the heart or the lungs
- Changes in kidney function tests which may lead to stopping docetaxel
- Low blood pressure
- Irregular heart beat (arrhythmias) or heart failure
- Skin irritation, redness, heat, swelling and pain at the site of injection of the medication
- Redness or irritation of the skin at a prior site of radiation therapy
- Irritation of the eye
- Low blood sodium level
- An abnormally high level of the simple sugar glucose in the blood
- Inflammation of the esophagus (the passageway between the mouth and the stomach)
- Inflammation of the pancreas (a small gland behind the stomach that secretes substances such as insulin)

Rare, but serious

- Seizures
- Blood clots
- Damage to the intestines which could be life-threatening
- Liver damage or failure

Rev. 7/08

LHRH agonists (such as leuprolide, goserelin, triptorelin)

Likely

- Hot flashes
- Swelling caused by fluid held in the tissues
- Abnormal enlargement of male breasts
- Bone pain, loss of bone density
- Blood clotting within blood vessels.
- Gastrointestinal (stomach) disturbances – diarrhea, nausea, vomiting

Less Likely

- The inability to achieve an erection or have sexual intercourse.
- Loss of sexual desire
- Weight loss
- Weight gain
- Depression
- Increase in liver enzyme
- Dizziness
- Headache
- Decreased number of red blood cells
- Increased thirst and urination
- Unusual taste in mouth

Rare, but Serious

- Rash due to allergic reaction
- Difficulty breathing

Antiandrogens (flutamide and bicalutamide)

Likely

- Abnormal enlargement of male breasts
- Diarrhea
- Nausea, vomiting
- Decreased number of red blood cells
- Breast tenderness and swelling
- Hot flashes
- High blood pressure

Less Likely

- Elevated level of liver enzymes indicating liver injury
- Too much bile in the blood causing a yellow color to the skin, gums, eye, and other tissues
- The inability to achieve an erection or have sexual intercourse.
- Loss of sexual desire
- Fatigue

Rare, but Serious

- Death due to liver failure
- A condition in which the skin is overly sensitive to light

Reproductive risks: You should not become pregnant or father a baby while on this study because the drugs in this study can affect a fetus. Women should not breastfeed a baby while on this study. It is important you understand that you need to use birth control while on this study. Check with your doctor about what kind of birth control methods to use and how long to use them. Some methods might not be approved for use in this study.

For more information about risks and side effects, ask your doctor.

ARE THERE BENEFITS TO TAKING PART IN THE STUDY?

Taking part in this study may or may not make your health better. While doctors hope chemotherapy and/or hormonal therapy will be more useful against cancer compared to the usual treatment, there is no proof of this yet. We do know that the information from this study will help doctors learn more about chemotherapy and/or hormonal therapy as a treatment for cancer. This information could help future cancer patients.

WHAT OTHER CHOICES DO I HAVE IF I DO NOT TAKE PART IN THIS STUDY?

Your other choices may include:

- Getting treatment or care for your cancer without being in a study
- Taking part in another study
- Getting no treatment

Talk to your doctor about your choices before you decide if you will take part in this study.

WILL MY MEDICAL INFORMATION BE KEPT PRIVATE?

The ECOG-ACRIN Cancer Research Group (ECOG-ACRIN) is conducting this study. ECOG-ACRIN is a cancer group that conducts studies for the National Cancer Institute. Your doctor is a member of ECOG-ACRIN or another group that is participating in this study. To help protect your privacy, ECOG-ACRIN has obtained a Confidentiality Certificate from the Department of Health and Human Services (DHHS).

With this Certificate, ECOG-ACRIN cannot be forced (for example, by court subpoena) to disclose information that may identify you in any federal, state or local civil, criminal, administrative, legislative or other proceedings. Disclosure will be necessary, however, upon request of DHHS for audit or program evaluation purposes.

You should understand that a Confidentiality Certificate does not prevent you or a member of your family from voluntarily releasing information about you or your involvement in this research. Note, however, that if an insurer or employer learns about your participation and obtains your consent to receive research information, then ECOG-ACRIN may not use the Certificate of Confidentiality to withhold this information. This means that you and your family must also actively protect your privacy.

Finally, you should understand that your doctor and ECOG-ACRIN are not prevented from taking steps, including reporting to authorities, to prevent serious harm to yourself or others and the Certificate does not prevent the review of your research records under some circumstances by certain organizations for an internal program audit or evaluation. Organizations that may inspect and/or copy your research records for quality assurance and data analysis include groups such as:

ECOG-ACRIN Cancer Research Group

National Cancer Institute

Food and Drug Administration

Other regulatory agencies and/or their designated representatives

Drug manufacturers, and/or their representatives

Central Laboratories

Cancer Trials Support Unit (CTSU), a research group sponsored by the National Cancer Institute (NCI) to provide greater access to cancer trials

[Note to Local Investigators: The NCI has recommended that the local institution address HIPAA regulations. The regulations may or may not be included in the informed consent form depending on local institutional policy.]

A description of this clinical trial will be available on <http://www.ClinicalTrials.gov> as required by U.S. Law. This Web site will not include information that can identify you. At most, the web site will include a summary of the results. You can search this web site at any time.

WHAT ARE THE COSTS OF TAKING PART IN THIS STUDY?

You and/or your health plan/ insurance company will need to pay for some or all of the costs of treating your cancer in this study. Some health plans will not pay these costs for people taking part in studies. The docetaxel will be provided free of charge, when starting hormone therapy but will not be provided free of charge if your disease worsens, as this would be standard of care. Check with your health plan or insurance company to find out what they will pay for. Taking part in this study may or may not cost your insurance company more than the cost of getting regular cancer treatment.

You will not be paid for taking part in this study.

For more information on clinical trials and insurance coverage, you can visit the National Cancer Institute's Web site at:

<http://www.cancer.gov/clinicaltrials/learningabout/payingfor>

You can print a copy of the "Clinical Trials and Insurance Coverage" information from this Web site.

Another way to get the information is to call 1-800-4-CANCER (1-800-422-6237) and ask them to send you a free copy.

WHAT HAPPENS IF I AM INJURED BECAUSE I TOOK PART IN THIS STUDY?

It is important that you tell your doctor, _____ [investigator's name(s)], if you feel that you have been injured because of taking part in this study. You can tell the doctor in person or call him/her at _____ [telephone number].

You will get medical treatment if you are injured as a result of taking part in this study. You and/or your health plan will be charged for this treatment. The study will not pay for medical treatment.

WHAT ARE MY RIGHTS IF I TAKE PART IN THIS STUDY?

Taking part in this study is your choice. You may choose either to take part or not to take part in the study. If you decide to take part in this study, you may leave the study at any time. No matter what decision you make, there will be no penalty to you and you will not lose any of your regular benefits. Leaving the study will not affect your medical care. You can still get your medical care from our institution.

We will tell you about new information or changes in the study that may affect your health or your willingness to continue in the study.

In the case of injury resulting from this study, you do not lose any of your legal rights to seek payment by signing this form.

Rev. 6/14

WHO CAN ANSWER MY QUESTIONS ABOUT THE STUDY?

You can talk to your doctor about any questions or concerns you have about this study. Contact your doctor _____ [name(s)] at _____ [telephone number].

For questions about your rights while taking part in this study, call the _____ [name of center] Institutional Review Board (a group of people who review the research to protect your rights) at _____ (telephone number). *[Note to Local Investigator: Contact information for patient representatives or other individuals in a local institution who are not on the IRB or research team but take calls regarding clinical trial questions can be listed here.]*

You may also call the Operations Office of the NCI Central Institutional Review Board (CIRB) at 888-657-3711 (from the continental US only). *[*Only applies to sites using the CIRB.]

Please note: This section of the informed consent form is about additional research studies that are being done with people who are taking part in the main study. You may take part in these additional studies if you want to. You can still be a part of the main study even if you say ‘no’ to taking part in any of these additional studies.

You can say “yes” or “no” to each of the following studies. Please mark your choice for each study.

QUALITY OF LIFE STUDY

We want to know your view of how your life has been affected by cancer and its treatment. This “Quality of life” study looks at how you are feeling physically and emotionally during your cancer treatment. It also looks at how you are able to carry out your day-to-day activities.

This information will help doctors better understand how patients feel during treatments and what effects the medicines are having. In the future, this information may help patients and doctors as they decide which medicines to use to treat cancer.

Rev. 2/07

You will be asked to complete 5 questionnaires: one on your first visit, one at 3 months (week 12), one at 6 months (week 24), one at 9 months (week 36), and one at 12 months (week 48). Each questionnaire has 4 parts. It takes about 15 minutes to fill out each questionnaire.

If any questions make you feel uncomfortable, you may skip those questions and not give an answer.

Rev. 7/08

If you decide to take part in this study, the only thing you will be asked to do is fill out the five questionnaires. You may change your mind about completing the questionnaires at any time.

Just like in the main study, we will do our best to make sure that your personal information will be kept private.

Please circle your answer.

Rev. 2/07

I choose to take part in the Quality of Life Study. I agree to fill out the five Quality of Life Questionnaires.

YES

NO

Rev. 12/11 **ABOUT PROVIDING SPECIMENS FOR RESEARCH**

Please read this form and ask about anything that is not clear to you. This is part of the **informed consent** process for research. This is to inform you of the possible risks, benefits, and limits of giving your samples for research.

You are being asked to give some of your samples (called specimens) and related information to be used in research studies defined in the protocol and to be stored (banked) for future research. This may help researchers learn more about how to prevent, find and treat cancer and other diseases.

The choice to have your samples used for research is up to you. No matter what you decide, it will not affect your medical care.

Below is some general information you should know before agreeing to allow the use of your specimens for research. After the general information there are descriptions of the research projects. Each project is described separately, including the types of samples requested and how they are collected. Each description is followed by questions concerning your participation in the project. Your samples will be used only for the projects in which you agree to participate.

WHAT ARE SAMPLES AND WHERE ARE THEY STORED?

A sample is any material taken from your body such as tissue, blood, urine and other fluids.

If you agree, your samples will be sent to laboratories to be used in research or will be stored for research in a Cooperative Group bank. These banks are supported by the National Cancer Institute. Cooperative Group banks contain samples and information. Your samples are kept along with those from other people in the banks. Researchers then ask for samples from the banks to study them.

WHAT INFORMATION WILL BE COLLECTED?

When your samples are sent by the institution treating you to any research laboratory or bank, some personal identifying information (such as your initials) will be sent with the samples. Any personal identifying information sent with the samples is not given to other researchers. The personal identifying information is used only by the laboratory or bank to identify your samples. Your privacy will be protected to the fullest extent possible. This will be discussed later in the section "How will information related to my samples be protected?"

Personal information about you and your health is collected when you participate in a clinical trial. Examples of other information that might be used for research include:

- Dates of medical procedures
- Any diagnosis and stage of your disease (if you have cancer)
- Your age and race
- Medical and family history
- Treatments you had

How you responded to treatments

WHAT WILL HAPPEN TO MY SAMPLES IF I AGREE TO GIVE THEM FOR RESEARCH?

If you agree to provide samples for the planned laboratory research studies, your samples will be sent to researchers who will study them to find answers to specific questions. These researchers may receive some personal identifying information but it will be used only to identify your samples.

If you agree to let your samples be kept for future research (research not yet defined), your samples will be stored in a Cooperative Group bank. The samples will be kept until they are used up or destroyed. The samples are given a code to protect your privacy before they are used. Any related information given to researchers will also be coded. Researchers will receive the code instead of any information that might directly identify you.

You or your doctor will not be given reports or other information about the research that uses your samples. This information will not be put into your health record. Results may be used for future research.

You will not be named or identified by other personal information if any results are published. Most publications contain results from many patients.

Your samples and related information will be used only for research and will not be sold. It is possible that research may help to create new products or treatments. If this should happen, you will not be paid.

Coded data from some research studies that use samples could be put into secure Internet databases that can be shared by other approved researchers. This could include genetic data. Current safety rules are followed to safeguard your privacy. Your name or contact information will not be put in the database.

WHAT KIND OF RESEARCH WILL BE DONE WITH MY SAMPLES?

Many types of research use normal or diseased (**cancerous**) samples. Researchers can study proteins, RNA and DNA (genes). The study of genes (DNA) is often called **genetic research**.

For example, your samples may be looked at:

- To see if a trait is passed down in families from one generation to the next (**inherited**). This type of research may help to explain why some cancers run in families or why some people have side effects from treatment while others do not. This is often studied through blood cells and DNA (**genes**).
- To learn about changes in the body that happen after you were born (**non-inherited**). For example, being in the sun too much can cause changes in cells that lead to skin cancer.

WILL IT HELP ME IF I GIVE MY SAMPLES FOR RESEARCH?

Using your samples for research will probably not help you. We do hope the research results will help people in the future. The best way to prevent, find or treat cancer and other diseases is by studying human samples and data.

WHAT ARE THE RISKS OF GIVING MY SAMPLES FOR RESEARCH?

There is a risk that your information could be misused. The chance of this happening is very small. We have many protections in place to lower this risk. See the next section, “How will the information related to your samples be protected?” Your privacy will be protected to the fullest extent possible.

There can be a risk in knowing genetic information. New health information about inherited traits that might affect you or your blood relatives could be found during a research study. Even though your genes are unique, you share some of the same genes with your blood relatives. Very rarely health or genetic information could be misused by employers, insurance companies, and others. For example, life insurance companies may charge a higher rate based on this information.

Some states have laws to protect against genetic discrimination [*list appropriate state information if your state has such laws*]. A new federal law called the Genetic Information Non-Discrimination Act, or GINA is in effect. This law helps to lower the risk of health insurance or employment discrimination. The law does not include other types of misuse by life insurance or long term care insurance. To learn more about the GINA Law, please check the Internet or ask the study staff.

Although we are not able to know all of the risks from taking part in research on inherited traits, we believe that the risks to you and your family are very low, because research results will not be returned to you or your doctor.

HOW WILL INFORMATION RELATED TO MY SAMPLES BE PROTECTED?

We have many ways to protect the information related to your samples:

Your samples and information receive a unique code. Researchers only receive coded samples and information, and will not be able to link the code to you. Only approved people in the ECOG-ACRIN Cancer Research Group can match you to the code on your samples and related information.

Strict security safeguards are in place to reduce the chance of misuse or unplanned release of information. Steps we take include password protected access to databases and restricted access to freezers or rooms that contain samples.

Before samples are given to researchers, studies are reviewed for the quality of the science and for patient protection. Records from research studies can be reviewed by the Cooperative Group, by the sponsor, and by government agencies. This is to make sure the research follows the rules of the Cooperative Group and state or federal laws.

Research results will not be returned to you or your doctor. If research results are published, your name and other personal information will not be given.

ECOG-ACRIN also has a Certificate of Confidentiality from the U.S. Department of Health and Human Services. The Certificate protects against the forced release of personal information from the Cooperative Group bank or database.

What this means is that ECOG-ACRIN cannot be forced to disclose your identity to any third party. It is possible that for some legal proceedings, the Certificate of Confidentiality could be over-ridden by a court of law

MAKING YOUR CHOICE

The choice to take part is up to you. You may choose not to let us store and use your samples. If you decide not to let us store and use your samples, you will still receive the same medical care and you may still participate in the treatment part of this clinical trial. You may also take part in other research studies.

If you decide that your samples can be kept, you may change your mind at any time. Contact the study staff at your hospital and let them know that you do not want your samples used for research [*Insert contact number*]. Then, any sample that remains in the bank will no longer be used. Samples that have already been given to or used by researchers cannot be returned or destroyed.

Please read the research study descriptions below, review the questions carefully and circle “Yes” or “No”. **If you have any questions, please talk to your doctor or nurse, or call the institution’s research review board at [*IRB’s phone number*].**

Laboratory research studies

This study includes one or more laboratory tests that will analyze a small sample of blood and tissue. The tissue sample will be from your diagnostic biopsy or when your prostate was surgically removed, an additional biopsy will not be needed. Blood will be collected by inserting a needle into a vein and drawing 2 to 4 tablespoons of blood. This will be done before you begin treatment, 6 months into your treatment, and if your disease progresses. The tissue and blood will be sent to central laboratories, where researchers will perform tests in order to better understand prostate cancer. The results from these tests will not be sent to you or your doctor, and they will not be used in planning your care. These tests are only for research purposes.

Making Your Choice

Please read the sentence below and think about your choice. After reading the sentence, circle "Yes" or "No."

I agree to participate in the scientific laboratory research studies that are being done as part of this clinical trial.

Yes No

Using specimens for future research

We would like to keep some of your specimens for future research.

If you participate in the laboratory research studies associated with this protocol, this means any specimens left over from the laboratory studies will be stored. No additional samples are being requested. If you choose not to participate in the laboratory research studies, we would still like some of the tissue from your biopsy or surgery for future research.

Although most future research studies will focus on cancer, some research projects may also include other diseases, such as heart disease, diabetes or Alzheimer's disease.

As indicated above, the specimens will only be given to researchers approved by scientific reviewers appointed by the ECOG-ACRIN Cancer Research Group. Any research done on the specimens must also be reviewed by the researcher's Institutional Review Board.

Please read the questions below carefully and circle "Yes" or "No".

Rev. 7/08

May we have your tissue and blood samples leftover from the laboratory research studies (if submitted) for future research about cancer?

My specimens may be kept for use in research to learn about, prevent, treat, or cure cancer.

Yes No

Rev. 7/08

My specimens may be kept for use in research about other health problems (for example, causes of diabetes, Alzheimer's disease, and heart disease).

Yes No

Rev. 7/08

Someone from this institution may contact me in the future to ask me to take part in more research.

Yes No

WHERE CAN I GET MORE INFORMATION?

You may call the National Cancer Institute's Cancer Information Service at:
1-800-4-CANCER (1-800-422-6237) or TTY: 1-800-332-8615

You may also visit the NCI Web site at <http://cancer.gov/>

For NCI's clinical trials information, go to: <http://cancer.gov/clinicaltrials/>

For NCI's general information about cancer, go to
<http://cancer.gov/cancerinfo/>

You will get a copy of this form. If you want more information about this study, ask your doctor.

SIGNATURE

I have been given a copy of all _____ *[insert total of number of pages]* pages of this form. I have read it or it has been read to me. I understand the information and have had my questions answered. I agree to take part in this study.

Participant _____

Date _____

Background

This study's statistical analysis plan was contained within the protocol document in the section entitled "Statistical Considerations". This is Section 9 of the protocol and can be found on page 40 of the original protocol. As described below, there were significant changes to the statistical analysis plan in Addendum 3, dated July 9, 2008, and in Addendum 5, dated December 7, 2011. The final version of the protocol reflects the statistical analysis plan after that addendum.

The protocol describes the interim analysis plans for the study, which were timed to correspond to semiannual meetings of the ECOG Data Monitoring Committee. Information for these analyses is extracted from the clinical database exactly eight weeks prior to each meeting. Thus, the information proportions would not align exactly with the hypothetical table provided in the protocol.

At the time data were extracted for the Data Monitoring Committee meeting of October 2013, there was 53% information for the efficacy analysis. Pre-specified criteria for significance were reached, and the Data Monitoring Committee released the data for presentation. Once this decision was made, the study had essentially achieved "full" information, so the data was finalized soon thereafter in terms of the primary endpoint. This report thus represents data with a cutoff for survival of December 23, 2013. No further updates were made in order to preserve the Type I error. Data cleaning continued for other aspects of the study, including treatment, adverse events, and secondary endpoints. All other data in this report apart from survival reflect the database as of December 23, 2014.

Original design

The primary endpoint of this study is overall survival. The study population consists of prostate cancer patients with extensive metastatic disease (high volume disease). Patients were stratified based on age (≥ 70 versus < 70), performance status (0-1 versus 2), prior adjuvant hormonal therapy (> 12 months versus ≤ 12 months), intended use of combined androgen blockade > 30 days from start of treatment, and bisphosphonate use (bisphosphonate was then changed to "FDA approved drugs for delaying skeletal related events" in addendum 5).

Survival time was to be compared between the chemohormonal therapy arm (Arm A) and the androgen ablation arm (Arm B) by the log-rank test. This study was designed to detect a 33.3% improvement in median survival time from 24 months in the androgen ablation arm to 32 months in the chemohormonal therapy arm, corresponding to a hazard ratio of 1.33, with one-sided Type I error of 0.025. Planned full information was 408 deaths, which provided 80% power, allowing for the interim analysis schedule described below.

Accrual of 12 eligible patients per month for 45 months (568 total patients allowing a 5% ineligibility rate) followed by 36 months of follow-up was anticipated to achieve full information. Interim analyses were to be performed for all semi-annual ECOG Data Monitoring Committee meetings beginning when approximately 25% of the planned full information events had occurred, about 29 months after the study opens, continuing every 6 months until either the criterion for early stopping was met or full information was reached. The truncated O'Brien-Fleming group sequential boundary was used to adjust for the sequential testing.

At each interim analysis, the nominal confidence interval on the overall hazard ratio comparing the androgen ablation arm to the chemohormonal therapy arm was to be computed. If the confidence interval

did not contain the target alternative of 1.33, then the Data Monitoring Committee would consider closing an arm of the trial or stopping the study early for overall lack of treatment differences.

Addendum 3

Addendum 3, activated in July 2008, expanded study eligibility to all patients with both low and high volume metastatic disease; eligibility was no longer restricted to patients with high volume disease. Extent of disease (high vs. low volume disease) was also added as an additional stratification factor in this addendum.

Besides the change in eligibility criteria, the planned accrual rate was changed from 12 patients per month to 8 patients per month. The proportion of patients with high volume disease was assumed to be 50%. Median survivals in the androgen ablation arm were expected to be 24 months and 48 months for patients with high volume disease and low volume disease, respectively. Based on the assumptions above, the sample size was slightly increased from 568 patients to 600 patients (300 patients per arm). With the projected accrual rate, the accrual goal was to be reached in 75 months with a follow up period of 30 months. The study duration was approximately 8.5 years.

As with the original design, the study as revised had 80% power to detect a 33.3% improvement in median overall survival across treatments, using a stratified log-rank test with a one-sided 0.025 significance level. The primary comparison was an intent-to-treat analysis including all randomized patients. Due to the changes above, the interim analysis schedule was also revised and full information was 402 deaths.

Addendum 5

The second major revision, addendum 5, increased the number of patients to be accrued from 600 patients to 780 patients. This was based on data from S9346 demonstrating that the overall survival of men with prostate cancer starting androgen deprivation therapy in the modern era is longer than the historical data which was used to develop the assumptions of addendum 3. In addition, the estimated proportion of patients with high volume disease was also changed from 50% to 70% based on the preliminary data released from the DSMC meeting in September 2011. The accrual goal was changed from 600 patients to 780 patients to maintain sufficient power to detect the original hypothesized hazard ratio.

Survival time was to be compared between the chemohormonal therapy arm and the androgen ablation arm by the stratified log-rank test. Median survivals in the androgen ablation arm were expected to be 33 months and 67 months for patients with high volume disease and low volume disease, respectively. Again, the study's modification was designed to detect a 33.3% improvement in median survival time across treatments, corresponding to a hazard ratio of 1.33, with one-sided Type I error of 0.025. Planned full information was 399 deaths, which was to provide 80% power, allowing for the interim analysis schedule described below.

After the activation of addendum 3, E3805 was enrolling about 12 patients per month and exceeded the anticipated accrual rate of 8 patients per month detailed in addendum 3. Therefore, the anticipated accrual rate was changed to 12 patients per month. Accrual of 12 patients per month for about 5.5 years (780 total patients) followed by 1.5 years of follow-up was anticipated to achieve full information. Interim analyses were to be performed for all semi-annual ECOG-ACRIN DSMC meetings beginning when approximately 25% of the planned full information has occurred, about 35.5 months after the study opened, continuing

every 6 months until either the criterion for early stopping was met or full information was reached. The truncated O'Brien-Fleming group sequential boundary was used to adjust for the sequential testing.